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Graduation Design Project

Final Report

Topic : **A Sustainable and Inclusive Agricultural
Business Model Design**

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1. INTRODUCTION

1.1 Background of the Study

Urbanization and industrial agricultural practices have drastically altered how people access fresh and natural food. While individuals and families increasingly seek healthy, organic, and sustainable food options, existing solutions often fail to meet these needs. Alternatives such as hobby gardening, organic food markets, and e-commerce platforms present significant challenges, including the time, effort, and expertise required for consistent agricultural practices. This highlights the need for an innovative and inclusive model that bridges urban consumers with sustainable agriculture.

Social cooperatives have emerged as a vital tool for addressing these challenges. They combine economic efficiency with a mission to solve social issues. By integrating disadvantaged groups, particularly women living in rural areas, into the workforce, social cooperatives create sustainable opportunities for economic independence and social inclusion. In this context, agriculture provides a unique platform that unites these principles, enabling communities to access fresh products while empowering marginalized individuals.

1.2 Problem Definition

Urban consumers face significant barriers to accessing fresh, natural, and healthy foods. The high cost of organic products, the time commitment required for hobby gardening, and the lack of transparency in production processes limit access to quality products. On the production side, traditional agricultural systems fail to integrate disadvantaged groups into sustainable and meaningful roles sufficiently.

Women living in rural areas, despite playing a vital role in agricultural production, often face limited opportunities, low income levels, and systemic social inequalities. The lack of sustainable models that empower these women to achieve economic independence exacerbates social and economic challenges in rural communities.

In addition to these social and economic issues, modern agricultural practices pose serious environmental sustainability risks. From water-intensive irrigation methods to the use of chemical fertilizers and pesticides, conventional farming practices contribute significantly to ecological degradation. The absence of integrated agricultural business models that balance consumer needs, environmental sustainability, and social responsibility underscores the urgent need for innovative solutions.

1.3 Project Scope and Goal

This project aims to design a fully manageable, customizable, and organic garden model based on the principles of sustainable agriculture and social inclusivity. By integrating women living in rural areas into the agricultural production process within a cooperative structure, the project seeks to achieve dual impacts: providing urban consumers with access to healthy and organic food and empowering rural women to achieve economic independence.

The scope of this project includes identifying the technical, legal, and operational requirements for implementing this model, as well as analyzing its sustainability and long-term viability. The ultimate goal is to develop a roadmap that promotes sustainable agriculture, strengthens social ties, and fosters economic empowerment through an innovative agricultural business model.

2. METHODOLOGY

In this study, an innovative agribusiness model that prioritizes sustainability and social impact was developed. A systematic methodology was adopted to understand user demands and a combination of both quantitative and qualitative data collection techniques were used. An online survey conducted between April and May 2025 provided information on the demographic characteristics of the participants, their interest in organic products, the importance they attach to transparency in production and their willingness to pay. Those who completed the survey also sent the survey link to their circle, which led to a snowball effect and increased the sample size, thus reaching the profile of the target audience suitable for the project. In addition, semi-structured interviews were conducted with five participants to gain more in-depth insights into consumer motivations, perceptions of trust and expectations of social benefits.

In line with the collected data, user needs were identified, these needs were evaluated according to their importance and ranked using the QFD (Quality Function Distribution) method. Nonparametric tests (Mann-Whitney U, Kruskal-Wallis and Spearman Correlation) were preferred since the survey results did not show normal distribution. These analyses were conducted in R Studio software and strong positive relationships were found especially between transparent production and social responsibility awareness. The results are instructive in supporting both the economic and social impacts of project design.

During the concept development phase, systematic comparisons were made with the Concept Screening method; in this context, a benchmark analysis was carried out comparing similar business models in the sector. In addition, a method based on a probability and impact matrix was used in the project risk management process, risks were categorized and a weighted scoring method was used for prioritization. The production, marketing, logistics and operational structures were based on realistic assumptions, and a

system design that focuses on customer experience was created by visualizing the process flow.

In the financial feasibility analysis, Net Present Value (NPV), Internal Rate of Return (IRR), Return on Investment (ROI) and Payback Period were calculated to assess the return of the project. These analyses were performed over a total of 9 different scenarios based on different capacity utilizations (66%, 83% and 100%) and discount rates (49%, 44.08% and 39.5%). The probability of each scenario was analyzed through Monte Carlo simulation with 10,000 iterations. With this method, the expected values of financial metrics were calculated by weighting them with probabilities and the impact of uncertainties was taken into account when making decisions about the project. As a result, the financial sustainability of the project has been established on a strong basis through multiple scenario analysis. All methods used in the project are listed in Table 1.

ID	Method	Section Where the Method is Used	Explanation
1	Systematic Literature Reviewing	Literature Review	Academic and sectoral resources on organic agriculture, social cooperatives and women's employment were reviewed.
2	Benchmark Analysis	Preliminary Analysis	Comparisons were made with existing similar business models (Çiftçiden Eve, Ovacık Doğal, Ekozel Organik).
3	Stakeholder Analysis	Preliminary Analysis	Internal and external stakeholders were systematically classified and assessed.
4	Risk Analysis	Preliminary Analysis	The probability and impact levels of risks were scored and weighted risk priorities were determined.
5	Process Flow Designing	Preliminary Analysis	Production and delivery processes were modeled with detailed process flow charts.
6	Survey Researching	Market Research and Survey Analysis	An online survey was conducted with N=59 respondents, measuring customer preferences, behavior and the price levels they are willing to pay.
7	Normality Tests and Distribution Analysis	Market Research and Survey Analysis	Normality tests were conducted to assess whether the data followed a normal distribution, which is essential for selecting appropriate statistical tests.
8	Descriptive Statistics	Market Research and Survey Analysis	Descriptive statistics were used to summarize the central tendency, variability, and overall patterns within the collected
9	Difference Tests and Hypothesis Tests	Market Research and Survey Analysis	Hypothesis tests were applied to examine whether there were statistically significant differences between groups or variables within the dataset.
10	Correlation Analysis	Market Research and Survey Analysis	Correlation analysis was employed to explore the strength and direction of relationships between different variables.
11	Non-Parametric Tests	Market Research and Survey Analysis	Shapiro-Wilk normality test, Mann-Whitney U test, Kruskal-Wallis test, Spearman Correlation analysis were applied.
12	In-depth Interviewing	Market Research and Survey Analysis	In-depth interviews were conducted to gather detailed qualitative insights from participants regarding their preferences, motivations, and concerns related to organic production and
13	Interview Coding	Market Research and Survey Analysis	The interview data were systematically coded to identify recurring themes and categorize participants' responses for further thematic analysis.
14	Thematic Frequency Analysis	Market Research and Survey Analysis	Themes were extracted from the in-depth interviews and a frequency table was created by coding.
15	QFD (Quality Function Deployment)	Business Model and Operational Structure	User needs were prioritized and matched with design decisions.
16	Financial Feasibility Analysis (NPV, IRR, ROI, PP)	Financial Analysis	9 scenario financial analyses were conducted; cash flow projections and return on investment analyses were calculated.
17	Monte Carlo Simulation (10,000 Iterations)	Financial Analysis	It was used to measure the impact of financial uncertainties.
18	Index Scoring Model	Financial Analysis	Capacity estimates were made based on market conditions, price alignment and customer interest.
19	Scenario Analysis	Financial Analysis	Alternative scenarios were developed for situations such as organic certification risk and financing problems.

Table 1: Methodologies used

3. LITERATURE REVIEW

3.1 Organic Agriculture

Conventional farming methods that use lots of chemicals and pesticides lose the fertility of the earth, hurt people, plants, and animals' health, and throw off the natural balance. In recent years, people have become more aware of the need to protect the environment. This has led to the creation and use of agricultural goods that are safe for people, animals, and plants, do not hurt the environment, and do not hurt living things. In this case, the goal was to improve the way standard farming is done while also making a more environmentally friendly way of farming (Özkan, 2022, as cited in Çiğ & Özek 2023).

Organic farming emerged as a novel agricultural production technique in the early 20th century, established by an organization of pioneers who methodically observed ecological disruptions. The inception of organic farming operations, initially diverse movements in wealthy nations, was consolidated under the International Federation of Organic Agriculture Movements (IFOAM) in 1972 (Aksoy et al., 2020 as cited in Çiğ & Özek, 2023).

Organic farming is a production system that forbids the use of chemical pesticides, hormones, and synthetic fertilizers, with the objective of enhancing both the quantity and quality of goods while maintaining ecological balance. Organic farming not only mitigates environmental damage but also guarantees the production of nutritious food and the conservation of biodiversity (Çiğ et al., 2022, as cited in Çiğ & Özek, 2023)

IFOAM asserts that organic agriculture is founded on four fundamental concepts. The primary premise is health, which promotes preserving and enhancing the health of the entire living ecosystem, encompassing soil, water, humans, animals, plants, and the environment. The second concept is ecology, which underscores the sustainable and efficient utilization of natural resources and the preservation of ecosystems. The third

principle, equity, is to ensure that all individuals have access to nutritious food and to improve their quality of life without bias. The final concept, care, emphasizes the imperative of safeguarding human health and preserving the environment in agricultural methods associated with organic farming. In organic agriculture, the utilization of genetically modified organisms (GMOs) and an array of chemicals is forbidden. These concepts are regarded as the foundational pillars of organic farming (Bulut & Şen, 2023, as cited in Çığ & Özek, 2023).

3.1.1 Organic Agriculture in the World

As reported by Research Institute of Organic Agriculture (FIBL), the global organic agriculture study indicates that in 2021, organic agricultural practices were conducted on 76.4 million hectares, reflecting a 1.6% annual increase in global organic agricultural land compared to 2020. In the 2021 ranking of countries by organic agricultural area, Australia occupies the leading position (35.7 million hectares), followed by Argentina (4.1 million hectares), and France (2.5 million hectares). The top 10 countries with the greatest organic agricultural lands encompass 59.6 million hectares, or about 80% of the global total for organic agriculture. Türkiye is positioned twenty-sixth globally, with 327,583 hectares designated for organic agriculture (2023).

In 2021, organic agriculture accounted for 1.6% of the global agricultural land. In EU nations that have established policies and framework agreements to expand organic farming and diminish chemical usage in agriculture, the proportion of organic farming areas relative to the total agricultural area is 9.6%. In other continents and regions globally, this rate persists below 1%. In 20 nations where organic agriculture is implemented, the proportion of organic farming land constitutes 10% or more of the total agricultural area, with the majority of these nations situated in Europe. Liechtenstein possesses the greatest proportion of organic farming areas globally, accounting for 40.2%. In Türkiye, organic farming constitutes 0.9% of the total agricultural land. The expanse of organic farming regions has been increasing globally throughout the years. In 2021, the latest available

data indicates that global organic farming areas expanded by 1.7% compared to 2020. Figure 1 illustrates that the area dedicated to organic farming has quintupled between 2000 and 2021, rising from 15 million hectares in 2000. Globally, organic farming areas have increased by 108.4% between 2011 and 2021. (FIBL, 2023).

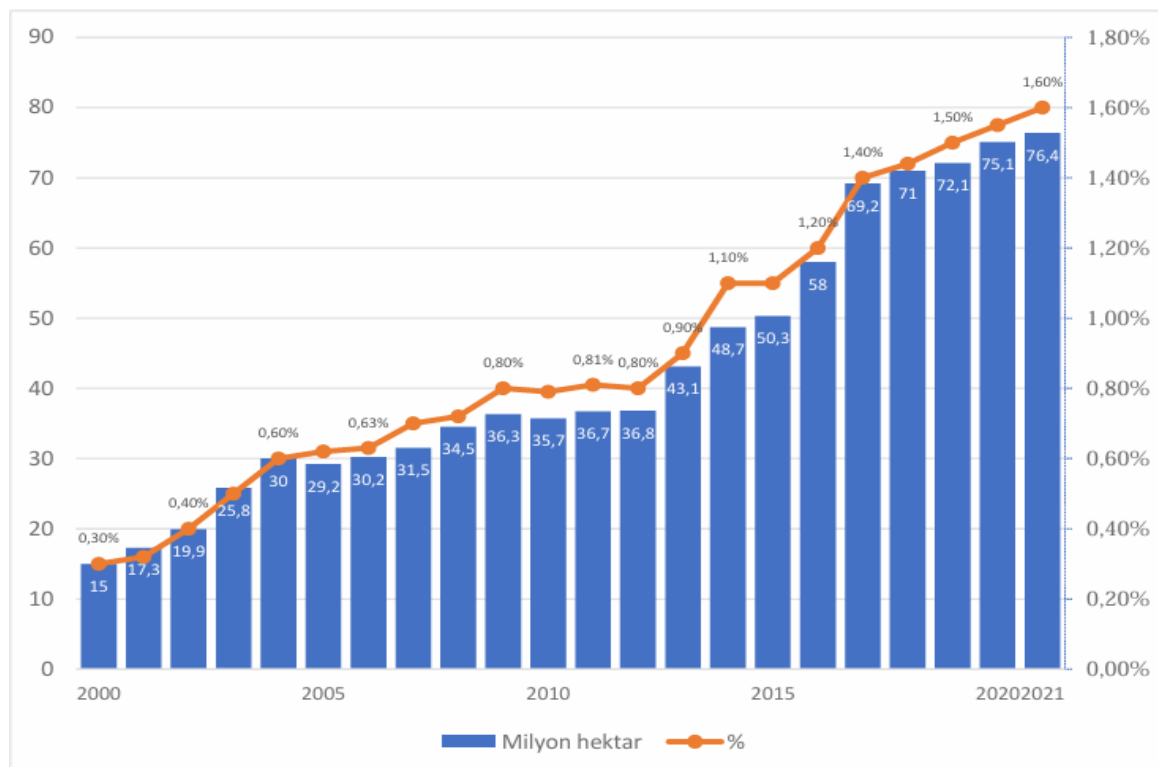


Figure 1: Changes in organic farming areas over the years and their shares in total farming areas (FIBL, 2023)

In 2021, the global count of producers involved in organic agriculture was reported to reach 3.7 million individuals. In 2021, the global count of organic agricultural producers rose by 170,000 individuals, representing a 4.9% rise compared to 2020. Although the number of organic agricultural producers rose in Africa, Oceania, Latin America, Europe, and North America, it declined solely in Asia. In comparison to 2011, the global count of

organic agricultural producers rose by 1,902,412 individuals, representing a growth of 107.7% (FIBL, 2023).

3.1.2 Organic Agriculture in Türkiye

The first statistics published by The Republic of Türkiye Ministry of Agriculture and Forestry (MoAF) on its website concerning organic farming in Türkiye date back to 2002. Analysis of these numbers reveals that 12,395 farmers cultivating organic products in Türkiye produce 310,125 tons of organic goods across 89,827 hectares of land, encompassing 150 product varieties as of 2002. Organic agricultural production in Türkiye has varied over the years due to climatic conditions, economic circumstances, support for organic farming, and the policies of MoAF. In 2008, a record was established for the variety of goods cultivated, although 2018 marked the year with the highest number of organic farmers, totaling 79,563 individuals (MoAF 2022).

As of 2002, when official numbers were first released, organic agricultural output in Türkiye occurred throughout 42 provinces with 12,395 farmers. Erzurum secured the top position with 1,734 farmers, followed by Aydın with 1,406 farmers. Figure 2 illustrates the provinces (NUTS1) where one thousand or more farmers cultivate organic crops, based on 2021 statistics. The Eastern Black Sea, Central Black Sea, and Aegean Regions are prominent in the number of farmers engaged in organic crop production. At the NUTS3 level, in 2021, Rize province, which had no registered organic producers in 2002, ascended to first place with a total of 10,542 producers, while Aydın province secured second place with 7,380 producers. According to 2021 data, Artvin (2,258), Trabzon (1,796), İzmir (1,597), and Manisa (1,597) follow these two provinces (MoAF 2022).



Figure 2: Provinces with the highest number of organic farmers (NUTS3) (2022)
(TurkStat, 2023)

Table 2 presents the data of the provinces with the highest number of farmers engaged in organic agricultural production in Türkiye according to the average of the last five years, with Aydın leading the way with 11,849 producers. Aydın's neighboring regions, İzmir and Manisa, occupy the third and fourth positions with 2,833 and 2,711 producers, respectively. Rize, located in the Eastern Black Sea Region, ranks second with 10,509 farmers, while Van, situated in the Eastern Anatolia Region, ranks fifth with 2,671 farmers. Over the past five years, the cumulative count of farmers in these five provinces is 30,573, representing over half (47.6%) of the national total of farmers involved in organic agriculture, which is 64,105 throughout the same period (MoAF, 2022).

CITIES	2002	2011	2017	2018	2019	2020	2021	5 year average
Aydın	1.406	4.739	13.588	14.388	14.011	9.877	7.38	11.849
Rize	0	188	10.225	10.711	10.616	10.453	10.542	10.509
İzmir	1.247	1.409	3.416	3.707	3.226	2.217	1.597	2.833
Manisa	1.032	1.677	4.143	2.387	3.172	2.258	1.597	2.711
Van	0	7.34	5.541	4.595	2.815	342	60	2.671

Table 2: The average number of farmers doing organic farming in Türkiye over the last five years
(MoAF, 2022)

In Türkiye, the number of farmers involved in organic plant production reached its highest point in 2018, totaling 79,563. Following this peak, there was a notable decline in the number of organic producers, resulting in a final count of 38,748 in 2021. The data indicates that the number of farmers involved in organic plant production experienced a proportional decline of 51.3% from 2018 to 2021. The decline in the number of farmers involved in organic production in recent years can be attributed to rising production costs, diminishing market opportunities, and inadequate state support for organic agriculture (MoAF, 2022). During the specified period, there was a decline in the number of farmers involved in organic plant production across 68 provinces. Aydın, Van, and Kastamonu experienced the most significant declines in the number of farmers. The decline in the number of farmers involved in organic plant production in Aydın province was 48.7%, whereas, in Van and Kastamonu provinces, the reductions were significantly higher at 98.7% and 99.6%, respectively. Figure 3 illustrates the provinces that experienced a decline of a thousand or more farmers involved in organic farming in 2021 relative to 2018 (MoAF, 2022).

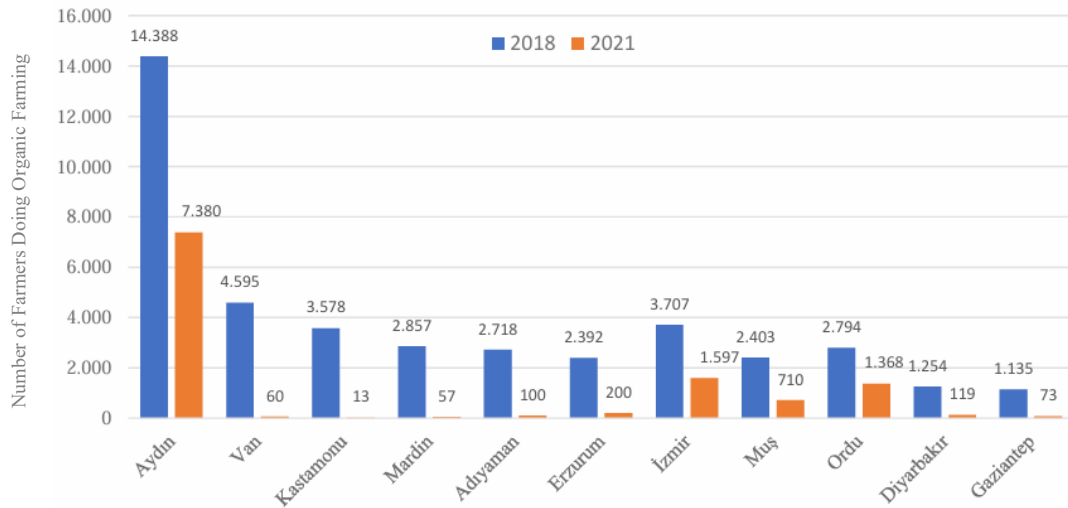


Figure 3: Provinces with the biggest decrease in the number of farmers doing organic farming in Türkiye in 2021 compared to 2018 (MoAF, 2022)

A key consideration in analyzing the evolution of organic agriculture in Türkiye and its geographical distribution is the allocation of organic crop production areas by province. Table 3 displays that Aydın province, which has the greatest average number of farmers involved in organic production over the past five years, also leads in organic plant production area over the same period. Aydın province, with a total production area of 50,961 hectares, is succeeded by Kars (39,417 hectares) and Ağrı (30,158 hectares). Based on the last five-year (2017-2022) average, 41 provinces possess an organic plant-producing area of 1,000 hectares or more. Accordingly, the first fifteen provinces with the biggest organic plant production area account for 79.40% of the total organic production area in the country (MoAF, 2022).

CITIES	2002	2017	2018	2019	2020	2021	5 year average
Aydın	8.739	55.125	57.328	56.191	44.809	41.351	50.961
Kars	0	46.429	50.509	46.38	28.375	25.392	39.417
Ağrı	0	33.46	31.959	32.961	28.273	24.138	30.158
Van	0	56.65	50.525	31.041	3.744	739	28.54
Muş	0	28.217	28.965	24.814	15.743	11.901	21.928

Table 3: *Distribution of organic plant production areas by province in Türkiye (hectares)*
(MoAF, 2022)

In 2021, organic plant production in Türkiye occurred across a total area of 243,778 hectares, yielding 1,101,236 tons of goods. During the year, a total of 230 commodities were produced, with olive and wheat occupying the first and second positions in both production area and quantity. Table 4 shows that, in 2021, olive grove areas comprised 17.46% of the total organic plant production, covering an area of 42.5 thousand hectares. 66.37% of organic olive cultivation areas are located in Aydın province. In 2021, wheat occupied the second position in the organic production area with 31.7 thousand hectares, while barley held the third position with 18.8 thousand hectares. The proportion of the initial fifteen goods on the list (olive, wheat, barley, hazelnut, fig,

oat, apple, clover, meadow grass, pine nut, grape, apricot, vetch, and tea) constituted 75.79% of the overall organic plant production area (MoAF, 2022).

Crops	Planting Area (Hectares)	Percentage (%)
Olive	42.563,24	17,46
Wheat	31.775,04	13,03
Barley	18.872,76	7,74
Hazelnut	14.427,26	5,92
Fig	11.826,78	4,85
Oats	9.127,50	3,74
Apple	8.306,55	3,41
Alfalfa	8.272,50	3,39
Meadow Grass	6.886,30	2,82
Pine Nuts	6.797,18	2,79
Sainfoin	6.411,28	2,63
Grapes	6.145,92	2,52
Apricot	4.982,22	2,04
Vetches	4.373,12	1,79
Tea	3.825,86	1,57
Other	59.185,24	24,21
Total	243.778,85	100

Table 4: *Distribution of organically produced plants in Türkiye according to production area size.*
(MoAF, 2022)

3.1.3 Marketing of Organic Agriculture

As stated by FIBL (2022),

“The global organic agriculture market, valued at 120.6 billion Euros in 2020, is experiencing annual growth. The rise in health issues and changing dietary patterns, together with the adverse impacts of conventional agriculture, facilitate the expansion of the organic agricultural and organic food products market. Despite being one of the main nations in terms of the number of organic agriculture

producers globally, Türkiye's market share in this area remains considerably restricted. Türkiye's market volume increased from 2 million Euros in 2006 to 46 million Euros in 2020. Also, as per 2020 data, Türkiye ranks as the seventh largest exporter of organic products to EU nations.’’

Marketing is one of the biggest requirements of sales. When looking at the marketing of organic products, as stated by Bulut (2011, cited in Saygı & Bulut, 2024), “No product that is not produced in accordance with the provisions of the Regulation and does not have a “Product Certificate” indicating that it is an organic raw material and/or an organic processed product can be marketed as an organic product. Entrepreneurs trading organic products must document each sales party with a “Product Certificate”. Organic products are sold by clearly stating that they are organic products. It is the responsibility of the entrepreneur selling organic products to ensure that organic products are not confused with conventional products and that their organic structure is preserved. Organic products must definitely carry one of the logos specified in the Regulation in the Turkish domestic market.”

The transportation of organic products from producer to consumer can be done through many different methods. In what follows, we will discuss the marketing methods of organic products and the content of these methods (Bulut, 2011, as cited in Saygı & Bulut, 2024):

- Large Supermarket Chains: Large supermarkets serve as a significant conduit for marketing organic products due to their ability to access a diverse consumer base. Consumers can readily access the products they require due to the organic product categories in these markets. The use of natural materials in product packaging and the availability of small-sized packages favorably influence consumer preferences. Some prominent grocery chains in Türkiye have begun to establish areas for organic products.

- Farmers' Markets: Farmers' markets provide consumers the chance to purchase goods directly from the producer. This scenario fosters trust and delivers a gratifying experience for consumers. The diversity of products is a significant determinant in the preference for farmers' markets. These marketplaces are venues where various natural products are offered alongside organic products.
- Farm Shop: Farm markets provide people with the chance to purchase items directly from the farms of origin. This approach can be highly effective if consumer happiness is attained. Factors such as transit accessibility, parking facilities, children's play spaces, and security significantly influence the success of agricultural markets. Moreover, product diversity, eco-friendly and recyclable packaging (such as cloth bags and baskets), along with efficient promotion and skilled staff assistance, enhance the efficacy of this marketing method.
- Mail Order: The mail order delivery system provides consumers with remote access to organic items. Orders must be received consistently through telephone, email, or online forms. The order forms must explicitly delineate product variety, pricing, and delivery alternatives (freight, courier, on-site delivery). This approach necessitates robust communication links to guarantee user satisfaction.
- Online Sales: Online sales are a rapid and accessible approach to selling organic products. An intuitive, rapid-loading, and straightforward website design streamlines the ordering experience for consumers. Products must be supplied via freight, mail, or courier; uncertainties around payment security must be resolved. A uniform return policy will enhance consumer confidence.
- Tourist and Service Sectors: Establishments in the tourism and service sectors, including restaurants, hotels, school cafeterias, and hospitals, serve as an efficient medium for promoting organic products. Organic products are

marketed and sold collectively in these establishments. Consumers exposed to organic products generally exhibit a preference for them.

- Dealers, Wholesalers, and Intermediary Firms: Utilizing dealers, brokers, wholesalers, and traders for marketing is an effective strategy for products in huge quantities. Production schedules, timelines, and projected volumes must be communicated to intermediary firms, and the pricing structure and payment conditions should be elucidated. This strategy is particularly significant for cultivators who are novices in organic farming.
- Organic Products Processing Facilities: Marketing to organic food processing companies is an excellent strategy for organic agriculture farmers. Contractual agreements must be observed, and confidence must be cultivated between the parties. The contractual wheat production executed by the Istanbul Metropolitan Municipality in collaboration with Eastern Anatolia Producers exemplifies the efficacy of this approach.
- Cooperatives and Producer Unions: Cooperatives and producer unions are essential entities that facilitate the large-scale sale of organic products. Major producer unions enhance the distribution of organic products by offering appropriate storage and infrastructure facilities. The formation of new producer unions will yield beneficial impacts on the sector.
- Small Retailers (Grocery Stores, Fruit and Vegetable Vendors, Kiosks): Small-scale independent merchants serve as an appropriate marketing channel for small firms that manufacture limited quantities of organic products. Supplying processed organic items to these businesses can enhance product diversity. Moreover, customer confidence can be bolstered and demand for organic products can be augmented through the utilization of natural packaging materials (Bulut, 2011, as cited in Saygı & Bulut, 2024).

3.2 Social Cooperatives

Social cooperatives, a model originated in Europe, are a legal and organizational framework initially established in Italy. Social cooperatives, as stipulated by law, aim to promote personal development and serve the collective interests of society through various activities that deliver social welfare, educational services, and employment opportunities for marginalized individuals (Borzaga et al., 2014:11, as cited in Demircan Yildirim, 2023).

Social cooperatives are organizations focused to generating social value. Therefore, their principal objective is not to produce economic value but to cultivate riches for the benefit of the public. However, this does not imply that social cooperatives should refrain from using tactics that ensure net revenue increases. Conversely, to sustain their objective, they must endure over time and perpetually generate economic value. Profit-oriented companies perceive wealth creation as an indicator of value generation, whereas social cooperatives regard wealth creation as a mechanism for self-sustainability. In this context, the main purpose of social cooperatives is to create social value and “creating economic value in the form of earned income is necessary to ensure the sustainability of the enterprise and its financial self-sufficiency” (Mair & Marti, 2006, as cited in Carini & Costa, 2013:5). In this sense, social cooperatives are considered as dual-profit entities capable of generating both social and economic benefits.

The main characteristic of social cooperatives is their commitment to the public interest, which they fulfill by directly creating commodities and services for the benefit of the community. Public benefit refers to fundamental human necessities within a certain region or community and includes all inhabitants residing there. Social cooperatives typically oversee activities of broad interest that are either inadequately supplied or absent from the public sector. However, regardless of the goods and services they provide, the main goal of social cooperatives is integrating disadvantaged groups into the workforce, and it is considered as public benefit service. (CICOPA, 2004:2, as cited in Demircan

Yıldırım, 2023). Social cooperatives specialize in providing services such as social care, health, and education, as well as the integration of disadvantaged groups into the workforce, so they have a distinctive feature from other cooperatives (TÜSEV, 2018: 3).

Social cooperatives are independent and autonomous entities defined by cooperative principles and operate as non-state actors. In this context, they need to operate autonomously from the public sector and other entities, including defining the nature and extent of the support they may receive, the partnership agreements they may form with governmental authorities, and the involvement or representation of governmental authorities (CICOPA, 2004:2, as cited in Demircan Yıldırım, 2023).

Social cooperatives have a multi-stakeholder collaboration framework. Social cooperatives may include stakeholders such as employees, beneficiaries, local authorities, and various legal organizations. The multi-stakeholder partnership framework is a governance paradigm that allows many stakeholders to collaborate as partners under democratic oversight (CICOPA, 2004:3, as cited in Demircan Yıldırım, 2023).

The representation of employee-members is crucial in social cooperatives. The governing framework of social cooperatives must be represented at every level. The representation of employee members must exceed one-third of the votes in each governance structure. For complete job integration, a minimum of 51% of the members in social cooperatives, which combine disadvantaged workers with other workers, must be employed as workers. In both cases, a minimum of 51% of the employees must be partners (CICOPA, 2004:3, as cited in Demircan Yıldırım, 2023).

Social cooperatives neither transfer profits to their members nor engage in limited profit distributions. On the other hand, other cooperatives allocate earnings based on the material contributions made by the members to the cooperative. In this perspective, while considering the public benefit goals of social cooperatives, members can derive greater advantages by lowering prices or enhancing the quality of services offered by the

cooperative rather than through profit distribution. The limited allocation of earnings or the absence of profit distribution serves as a significant indicator that the principal objective of social cooperatives is a public benefit (İhtiyaç Haritası, 2020: 21). Figure 4 below illustrates the Basic Distinctive Features of Social Cooperatives.

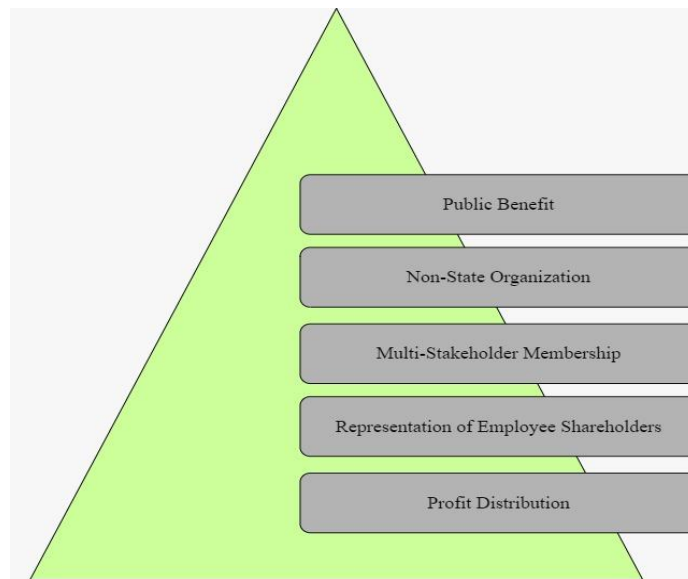


Figure 4: Basic distinctive features of social cooperatives (CICOPA, 2004)

As stated by Borzaga et al. (2014),

“A recent Euricse survey carried out on a sample of more than 4,000 workers in Italian social enterprises highlights that employee are strongly attracted by altruistic motivations (social relatedness and usefulness of the job), followed by some extrinsic motivations (job stability at first) and intrinsic motivations such shared ideals and values with the enterprise and colleagues. The employees interviewed for this survey placed wages and other economic incentives only in the middle of the ranking of their motivations, and other extrinsic aspects were assigned even lower relevance. The analysis demonstrates that other regarding or altruistic but also relational motivations appear to be the most relevant motivations in attracting people toward the social enterprise sector, since

most workers agree on the importance of carrying out activities that help other people and, on the necessity, to work in an environment where relations are good. Also, the search for variety and creativity and for shared values and objectives appear dominant. Extrinsic aspects of the job receive a high degree of attention too, but people are especially motivated by job stability and accomplishment in terms of career and self-realization (Gagné and Deci, 2005). The most important consequence of these strong intrinsic motivations is the high level of commitment on the part of the worker (see also Leete, 2000; Preston, 1989; Valentinov, 2007). Econometric findings suggest that the higher the intrinsic motivations and the lower the extrinsic motivations of workers, the higher their job satisfaction and their loyalty to the organization.” (p.11) Table 5 categorizes and ranks the sources of motivation of social cooperative employees in their work.

	Average (1-12)	Percent of scores 10 to 12
<i>Altruistic motivations</i>		
Helping disadvantaged people	9.48	62.3
Relatedness on the job	9.50	61.3
Relatedness with people outside the job	8.73	52.0
<i>Intrinsic motivations</i>		
Autonomy, variety and creativity of the job	8.48	45.6
Job coherent with individual training	7.06	33.0
Social visibility of the job	7.20	32.5
Physical working environment	7.44	38.4
Sharing common ideals and values	8.77	52.5
<i>Extrinsic motivations</i>		
Flexibility of working hours	8.00	37.4
Wages and economic incentives	8.63	49.2
Self-realization and career prospects	8.38	44.8
Job stability	9.52	61.9

Table 5: *Work and motivations in the social economy*
(Euricse, 2014)

In addition, social cooperatives stand out from other types of enterprises because of their resilience in times of economic crisis. As stated by Euricse (2014), “focusing on cooperatives, several studies have shown that, during an economic crisis, cooperatives demonstrate a greater resilience than other forms of enterprise.” (p.13)

3.2.1 Social Cooperatives in the World

As stated by CICOPA (2018), “owing to different national and regional contexts, the classification of social cooperatives within the cooperative movement differs from country to country.” (p.1)

Despite such differences, social cooperatives fundamentally share all the commonly agreed standards of the cooperative model, namely the definition, values and operational principles enshrined in the ICA Statement on the Cooperative Identity (Manchester, 1995) and in ILO Recommendation 193 on the Promotion of Cooperatives (Geneva, 2002). At the same time, they also possess important distinctive characteristics. (CICOPA, 2018, p.1)

“In terms of financial data, in 2011 Euricse (2013) registered the total value of the production generated by Italian social cooperatives at about 10.1 billion EUR, with 28.3 per cent of the social cooperatives with a value of production no higher than 50,000 EUR and only 16 per cent that could be considered large enterprises (with the value of production over 1 million EUR).” (Borzaga, Bodini, Carini, Depedri, Galera, & Salvatori, 2014, p.11)

“In France, for instance, the social economy (which accounts for 9.9 per cent of all salaried employment) has created 18 per cent of all new jobs between 2006 and 2008. Between 2008 and 2009, employment in the social economy has increased by 2.9 per cent with a net creation of 70,000 paid jobs, compared with a fall of 1.6 per cent in the rest of the private sector and of 4.2 per cent in the public sector (Fauer, 2012, as cited in Borzaga et al., 2014, p. 13).

3.2.2 Social Cooperatives in Türkiye

In this section, the general situation of social cooperatives in Türkiye will be analyzed.

In Türkiye, social cooperatives have a significant role in the contemporary cooperative movement, which has developed recently to deliver social services and enhance employment, particularly in regions where the state and private sector fall short (Act Human, 2021:27, as cited in Yıldırım, 2023). Turkish cooperative legislation lacks restrictions concerning the classification of social cooperatives. However, there exist instances of cooperatives that function in accordance with the social cooperative model and identify themselves as social cooperatives. Cooperatives implementing this model engage in awareness and advocacy initiatives, primarily to enhance the recognition and growth of social cooperatives (TÜSEV, 2018: 6).

Social cooperatives use the majority of their activities in society to achieve goals aimed at providing social benefits. In this context, there are successful examples of social cooperatives that create social impact and operate on a national scale. In this section, examples of social cooperatives in Türkiye will be presented and the activities and operational purposes of social cooperatives will be discussed (Demircan Yıldırım, 2023)

- İhtiyaç Haritası: It is a social cooperative that brings together individuals in need with governmental and non-governmental organizations as a non-profit organization established in 2015. It provides assistance and solidarity in areas such as education, health, culture and art through the map technology and community-based verification function it uses. İhtiyaç Haritası first collects needs from individuals, educational institutions and NGOs, determines their accuracy, and then facilitates the meeting of supports with those in need with its map-based technology. While meeting 9.88 million needs through this platform, a high social impact difference has been created by creating a sharing

economy equivalent to an average of 490 million Turkish liras (İhtiyaç Haritası, 2015).

- Genç İşi Kooperatif: It is the first Scientific Research and Development Cooperative in Türkiye, established in İzmir in 2015. “It is a non-profit social cooperative that is based on widespread education practices and strives to provide disadvantaged groups with access to sustainable employment and livelihoods; it carries out education, research and consultancy activities for this purpose.” This cooperative carries out community-oriented work for individuals, women and young people who have difficulty accessing labor, employment and socio-economic rights. In addition, it provides support to local solidarity economic organizations in the areas of production, consumption and democratic participation for disadvantaged groups of working age (Genç İşi Kooperatif, 2015).
- Harmoni Kadın Kooperatifi: It was founded in 2015 by 20 women and its main purpose is to ensure women's participation in the workforce. In addition, the women members want to develop and grow socioeconomically in order to build a good future for their children. Harmoni, which works to emphasize the value and reputation of women in Türkiye's workforce, provides women with opportunities to produce and encourage them, contributing to the local economy and production, as well as offering solidarity, socialization, learning, innovative product development and cooperation opportunities (Harmoni Kadın Kooperatifi, 2023).

In addition to the above-mentioned cooperatives, many cooperatives have achieved great success and have managed to make a name for themselves in the public sphere. Examples of these cooperatives are as follows: Üretkeniz Biz, Uluslararası Mucizevi Eller Sosyal Kooperatifi, Başka Bir Okul Mümkün Eğitim Kooperatifi, Yeryüzü

Kalkınma Kooperatifi, Gökova Kadın Çevre ve Kültür İşletme Kooperatifi (Demircan Yıldırım, 2023).

According to the data, social cooperatives in Türkiye serve as an alternative model that actively contributes to the resolution and transformation of social events, as well as economic and social issues. The needs analysis report by İhtiyaç Haritası on Social cooperatives indicated that 144 individuals give positive feedback towards their objectives aimed at addressing a social need. Therefore, social cooperatives enhance the social and economic fabric of their communities while also playing a crucial part in the advancement and development of their target demographics (İhtiyaç Haritası, 2020: 12). According to Needs Map's needs assessment report on social cooperatives in Figure 5, 144 people responded positively to the objectives of social cooperatives to meet a social need.

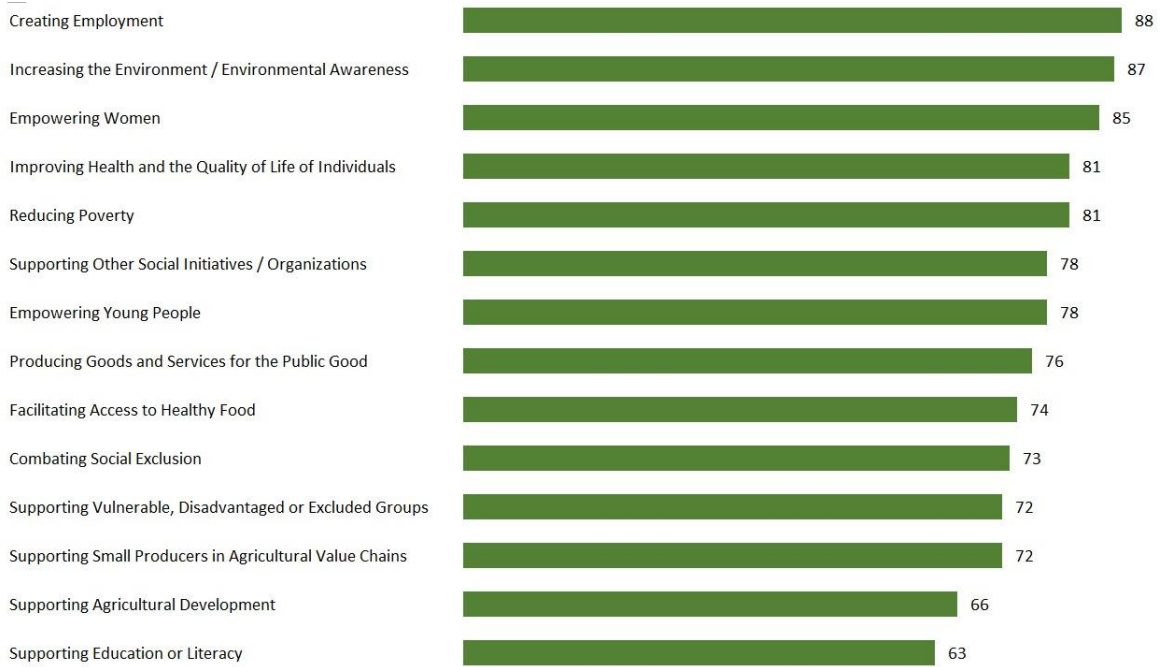


Figure 5: Purpose of establishing social cooperatives
(İhtiyaç Haritası, 2020)

3.3 Women Employment

In this section, we will discuss the global and Türkiye-specific situation regarding women's employment. In addition, we will comprehensively describe the current problems faced in women's labor force participation. In this way, we aim to make important observations by focusing on both the global situation and the dynamics of the situation in Türkiye.

According to Hakim (1995) (as cited in Gözüml, 2024),

“Employment refers to individuals actively participating in the labor market, either paid or unpaid, for a certain period. It encompasses all sectors and professions where economic activities are conducted and includes individuals contributing to the economy by participating in the production process of goods or services” (p. 431).

As stated by Crompton & Harris (1998) (as cited in Gözüml, 2024), employed individuals contribute to production processes and help the growth and development of the economy. The size and quality of employment are influenced by many factors, such as a country's economic structure, education level, technological development, and social policies (p. 119).

This project's main focus here is to increase labor force participation and employment of disadvantaged women through social cooperatives. Analyzing issues around women's employment allows us to understand larger-scale social and economic problems.

In the words of Rubery (2005) (as cited in Gözüml, 2024) Increasing the labor force participation of women and disadvantaged groups promotes gender equality and social inclusion (p. 45).

3.3.1 Women Employment in the World

Women's employment in the world has improved significantly over the last century. Although women can now actively participate in the labor force in most countries of the world, they face a variety of challenges. As women's participation in the labor market has increased, problems such as unequal pay, inequality of opportunity, and discrimination between the sexes have become more pronounced. Cultural habits, social taboos, and economic and political barriers prevent women from fully integrating into the labor market. Therefore, there is a huge loss of potential for global economic development. Ensuring women's fair participation in the labor force and preventing discrimination are of great importance for our future.

As stated by Fine (2012) (as cited in Gözümlü, 2024) “According to reports by organizations such as the United Nations and the World Bank, women's labor force participation rate is lower than men's worldwide. Women generally work in lower-paid, insecure and part-time jobs” (p. 92).

Gözümlü (2024) claims that “While women's employment is lower than men's across the world, women's labor often faces many challenges, including low wages, gender-based discrimination, and the invisibility of domestic labor” (p. 117).

The most prominent problems related to women's employment are low employment rate, gender wage inequalities, and invisible labor.

Women's labor force participation rate remains far behind that of men all over the world. According to 2023 data, while the labor force participation rate for men is 67.9% worldwide, it is 43.8% for women (Dyvik, 2024). Figure 6 is the comparison of female and male employment rates between 2020-2023.

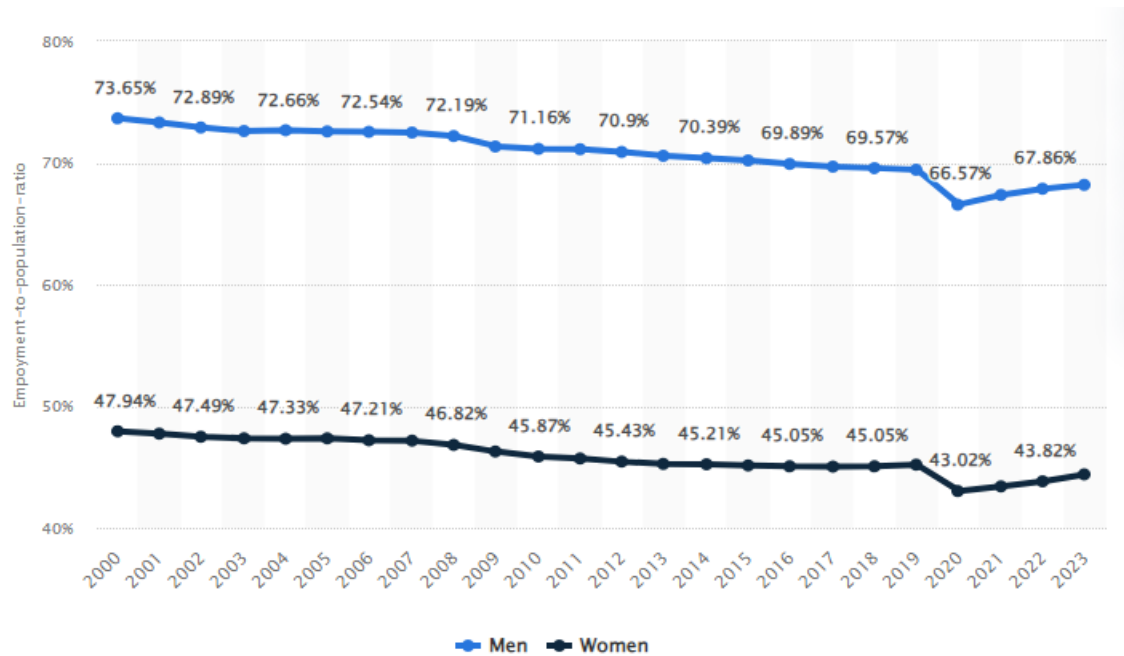


Figure 6: The comparison of female and male employment rates between 2020-2023 (Dyvik, 2024)

According to Gözüml (2024) “Although women's employment rate varies according to many factors, such as economic, cultural, and social factors, this rate is lower than that of men in the world” (p. 117).

Worldwide, there is a problem of gender inequality in relation to women's employment. Even when women have the opportunity to work, they are paid lower salaries and are less likely to be promoted than men. For every dollar man earn, Women earn only 0.83\$ in the U.S.A. in 2024 (Dyvik, 2024). Family responsibilities and gender discrimination in the workplace also negatively affect women's careers. Thus, women's economic independence is endangered. Achieving gender equality in the workplace is essential for economic growth, justice, and social peace. Figure 7 shows how much a Women earns for every 1\$ a man earns between 2015-2024.

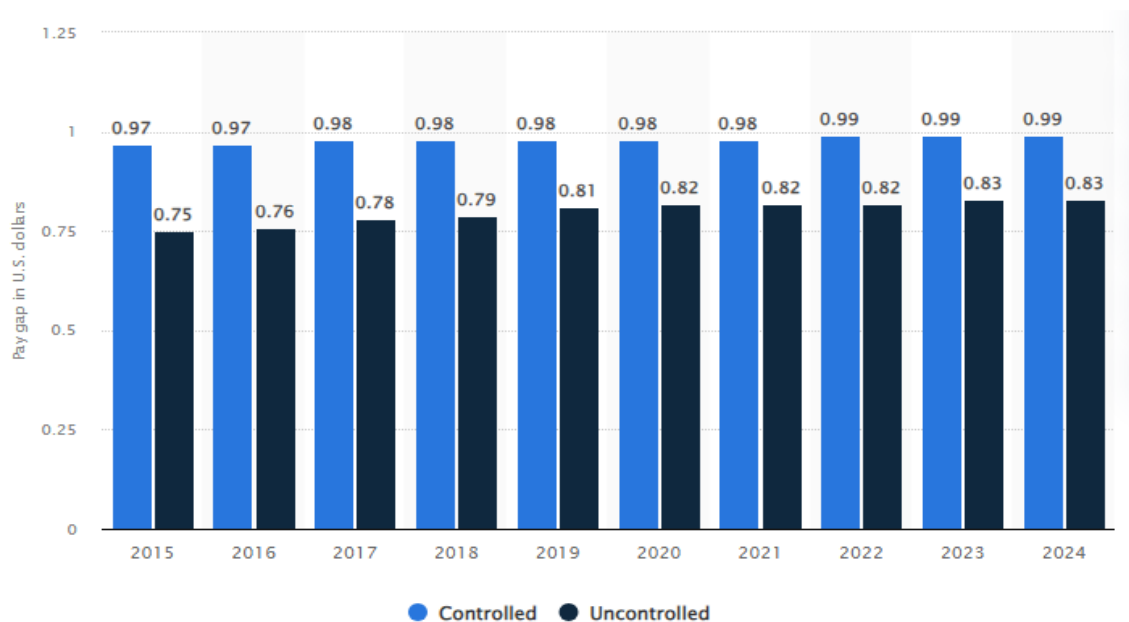


Figure 7: How much a woman earns for every 1\$ a man earns between 2015-2024
(Dyvik, 2024)

As stated by Kılınç (2015), “although women's employment rates have increased with globalization, the majority of these jobs are low-paid, without social security and legal protection” (p. 122).

Invisible labor in women's employment is usually unpaid work/labor such as domestic work and childcare. Domestic work, cleaning, and childcare, which are generally seen as belonging to women due to gender roles and are mostly performed by women, are unpaid. Although this work is a large part of the economy, it is often overlooked and ignored. This invisible labor restricts women's participation in the labor market and increases the problem of gender inequality in working life. Figure 8 compares how often women and men are exposed to the problem of invisible labor

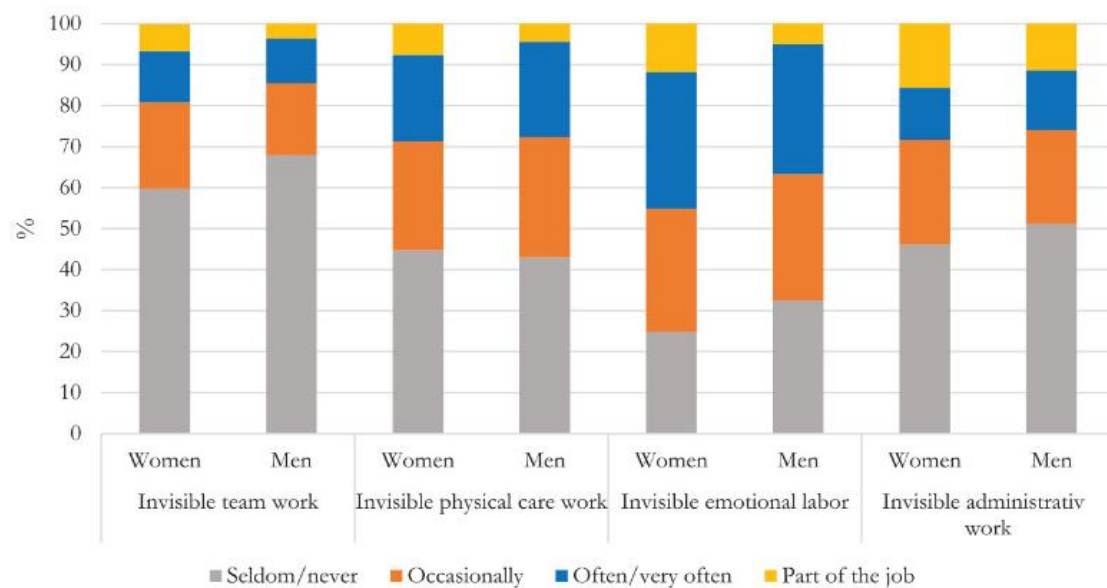


Figure 8: How often women and men are exposed to the problem of invisible labor
(Dyvik, 2024)

In the words of Gözümlü (2024) “In Türkiye and around the world, women play important roles at work, at home, and in all aspects of society, yet these roles often involve forms of labor that are devalued, invisible, or unfairly compensated” (p. 117).

3.3.2 Women Employment in Türkiye

In Türkiye, the problem of women's employment is especially significant in the low-income class and in rural/undeveloped regions. As stated by Jayachandran (2021), the biggest obstacles to women's labor force participation are family responsibilities, education level, social norms, and inequality of opportunity. In the words of Jayachandran (2021), particularly in underdeveloped regions, many women continue to remain in domestic work due to the impact of gender roles, thus limiting their participation in the labor force. Even when they do work, they are forced to work in low-paid and precarious jobs. While steps have been taken to improve labor force participation and conditions for women, there is still a long way to go.

As stated by Kılınç (2015), “As in the rest of the world, labor market participation rates of women in Türkiye have remained lower than those of men” (p. 128). This ongoing disparity is further emphasized by Gözümlü (2024), who states, “When the labor force is analyzed from a gender perspective, it is seen that the labor force participation rate of women in Türkiye is considerably lower than that of men” (p. 115). Additionally, Gözümlü (2024) highlights the broader international context by noting, “Although women's employment is lower than men's in Europe, there is a large gap between Türkiye and European Union (EU) and OECD countries in terms of women's labor force participation and employment rates” (p. 117).

As stated by Şahin & Uslu (2024, p. 87),

“Türkiye ranks last among OECD countries with a female employment rate of 35.5%, and women's participation in the labor force is seriously insufficient. In order to increase women's participation in the labor force, it is recommended to raise the level of education, carry out awareness-raising activities for women's employment, and to develop policies in this direction”.

According to the 2023 Labor Market data published by TurkStat (2023), 7 out of every 10 workers in Türkiye are men, while only 3 are women (p. 5). The figure 9 shown below.



Figure 9: *In Türkiye, only 3 out every 10 workers are women*
(TurkStat, 2023)

When the gender distribution of employment rates is analyzed, Aydın province with 32.1% female employment rate is above Türkiye with 30.7% employment rate, but the gender gap is significant. (TurkStat, 2023)

“In many societies, women and men are assigned different roles due to gender inequality and gender-based division of labor.” (ILO & TurkStat, 2020, p. 66). “In addition, in Türkiye, gender-based wage inequality is widely observed in various sectors and occupational groups. According to 2018 data from ILO and TurkStat (2020), the gender pay gap in Türkiye was 15.6%.” (ILO & TurkStat, 2020, p. 67).

The outputs of our literature review show that although female and male employment rates in the agricultural sector are more balanced than in other sectors, female workers earn less than male workers doing the same job. Figure 10 shows the daily earnings of seasonal agricultural workers in 2023, disaggregated by gender and city. This is where our social cooperative model makes sense. In addition to increasing women's employment, efforts will be made to ensure that women earn the income they deserve.

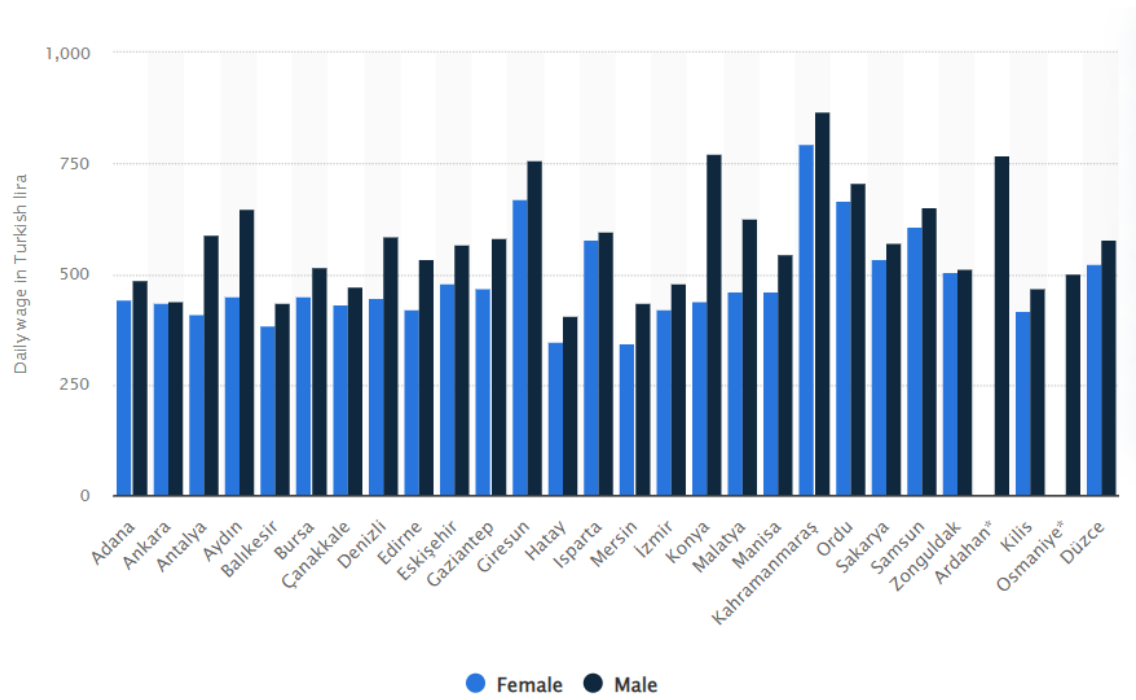


Figure 10: Daily wages of seasonal agricultural workers in Türkiye in 2023, by province and gender
(Statista, 2023)

4. OUTPUTS OF LITERATURE RESEARCH

4.1 Outputs of Organic Agriculture Literature Research

4.1.1 Implementation Province of Our Business Model

Aydın is a strategically important city in terms of organic agriculture practices in Türkiye. Data from the “Tarım Orman Bakanlığı” (MoAF) shows that Aydın is the leader in Türkiye in terms of the number of organic farmers and production area. Data for 2021 shows that there are 7.380 organic farmers in Aydın, and 50.961 hectares of land are used for organic farming. In the light of this data, Aydın has a high potential in organic agriculture, both in terms of production and product range. For this reason, Aydın was chosen as the pilot province of our project.

One of the most important reasons why Aydın has high productivity in terms of production is its climate. Aydın has a Mediterranean climate, making it ideal for organic vegetable production. It has a lot of sunny days all year round, and the soil is suitable for agriculture. This allows vegetables to be grown naturally and efficiently.

Aydın has been prominent in traditional agricultural products such as olives and figs for many years and has a long-standing experience. This agricultural background allows us to utilize the labor, know-how and local networks required for our business model.

In addition to its agricultural advantages, Aydın's potential also offers significant opportunities for social and economic development. Our business model will provide employment opportunities for unemployed women, especially those living in the rural areas of Aydın, thus creating social benefits. When unemployed women are included in our agricultural production process, they will also gain economic independence. Aydın's above-mentioned existing potential provides us with a strong opportunity and foundation to realize these social and economic goals. In addition, organic farming is known to stimulate economic growth in agriculture-oriented regions such as Aydın.

4.1.2 Eliminating the Deficiency in Vegetable Production

Most agricultural practices in Türkiye focus on perennial plants and tree farming. For example, 66.37% of Aydın's total organic farming area is dedicated to organic olive production. In addition, agricultural products such as wheat, barley, and other grains are also widely cultivated. However, the low rates of annual plant and vegetable production indicate a potential development opportunity in this area.

These imbalances present a great opportunity in the organic food market, especially in terms of fresh vegetable production. As we mentioned in the literature review, the demand for healthy and organic foods on the consumer side is rapidly increasing. However, the current production infrastructure is not at a level to meet the expected demand. Fresh products like vegetables hold an important place in the organic food market. Our business model will operate with the aim of filling this gap.

Organic vegetables meet our consumers' need for healthy nutrition with their high nutritional values. Our business model will focus on meeting this need in both the local and national markets by producing these products organically.

Accordingly, our business model has developed a strategic production plan to meet consumer demand while at the same time addressing gaps in the sector. Our business model will aim to harness the potential in the current production imbalance and will focus on organic vegetable production. In this context, vegetables such as tomatoes, peppers, zucchini, and lettuce will be our priority product groups. The products we mentioned are important food groups that are both suitable for Aydın's climate and soil structure and meet consumers' daily healthy nutrition needs.

However, our business model aims to make long-term contributions and create a sustainable balance in the organic farming sector. Our business model aims to increase the product variety in organic agricultural products and eliminate the existing imbalances in the organic agriculture sector.

Aydın will not be limited to olive and fig production but will become a center for organic vegetable production. This situation will make Aydın more competitive in organic agriculture across a wider range in Türkiye.

4.1.3 Production and Delivery Process Planning

Transparency has been adopted as a fundamental principle in our project's marketing strategy. One of the most effective ways to build trust in organic farming is for consumers to see and verify the production processes of our products. Based on the marketing methods specified in the literature, we will strengthen our project's transparency approach as stated in the following paragraphs.

In this regard, we are taking some concrete steps to implement the principle of transparency in our business model. In our business model, we offer our consumers the opportunity to visit our fields and even directly observe the processes of growing our products on-site. This approach of ours is a practical application of the "Farm Shop" marketing model mentioned in our literature review. In farm shops, consumers have direct access to the production area. Thus, both the trust in product quality increases, and consumers are offered a unique experience. In addition, we plan to make some arrangements such as play areas for children and social event spaces during farm visits, turning these visits into family-oriented activities. Thus, in addition to fulfilling the requirements of our transparency principle, we aim to increase customer loyalty by providing an experience that customers will remember in the future.

In addition, innovative solutions to increase transparency in our business model through digital channels are also included as part of our business model. The "Mail Order" and "Online Sales" methods emphasized in our literature review offer digital solutions to our customers in terms of transparency. A platform can be created for our consumers that shows the production stages of our products through live broadcasts. In addition, each stage of the farm-to-table process (planting, harvesting, packaging, distribution) can be

recorded and shared digitally. Thanks to this method, the production processes will become visible, especially for our consumers who shop online, and our customers' trust in our business model will increase.

In addition, we considered various certification and labeling methods to increase the reliability and traceability of our products as an important part of our project. All the products we will sell in our project will have the "Product Certificate" and "Organic Farming Logo," which are mandatory for organic farming in Türkiye. Thanks to these requirements mentioned in our literature review, we will be able to prove to our consumers that our products are organic and also increase the traceability and reliability of our products. In addition, by adding innovative labeling methods such as QR codes to our product packages, we will enable consumers to scan these codes and access detailed information about the cultivation process of the products.

In addition, we will implement effective marketing and promotion strategies to effectively communicate with our customers and ensure their trust. To establish good communication with our consumers and to increase their trust in our project, we will regularly organize information campaigns. Additionally, social media content and promotional events can be planned. As we mentioned in our literature review, an effective communication strategy in the marketing of our organic products plays a critical role in increasing consumer loyalty. In this context, consumer comments and experiences obtained from farm visits can be shared on digital platforms, thus reaching a wider audience.

4.2 Outputs of Social Cooperative Literature Research

4.2.1 Impact on Regional Development and Local Economy

Social cooperatives not only produce social benefits but also add sustainable value to the local economy. Our business model will support rural development in a region with high agricultural production potential, like Aydın, through the social cooperative model. The participation of women in the workforce in the region will contribute to the increase

in family incomes and the revitalization of the local economy. Thus, Aydın will have the opportunity to rise to a stronger position in both national and international markets for many agricultural products.

Our business model aims to reduce agricultural production costs by enabling agricultural producers in the region to come together and benefit from economies of scale through cooperative structures. Thus, competition will increase, and as a result, consumers will have more affordable access to food. As a requirement of the democratic governance structures of cooperatives, the participation of local stakeholders in the project will be strengthened, and long-term sustainability will be ensured.

4.2.2 Social Solidarity and Increased Trust in Society

Our social cooperative, which will be established within the framework of our business model, has the potential to not only encourage women's participation in economic life but also to enhance social solidarity in the community. In our literature review, it was emphasized that social cooperatives are not merely an economic structure, but also a tool that supports the social integration of disadvantaged groups. Our business model will strengthen social trust and serve as a bridge between producers and consumers.

The ability for consumers to observe production processes through our cooperative and interact directly with producers will further strengthen the trust factor, and this model could start to spread in both Aydın and our other cities. Thus, our cooperative-based business model can be implemented in other regions and can be effective at the national level.

4.2.3 Training and Capacity Building

Another important benefit of our business model is that it also includes the education and self-improvement processes of individuals involved in the social cooperative structure. The training we will provide on topics such as organic farming techniques, marketing methods, and cooperative management will not only enhance the skills of women and other cooperative members but also enable them to participate more

effectively in the workforce. Thus, individuals will take on roles not only as producers but also as entrepreneurs and leaders.

4.2.4 The Situation of Social Cooperatives in Türkiye: Evaluation with Swot Analysis

Strengths: Social cooperatives promote women's employment and women's economic independence. They also promote gender equality by ensuring women's participation in decision-making processes. As stated by International Labour Organization (ILO) (2015), “The findings suggest that various manifestations of gender equity have been achieved across the co-operative movement, and that the co-operative model is particularly adept at addressing women’s empowerment and gender equality concerns.” (p. 4). They also develop local economies by involving women in production processes. Below, we list the weaknesses of social cooperatives:

- Contribution to Increase in Women's Employment Rate: Social cooperatives support women's participation in the workforce and contribute to their struggle for economic independence.
- Contribution to Gender Equality: The inclusion of women in decision-making processes contributes to gender equality.
- Contribution to Local Economy: The inclusion of women in the production process contributes to strengthening the local economy.

Weaknesses: Legal and institutional inefficiencies, limited resources and lack of public awareness can limit the ability of our social cooperative to create jobs for women. As stated by Şahankaya Adar, Dedeoğlu, & Kurtuluş (2023) “This study shows that despite the rapid increase in the number of women's co-operatives in recent years, women's co-operatives have had very weak impacts on increasing women's employment and promoting women's empowerment.” (p. 1172). Below, we list the strengths of social cooperatives:

- Legal and Institutional Deficiencies: Legal and institutional deficiencies for social cooperatives in Türkiye may hinder our efforts to increase women's employment.
- Lack of Resources: Lack of financial resources for social cooperatives can make it difficult to create women's employment.
- Lack of Awareness: Lack of awareness of the positive aspects of social cooperatives may hinder women's participation.

Opportunities: Government and NGO support, national and international incentives and funding provide growth opportunities for social cooperatives. Our social cooperative also enriches the labor market. As stated by ILO (2017), “Formed by producers, consumers, workers and businesses worldwide, cooperatives can and do create and consolidate employment opportunities, empower people, provide protection and alleviate poverty.” (p. 1). Below, we list the opportunities of social cooperatives:

- Government and NGO Support: Increasing incentives and projects for women's employment allow social cooperatives to grow easily.
- National and International Funds and Programs: Supporting social cooperatives with national and international funds and programs can accelerate our growth.
- Entrepreneurship and Innovation: Supporting women entrepreneurs through social cooperatives can stimulate the labor market.

Threats: Societal prejudices against women's work and economic problems in Türkiye can prevent our social cooperative from generating benefits. As stated by ILO (2022), “Gender-based stereotypes and economic barriers continue to pose significant threats to the sustainability and efficiency of women's cooperatives in Türkiye” (p. 15). Below, we list the threats of social cooperatives:

- Social Prejudices: Societal prejudices against women's work, especially in rural and underdeveloped areas, can limit the benefits we plan to generate through the social cooperative.
- Economic Challenges in Türkiye: Economic problems and instability in Türkiye may limit the sustainability of our social cooperative and its capacity to create social benefit.

4.3 Outputs of Women Employment Literature Research

4.3.1 The Economic Empowerment and Increased Social Status of Women

The contributions of our business model to women's employment will enable women to gain economic independence and elevate their social status. Especially, the participation and integration of women living in rural and underdeveloped areas into economic life will help break the cycle of poverty by increasing the family's income. Thus, it will contribute to the establishment of an egalitarian structure not only at the individual level but also at the societal level.

The active participation of women in production processes not only boosts their self-confidence but also directly contributes to economic development. According to Rubery (2005), increasing women's employment is one of the key factors that strengthen social inclusion and gender equality. The success of our business model in Aydın will create and encourage a model to be followed in other cities as well.

4.3.2 New Opportunities for Regional Women's Employment

Our business model presents an innovative approach to agricultural production by enabling women in an agriculture-focused region like Aydın to play a more active role in organic farming processes. According to our literature review, the intensive involvement of women, particularly in the organic farming sector, proves to be critically important for environmental and economic sustainability (ILO, 2021).

In our business model, we plan to employ women not only in agricultural production but also in areas such as packaging, logistics, marketing, and cooperative management, creating new opportunities for them. Thus, we will ensure that women's employment spreads to more diverse fields and strengthen the dynamics of the local economy.

4.3.3 Increasing Women's Education and Leadership Capacities

One of the most effective ways to increase women's employment is to enhance their knowledge and skills. Within the scope of our business model, the training we will provide to women will not only focus on the agricultural production process but will also cover topics such as cooperative management, marketing, and business development. In these trainings, women's leadership skills will be enhanced, and they will be encouraged to take a more active and effective role in the economic sphere.

The inclusion of women in leadership positions will also strengthen the democratic structure of our cooperative and help us build a sustainable development model. In addition, the successes we achieve as a result of this.

4.3.4 Balance of Environmental and Social Benefits

The inclusion of women in the agricultural sector will benefit not only economically but also environmentally. We will encourage women to adopt sustainable agricultural techniques, promote them, and contribute to the conservation of natural resources. This situation will also support our ability to achieve environmental sustainability goals through organic farming and will enhance the long-term impact of our business model in Aydın.

5. PRELIMINARY ANALYSIS

5.1 Current Solutions and Competitor Analysis

5.1.1 ÇiftçidenEve

The source of the data in this section is the official website of ÇiftçidenEve (ÇiftçidenEve, 2024).

ÇiftçidenEve, founded in 2015, is Türkiye's first digital agricultural marketplace. Aiming to bring natural and organic food producers directly together with consumers, ÇiftçidenEve aims to provide advantages to both producers and consumers by offering an intermediary-free shopping experience and to contribute to supporting local producers and promoting sustainable agriculture. The products of producers who want to sell on ÇiftçidenEve are first subjected to a certain control and certification process. The products that successfully pass the control process can be put on the market at the price they want. In this way, while giving producers the opportunity to exhibit their products and make direct sales, they try to provide consumers with a reliable and transparent shopping environment.

ÇiftçidenEve tries to cater to different consumer needs by offering a wide range of products such as fruits, vegetables, pulses, meat products, dairy products, olive oil and honey. Producers can introduce their products in detail and provide information about their production processes, but this process is at the discretion of the producer, and if the producer does not follow this process, the trust environment that is tried to be created may be disrupted. In addition, the platform may occasionally experience stock shortages in some product categories (especially fresh fruits and vegetables). This is due to the fact that producers on the platform also sell their products in real markets, and stocks of fresh vegetables and fruits are often depleted before they receive orders from the platform.

The pricing model is one of the strengths of ÇiftçidenEve. The ability of producers to set prices directly is an important advantage of an intermediary-free system. This gives producers the opportunity to earn more, while consumers are often able to compare prices and access products at competitive prices. However, leaving prices to producers can sometimes lead to price inconsistencies, which can negatively affect the consumer experience.

In terms of marketing strategies, the platform tries to use digital tools effectively. It tries to reach consumers through social media, blog posts and email newsletters, and aims to increase brand loyalty by providing informative content on topics such as healthy living and organic products. However, its adoption of a digital-focused marketing strategy limits its capacity to reach offline consumers and may exclude the elderly or those without access to technology.

The platform's logistics processes aim to establish a direct connection from producer to consumer. Orders are packaged by the producers and sent via cargo companies, enabling the platform to provide services in a wide geographical area. However, the dependence of delivery times on cargo companies can occasionally lead to delays and quality problems. Especially for products that require cold chain, logistics processes need to be managed more precisely, but given the current cargo companies in Türkiye and their functioning, there are some question marks in this regard.

In conclusion, ÇiftçidenEve supports its leading position in Türkiye's digital agriculture market with a strong product-price relationship and digital marketing strategies, but some weaknesses in operational processes carry potential for improvement in terms of growth and customer experience.

In Figure 11, we have shown the process from the production of ÇiftçidenEve's products to their delivery to the consumer in a flowchart we have prepared. In this

diagram, the stages starting with the registration of the producer to the platform, listing of the products, selection by the consumer, payment confirmation, product delivery and customer feedback are presented in detail.

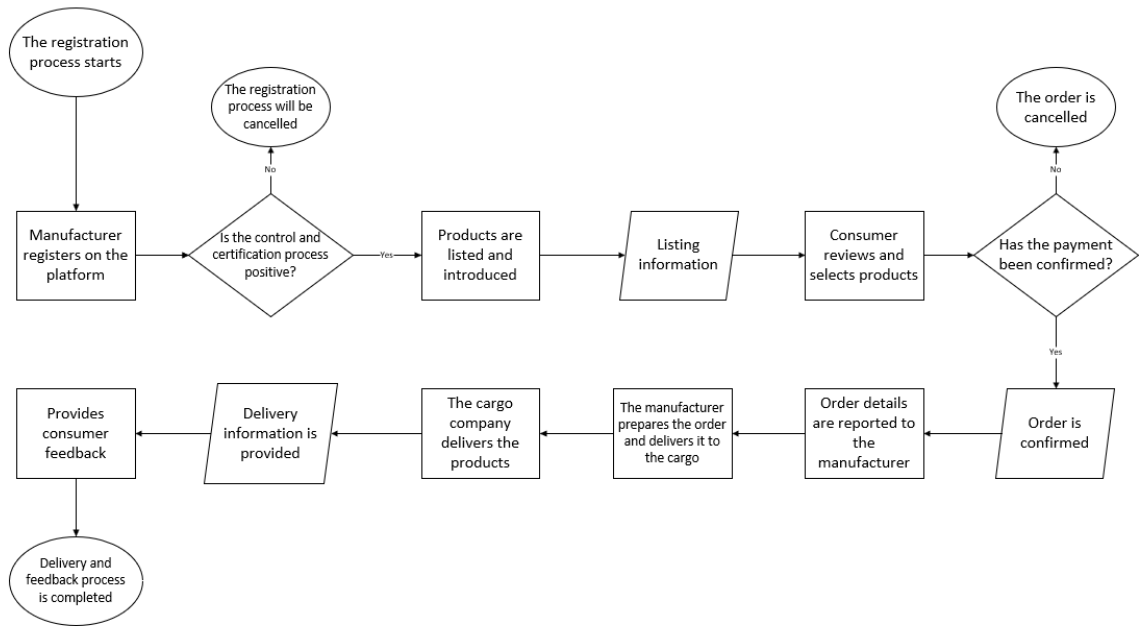


Figure 11: Process flow design of the ÇiftçidenEve platform

Figure 12 shows the distribution of “users ordering by region” for the organization ÇiftçidenEve. This map reveals the density of users ordering from the platform across Türkiye and some international locations. The image provides valuable information in understanding ÇiftçidenEve’s customer base, assessing their regional distribution, and understanding strategic decisions that can be made.

When you look at the map, it is clear that the majority of orders come from within Türkiye, especially from big cities like Istanbul. The fact that Istanbul is a central location in terms of both economy and population is one of the main reasons for this density. In addition, cities in the Aegean and Central Anatolia regions also show a remarkable

density. This shows that ÇiftçidenEve successfully serves regions where demand for organic products is higher. In addition, it is seen that the platform reaches a limited user base in Europe. This indicates that the organic agriculture sector in Türkiye has the potential to reach the international market.

However, there are some notable omissions on the map. The user density in the eastern regions of Türkiye, especially in Southeastern Anatolia and Eastern Anatolia, is quite low. This situation shows that ÇiftçidenEve is inadequate in reaching consumers in these regions. Factors such as regional economic difficulties, the socioeconomic structure of the region, lack of digital access and low demand for organic products may explain this situation.

In addition, users who place orders in Europe are concentrated in certain cities on the map. This density, especially seen in Germany, can be thought to reflect the interest of Turkish citizens living abroad who want to access organic products. In other words, it is difficult to say that the organization preferred by Turkish citizens abroad has achieved the same success for foreigners. At this point, ÇiftçidenEve and therefore the organic agriculture sector in Türkiye need to work to strengthen its presence in the international market and improve processes in these regions.

In conclusion, this visual is a useful tool for analyzing ÇiftçidenEve's user base and making strategic decisions. The map shows that the platform has a strong user base in the western part of Türkiye and its major cities, but there are opportunities for improvement in the eastern regions and the international market.



Figure 12: Map showing the distribution of users placing orders by region (ÇiftçidenEve website)

The image in Figure 13 shows the producer density of the ÇiftçidenEve platform by province across Türkiye. This map provides important information in understanding which regions the platform cooperates with producers, the geographical distribution of organic agriculture, and the areas where the producer base is strong and weak.

When the map is examined, it is seen that producers are densely located in the Marmara, Aegean and Mediterranean regions. There is a significant producer density especially in cities such as Istanbul, Izmir, Adana and Bursa and their surroundings. This may be due to the fact that these regions have developed agricultural infrastructure, have suitable conditions for organic farming and have a more conscious producer profile. In addition, the strong logistics networks of these cities allow such platforms to easily

cooperate with producers. The organic agriculture potential of the Aegean and Mediterranean regions stands out in olive, fruit and vegetable production. This allows the platform to increase its natural and organic product diversity.

When we look towards the inner regions of Anatolia, we see that there is a producer network in cities such as Ankara, Konya and Kayseri. The agricultural production capacity of these cities contributes to the platform offering a wide range of products. However, it is noteworthy that the producer density is lower in the Eastern and Southeastern Anatolia regions. The reason for this situation may be the rural structure of the region, the limited use of agricultural technology, the scarcity of regions suitable for organic agriculture and the scarcity of products, which limit organic production.

This visual also points to potential areas where the platform can increase the number of organic producers. For example, sustainable agriculture training in the Eastern Anatolia Region and increasing the awareness of producers about organic farming through state-supported incentive programs can strengthen the producer base in this region. Similarly, the potential in the Southeastern Anatolia Region can be evaluated by converting local products (such as pistachios, lentils and pomegranates) to organic production.

Overall, the map shows that ÇiftçidenEve has a strong producer network and is active in strategic regions in terms of organic agriculture. However, it is clear that the platform needs to make more efforts in the Eastern and Southeastern Anatolia regions to make its producer base more balanced. Such expansion will allow the platform to increase its product range and reach a wider consumer base. In addition, supportive strategies such as education and incentive programs can be implemented to establish stronger relationships with producers and reduce regional inequalities.

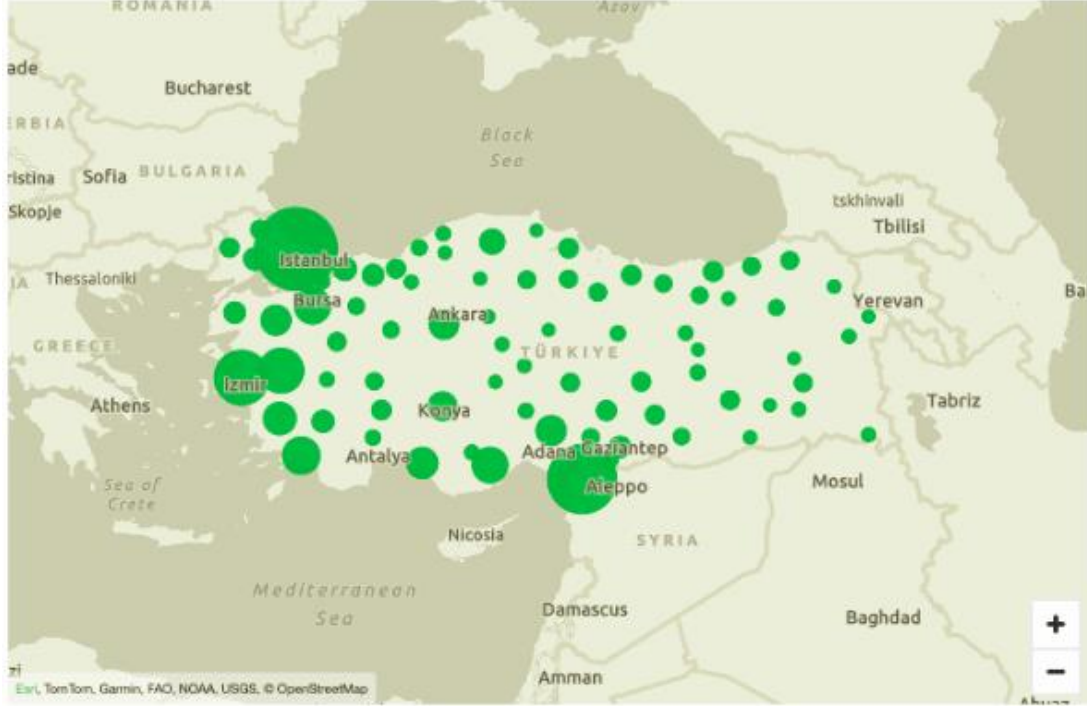


Figure 13: Map showing the distribution of producers by region
(ÇiftçidenEve website)

According to the information provided on ÇiftçidenEve's website, 87% of the orders were successfully delivered and the earnings were transferred to the producers. The rate of orders that were canceled despite payment was 6%. In addition, a total of ₺117,000 was refunded to its users due to incorrect stock or price information, shipment delays and shipping damages. However, the saddest situation for the company is that customers who placed an order for the first time lost their trust and did not place an order again. ÇiftçidenEve emphasized that it is the leading farmer community in Türkiye in direct sales and managed to increase its turnover by 36% this year, which grew by 200% last year. In addition, the website reached 354,000 visitors and 38,200 registered users (ÇiftçidenEve,

2025). Table 6 is the “Summary Transaction Report” on ÇiftçidenEve's website, prepared according to the order delivery date.

	2020	2021	2022	2023	2024
Manufacturers Selling	93	116	101	104	76
Different Customers	2,366	1,348	1,786	1,788	1,768
Order Quantities	2,465	1,415	1,856	1,887	1,842
Average Order Amount	109.4	158.52	302.42	1,417.08	1,290.45
Transaction Volume (GMV)	305,364.46	247,020.71	589,118.04	2,751,865.58	2,525,705.42
Manufacturer Earnings Rate	80.91%	83.12%	85.38%	87.18%	86.01%
Manufacturer Earnings	232,903	177,477.14	462,831.99	2,357,783.48	2,001,726.86

***Table 6:** Summary transaction report for 2020-2024*
(ÇiftçidenEve website)

The data on the operation of ÇiftçidenEve in Table 6 reveals both the strengths and the aspects of the platform that need to be improved. The platform, which started with 93 producers in 2020, grew by increasing the number of producers to 116 in 2021, but this number is decreased in the following years and decreased to 76 producers in 2024. This situation suggests sustainability problems on the producer side or that the platform may have focused on fewer but larger-scale producers. Considering this decrease, it can be thought that the platform needs to strengthen producer relations and make efforts to cooperate with new producers.

In terms of customer base, it is noteworthy that the number of different customers, which was 2,366 in 2020, decreased to 1,768 in 2024. This decrease suggests that the customer segment has narrowed or that a specific user group has been focused on. The information in the text that customers who placed an order for the first time did not place an order again reveals that there are deficiencies in creating customer loyalty and that improvements need to be made in this regard. However, the fact that order quantities have

remained constant in the range of 1,800-2,400 since 2020 can be interpreted as existing customers using the platform regularly and that the platform has a certain loyal user base.

It can be said that the significant increase in the average order amount since 2020 contributed to the growth of the platform. The inflationary environment and price increases in the country have a high share in reaching a high average order amount of ₺1,417 in 2023, but it can also be interpreted that consumer who placed orders were satisfied with their experiences and started to order more products. In addition, the 87% success rate and 6% cancellation rate mentioned in the text indicate that some problems continue in operational processes. In particular, shipping problems, stock information errors and damages experienced during transportation are among the factors that negatively affect customer satisfaction. The ₺117,000 loss caused by returns is not only a financial loss, but also has negative effects on customer trust.

Website data shows that the platform has a strong digital reach. The fact that the website, which has been visited by 354,000 people, has 38,200 registered users shows that the platform needs to develop more effective strategies to convert visitors into customers. Considering all this data, it is seen that ÇiftçidenEve has improvement opportunities for a more balanced growth on both the producer and customer side.

5.1.2 Ovacık Doğal

The source of the data in this section is the official website of Ovacık Doğal (Ovacık Doğal, 2024).

Ovacık Doğal is a cooperative established in the Ovacık district of Tunceli and aims to provide natural and healthy products directly from the producer to the consumer. The main goal of the cooperative is to revive the collective consciousness against the systems that devalue labor and nature, to ensure solidarity and to bring people together with healthy food by protecting nature. In addition, unlike industrial agriculture, it defends

traditional production and puts the producer and the consumer at the center of the system. It advocates that consumer should know under what conditions, how and where the food they consume is produced, and they deliver the product directly from the producer to the consumer by eliminating the middlemen and merchants. The cooperative only procures the products of the producers in its own region by conducting on-site inspections. It supports collective production against industrial production and solidarity-oriented marketing against industrial marketing. They carry out organic production by protecting the ecological balance without polluting the soil with chemical fertilizers and pesticides. In addition, it prefers environmentally friendly packaging by using cloth bags in legume packaging, and the labor of the women of the region is used in the production of these bags, thus supporting women's labor.

Ovacık Doğal has a wide range of products including honey, legumes, nuts, jam, dairy products and natural spring salt. While the cooperative tries to ensure that producers are rewarded for their labor, it also wants to contribute to public health by offering healthy and natural products to consumers. The cooperative's products are delivered to consumers through various sales offices throughout Türkiye. These offices establish a reliable bridge between producers and consumers, facilitating access to healthy food. In addition, production is encouraged by providing diesel support to producers with the income obtained.

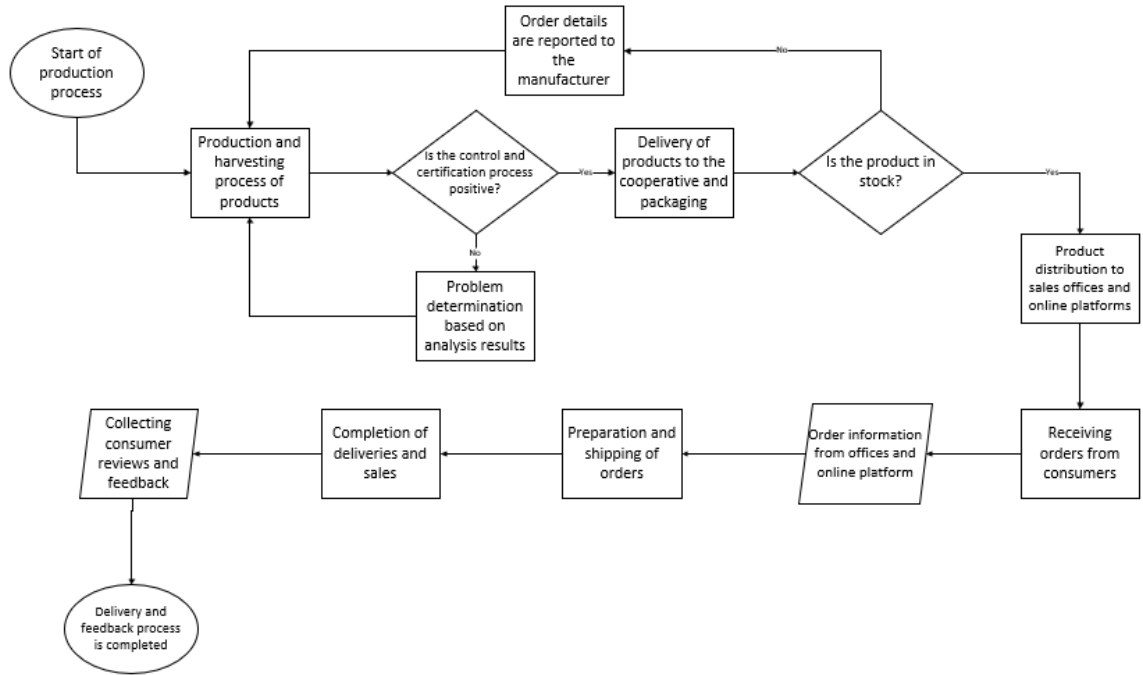


Figure 14: Process flow design of Ovacık Doğal Cooperative

In Figure 14, we have shown the process from the production of Ovacık Doğal's products to the consumer with a flow chart we prepared ourselves. Ovacık Doğal stands out as a cooperative that prioritizes transparency, quality control and sustainability in its processes from production to consumer. The process begins with the beginning of production and the certification phase begins after the products are harvested using natural methods. At this stage, the compliance of the products with organic farming standards is checked and problems are detected for products that cannot receive certification approval and the necessary arrangements are made. Products that successfully pass the certification process are delivered to the cooperative's facilities and packaged in environmentally friendly packaging. The cloth bags used during packaging and the support of local women's labor strengthen the cooperative's understanding of social responsibility.

Packaged products are distributed to sales offices and online platforms. The stock status is checked and the available products are offered to consumers. In case there is no product in stock, the production process is restarted and deficiencies are tried to be eliminated. Orders received from consumers are forwarded to the producers and after the orders are prepared, they are delivered to the cargo. The shipping process aims to deliver the products to the consumer on time and safely. After delivery, consumer feedback is collected and these feedbacks are evaluated for the cooperative to improve its processes.

This operating model of Ovacık Doğal enables natural and healthy products to be delivered directly from the producer to the consumer, while exhibiting an approach that is sensitive to nature and society. However, disruptions that may occur at critical points such as certification and stock management in the process may affect operational efficiency. The importance the cooperative attaches to feedback and the improvement efforts in the processes show the potential to increase customer satisfaction and reach a wider audience. In general, this process successfully reflects Ovacık Doğal's goals of creating a solidarity-oriented and sustainable business model.



Figure 15: *Locations of Ovacık Doğal Cooperative's sales points*

The map in Figure 15 is based on the locations of the sales points we found on the Ovacık Doğal Cooperative's website. The map provides important data to understand the geographical distribution and regional density of cooperative products across Türkiye. When the map is examined, it is seen that the vast majority of the sales points are concentrated in the Marmara, Aegean and Mediterranean regions. The high density in Istanbul and its surroundings in particular shows that these regions play a central role in terms of both population and economy, and that the demand for organic products is higher in the west. Similarly, the sales points in Izmir and its surroundings in the Aegean Region reveal that these regions are strategically located for the cooperative's target audience.

However, it is noteworthy that there are no sales points in Tunceli and its surroundings, despite the fact that Tunceli is located in that region and therefore distribution and logistics processes can be carried out more easily and in a shorter time. This situation can only be attributed to the fact that the demand for organic agricultural products in the Eastern Anatolia and Southeastern Anatolia regions is low compared to other regions when the geographical distribution of organic agriculture data in the country is considered. However, the fact that there are no sales points can also be interpreted as

the cooperative focusing on a broader national market rather than the local market. The fact that there are only a limited number of points in Central Anatolia suggests that the cooperative has difficulty reaching the consumer base in these regions or has not yet targeted these regions. The concentration of sales points on the coastline can be associated with the economic prosperity and high demand for organic products in these regions. In addition, the fact that the logistics infrastructure is more developed in the western regions also affects the cooperative's easier management of its operations in these regions.

Although Ovacık Doğal is a cooperative based in Tunceli, in general, the map reveals that the cooperative has reached a certain point in its market strategy, while opening more sales points, especially in the eastern regions, could offer a significant opportunity in terms of growth and reach.

5.1.3 Ekozel Organik

The source of the data in this section is the official website of Ekozel Organik (Ekozel Organik, 2024).

Ekozel Organik is an organization that adopts traditional organic farming methods and aims to protect the ecological balance. Its main purpose is to provide natural and reliable products by establishing a direct connection from the producer to the consumer. The organization's processes from production to the consumer and its marketing activities draw attention with both their strengths and aspects that need to be developed. Ekozel Organik's production process is shaped by completely natural methods, away from chemical inputs. Producers produce in accordance with organic farming standards and the products are inspected by certification institutions. In addition, quality control is prioritized during the harvest and processing of the products, and the processed products are packaged without using additives and made ready for consumption. This situation requires more attention in logistics and cost management, especially for a brand that targets wide-scale distribution and does not have a very high market share.

Ekozel Organik's process of reaching consumers is carried out through both online and physical sales channels. While sales made through the website allow the brand to connect directly with the consumer, physical sales points and ecological markets offer the opportunity to reach a wider consumer base. At this point, in terms of marketing strategies, they follow a strategy that highlights the themes of ecological life and healthy nutrition. While the aim is to increase the brand's awareness through awareness-raising activities and blog posts conducted through social media and digital platforms, transparent information about production processes is also tried to be provided to consumers. However, focusing solely on digital channels carries the risk of creating an obstacle in reaching consumer segments with less access to technology. In addition, the locations of physical sales points are also of great importance because access to consumers may be limited in regions where physical distribution is limited, and disruptions may occasionally occur in the supply chain and delivery times in logistics processes.

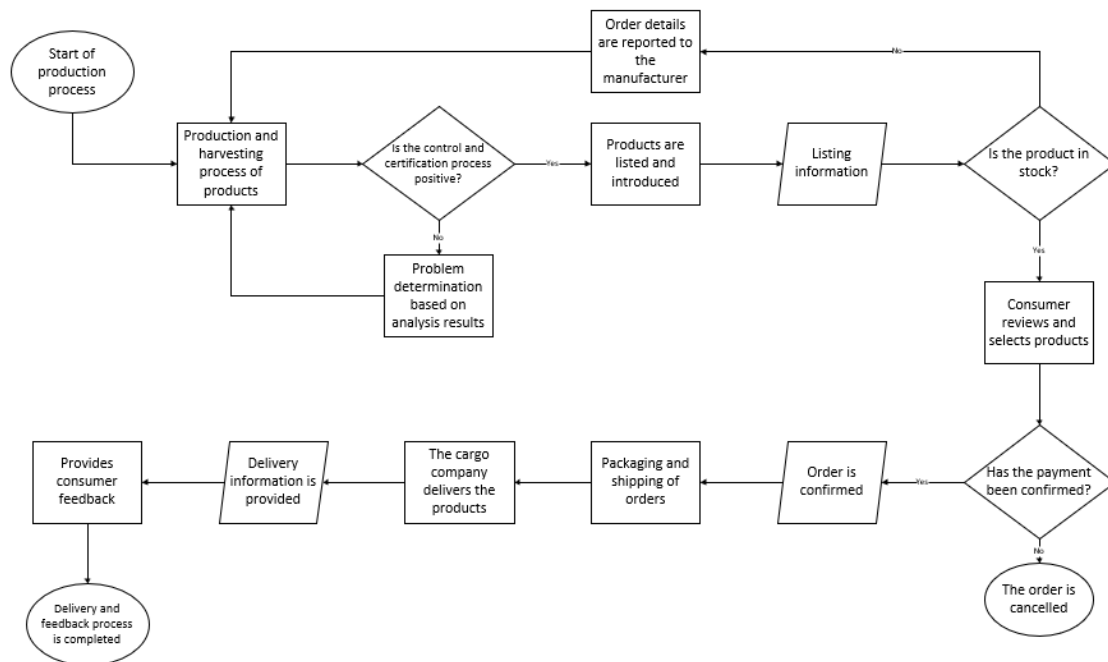


Figure 16: Process flow design of Ekozel Organik

The process of Ekozel Organik from production to the consumer is detailed in the flow diagram in Figure 16, which we prepared ourselves. The process begins with the production of products in accordance with organic farming principles. Following the production and harvesting process, the products proceed to the control and certification stage. At this stage, the compliance of the products with organic standards is checked. If a problem is detected during the certification process, analysis is performed, necessary corrections are applied and the process is restarted. Products that pass certification are listed through the cooperative or platform and offered to consumers according to their stock status. If there are products in stock, consumers examine the products and create orders. Payment approval is received after the order. Orders for which payment approval cannot be received are canceled, while approved orders are packaged and delivered to the cargo for shipment. The cargo company delivers the products to the consumers and feedback from the consumers is collected after this stage. Feedback is evaluated to improve the process and increase customer satisfaction. This structure shows that Ekozel Organik adopts a transparent, environmentally friendly and consumer-focused business model. However, disruptions that may occur at critical stages such as certification and inventory management are noteworthy as factors that may affect operational efficiency.

5.2 Senin Bahçen Model

The “Senin Bahçen” model offers two garden size options, 50 and 100 square meters, through its website. Consumers will first create a membership on the website and then select the garden size that best suits their needs. Consumers who become garden owners will be able to choose seasonal products from the product list provided. The planting, cultivation, and maintenance activities in their gardens will be carried out according to the calendar and guidance set by the expert agricultural team of our social cooperative. These tasks will be performed by the local women of the region, who are both members and employees of the cooperative. Once the harvest season begins, the

products will be collected on a designated day each week, subjected to quality control, packaged, and delivered to a shipping company for transportation to their owners. During this 12-15 week process, consumers can visit their gardens at any time, harvest their own products, and observe the production process firsthand. Additionally, through the website, they can access technical details such as the seeds used in production, how and when irrigation, fertilization, and maintenance are carried out, as well as the estimated harvest times.

We prioritize viewing labor not as an area of exploitation but as a space for organization and solidarity. Supporting women's labor, ensuring economic development, and creating social benefits for our region are among our key priorities. The income generated from the "Senin Bahçen" project will be used to grow and ensure the sustainability of the cooperative, establish it as an economically viable project, and distribute it equitably among the members of the social cooperative. We emphasize the importance of labor as a domain of solidarity and organization, rather than exploitation. Supporting women's labor, fostering economic development, and creating social benefits for our region are central goals.

In summary, the "Senin Bahçen" project can be defined as a "Sustainable and Inclusive Agricultural Business Model." It is a social cooperative model that uplifts agricultural labor, prioritizes the employment of women in the region, supports local development, respects nature, and prioritizes access to healthy and reliable food. The need for this project is clear to strengthen traditional agriculture, revive collective awareness, prevent the harmful practices of industrial agriculture that poison both people and nature, and avoid deceptive practices disguised as hobby gardening.

5.3 Benchmark Analysis

Above, the aims of ÇiftçidenEve, Ovacık Doğal and Ekozel Organik organizations in the organic agriculture market in Türkiye, their structures, how their processes from production to consumer work, what their marketing strategies are and other quantitative and qualitative data that can be accessed about the organizations were mentioned. Although the “Senin Bahçen” project has similarities with the Ovacık Doğal Cooperative with its social cooperative structure that prioritizes women’s employment, it adopts a business model that has no example in the Turkish organic agriculture sector in terms of making production specific to demand by renting land to consumers. Table 7 which shows benchmark analysis was created by considering the various similarities and differences between the existing solutions in the sector that we examined above and our business venture model “Senin Bahçen”.

Features	Senin Bahçen	ÇiftçidenEve	Ovacık Doğal	Ekozel Organik
Social Solidarity and Social Cooperative Structure	+	-	+	-
Support for Women's Labor and Employment	+	-	+	-
Support for Regional Development	+	-	+	-
Production Specific to Consumer Demand	+	-	-	-
Benefiting from Social Support Mechanisms	+	-	+	-
Transparency and Reliability in Production Processes	+	+	+	+
Support for Local Producers	+	+	+	-
Various Product Range	+	+	-	+
Environmental Awareness	+	+	+	+
Opportunity for Consumers to Participate in Production and Visit	+	-	-	-

Table 7: Benchmark analysis table

5.4 Stakeholder Analysis

Internal Stakeholders:

- Project Management Team: The team responsible for the planning, organizing, leading and controlling of our project. Their role is to ensure that our cooperative's operations run smoothly.

- Expert Agriculturalists: Professionals who will supervise the planting, maintenance and harvesting processes and create the calendar of the process. They ensure that the products are grown in accordance with organic standards. They will be members of our cooperative.
- Women Employees: Women who will work in our gardens in jobs such as planting, maintenance, harvesting, product sifting and packaging. In addition to generating economic income, their integration into social life will be ensured. The members of our cooperative will largely consist of our women employees.
- Technical Support Team: Our employees who are involved in the development, management and user experience of our website. They will be members of our cooperative.
- Marketing and Communication Team: Our employees who promote our project, manage our customer relations and aim to increase the brand value of our social cooperative. They will be members of our cooperative.

External Stakeholders:

- Customers (Garden Owners): They are individuals who rent their gardens through our website and manage the production process of their products. They are the main source of income for our project.
- Local Governments: These are the public institutions we will cooperate with for the allocation of agricultural land and local support for our project.
- Funders: National and international organizations that provide financial support to our project. They provide important resources for the sustainability of our project.
- Suppliers: They are the companies that provide organic seeds, saplings and agricultural equipment for our project. They are necessary for the start of our production process.

- Cargo and Logistics Companies: They are the service providers that ensure the timely and safe delivery of our harvested products to our customers.
- Civil Society Organizations (CSOs): In order to increase the social benefit of our project, we will receive support from organizations in areas such as women's labor, nature conservation and social cooperatives.
- Academic Institutions: Universities and research centers from which we will receive information support for the development of our project and the implementation of innovative agricultural techniques.
- Consumer Groups and Associations: Organizations that raise awareness about access to healthy and organic food and defend the rights of consumers.
- Local Society: Indirectly affected by the implementation of our project are the people of the region, which we plan to benefit in terms of economic and social development.
- News and Media Organizations: Press and media organs that we will cooperate with to promote our project and raise awareness.

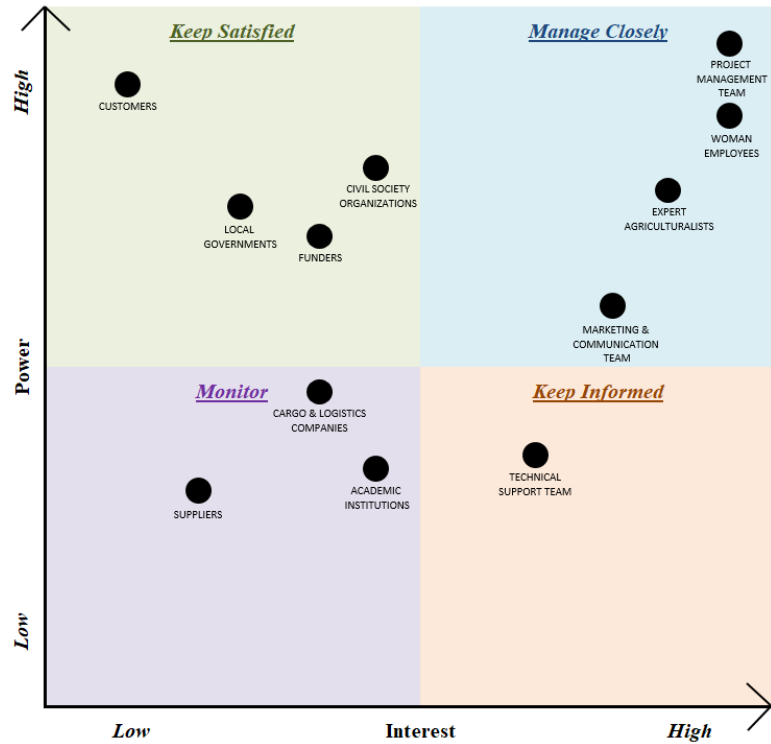


Figure 17: Stakeholder analysis

5.5 Risk Analysis

Risk management is a process that determines the risks and possible negative effects that may arise during project execution. It can be used as a guide by managers to minimize possible negative effects. When creating a risk management plan, risk identification is first necessary. At this stage, the brainstorming method was used to determine the risks. Then, the risks were divided into two categories; internal risks and external risks. Following this, the risks were rated between 1 and 5 according to both their probability of occurrence and their impact on the project. The importance levels were determined by multiplying the probabilities of the risks with their impacts. The importance levels were created using the probability effect risk assessment matrix in Table 8.

Probability	VH (5)	5	10	15	20	25
	H (4)	4	8	12	16	20
	M (3)	3	6	9	12	15
	L (2)	2	4	6	8	10
	VL (1)	1	2	3	4	5
		VL (1)	L (2)	M (3)	H (4)	VH (5)
		Impact				

Table 8: Risk analysis matrix

Table 9 shows the identified risks, probability and impact scores, possible responses, possible course of action and classification.

Risks	Likelihood	Impact	Risk Score	Priority	Possible Responses	Risk Actions	Classification
Seasonal conditions negatively affecting production (frost, drought, etc.)	4	5	20	Critical	Consult agricultural experts to plan crops according to seasonal risks	Mitigate	External
Garden rental demand falling below expectations	4	4	16	Very Important	Develop effective marketing strategies and conduct target audience analysis to increase demand	Mitigate	External
Lack of collaboration or conflict among social cooperative members	3	4	12	Important	Organize regular meetings and training sessions to strengthen communication among members	Mitigate	Internal
Insufficient funding or support during the financing process	3	5	15	Very Important	Strengthen partnerships with local governments, international organizations, and NGOs to secure funding	Accept	External
Delays or damages in the delivery process	4	5	20	Critical	Improve logistics processes by working closely with cargo companies and establish backup plans	Mitigate	External
Feedback not being effectively evaluated and applied	3	4	12	Important	Create a feedback tracking system and use the data to implement improvements in the process	Mitigate	External
Increasing shipping costs affecting economic sustainability	4	4	16	Very Important	Optimize shipping costs by partnering with contracted companies and conducting cost analyses	Mitigate	External
Lack of timely updates to consumers regarding the process	3	3	9	Evaluable	Regularly share process updates via email or SMS to keep consumers informed	Mitigate	Internal
Technical infrastructure issues reducing customer satisfaction	3	3	9	Evaluable	Strengthen technical infrastructure through regular maintenance and backup systems	Mitigate	Internal
Expenses higher than expected costs	4	4	16	Very Important	Review cost management processes	Accept	External
The impact of economic fluctuations on the budget	4	4	16	Very Important	Consider economic fluctuations in budget planning	Accept	External
Delays in meeting consumer-specific product demands	3	4	12	Important	Accurately calculate growth timelines for selected crops and offer alternative product suggestions	Mitigate	Internal

Table 9: Risk analysis table

5.6 Process Flow Design

The flowchart in Figure 18 is the process flow diagram for our “Senin Bahçen” project. The part of the process where the consumer creates a membership, chooses the garden size, the products they want to plant and completes the payment process takes place once at the beginning. In the continuation of the process, the products are regularly harvested, packaged and delivered to the cargo on a certain day of the week for 12-15 weeks. At the same time, the feedback received from consumers during these 12-15 weeks is analyzed and improvements are made as a result of these feedbacks.

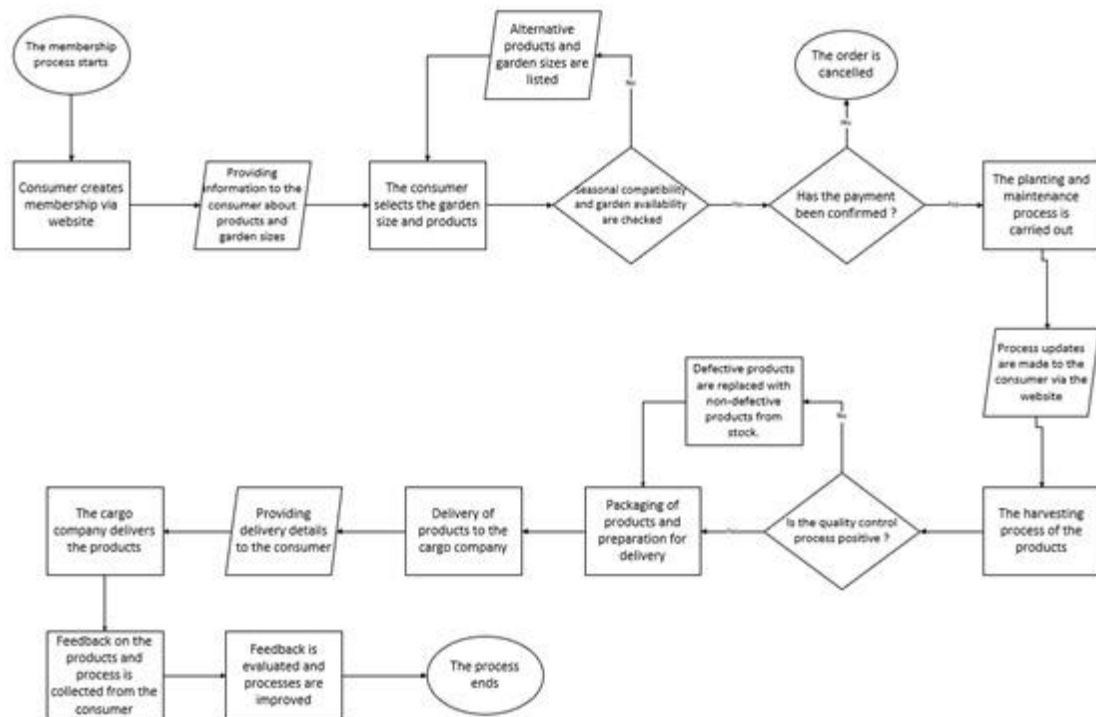


Figure 18: Process flow design of Senin Bahçen

6. MARKET RESEARCH AND SURVEY ANALYSIS

A market research survey was conducted to evaluate the suitability of the "Senin Bahçen" project carried out within the scope of this study for the target customer group and to base design decisions on data. The main purpose of the survey is to reveal the attitudes, behaviors and payment requests of potential customers towards themes such as healthy living, organic products, production transparency and social responsibility.

6.1 Data Collection Method

A market research survey was conducted to evaluate the suitability of the "Senin Bahçen" project carried out within the scope of this study for the target customer group and to base design decisions on data. The main purpose of the survey is to reveal the attitudes, behaviors and payment requests of potential customers towards themes such as healthy living, organic products, production transparency and social responsibility.

The survey was implemented online via the Google Forms platform and conducted in April 2025. Participants were selected from individuals reached via social media, personal communication channels and e-mail on a voluntary basis. A total of 59 valid responses were obtained.

The targeted sample profile within the scope of the research was determined as individuals living in the city center, between the ages of 25-55, belonging to the middle-upper socioeconomic class, and attaching importance to healthy and natural living. This profile was found to be appropriate in terms of representing the target audience of the project.

The survey form includes a total of 22 questions consisting of closed-ended questions. The majority of the questions are in multiple choice format, and some questions are aimed at measuring the attitudes of the participants with a 5-point Likert scale. In

addition, some questions were given multiple choice options, allowing participants to indicate more than one preference. The survey questions are structured around themes such as demographic information, consumption habits, perception of organic products, willingness to pay and social responsibility approach.

6.2 Survey Structure and Question Groups

The survey form was structured to test the basic assumptions of the project and to better understand the needs of the target audience. The questions were grouped under four main categories in terms of content:

- Demographic Information: Basic characteristics of the participants such as age, gender, education level, income level, region of residence and employment status were determined. This data is important in terms of evaluating the compatibility with the target audience profile.
- Healthy Lifestyle Habits and Digital Behaviors: Questions were asked about lifestyle indicators such as the participants' interest in healthy nutrition, regular exercise status, and plant-based nutrition preferences and actively used digital platforms. In this section, the participants' subjective interest in healthy living was also evaluated with a 5-point Likert scale.
- Consumption and Shopping Habits: This section covers how often and through which channels the participants shop for vegetables, the factors they consider important in product selection (freshness, organicity, price, place of production, etc.) and organic product preference. Both multiple choice and checkboxes questions were used in this section.
- Organic Product Perception, Trust and Social Values: The level of trust in organic products, the importance given to transparent production processes, the impact of social responsibility values (women's employment, environmentally friendly production, etc.) on consumption decisions and

willingness to pay were measured with 5-point Likert scales. In addition, participants' organic product consumption motivations and the elements they found attractive were questioned in a checkbox questions format.

Thanks to this structure, the survey provided a comprehensive data set in terms of both demographic and behavioral analysis. The results obtained were evaluated in order to verify the customer segmentation of the project and to guide design decisions.

6.3 Survey Findings

As a result of the survey, meaningful data was obtained about the target customer group of the “Senin Bahçen” project. This section summarizes the demographic profiles, consumption habits, attitudes towards organic products and perceptions about the project topic of the participants. The entire survey form is provided in Appendix 12.3.

- Demographic Distribution: 55.9% of the 59 participants in the survey were male, 40.7% were female. The age distribution was mainly between the ages of 25-44 (66.1%). 69.5% of the participants were undergraduate graduates or students, while 15.3% had a master’s degree or higher. In terms of monthly income distribution, 44.1% of the sample is in the range of 60,000 TL - 80,000 TL, 18.6% is in the range of 80,000 - 100,000 TL, and the majority of the sample represents the upper-middle income group. In terms of place of residence, it is seen that individuals living in the city center are the majority with a rate of 74.6%.

Percentage of Respondents (%)

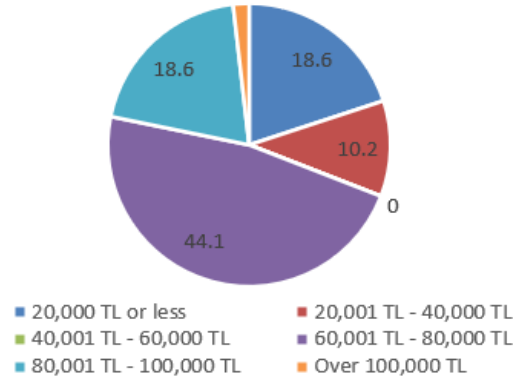


Figure 19: Monthly income distribution of survey respondents

- Healthy Living and Consumption Trends: 66.1% of the participants stated that they care about healthy nutrition, and 44.1% preferred organic/natural products. In the relevant questions of the survey measured with a 5-point Likert scale, it was observed that there was an average interest in healthy living and natural nutrition at a level of 4.1. In addition, it was seen that social media platforms such as Instagram (83.1%) and X (50.8%) were used intensively.

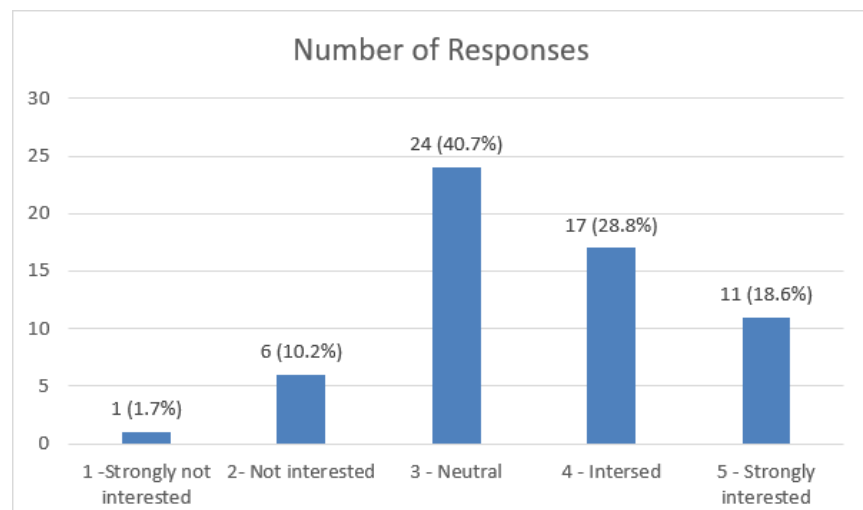


Figure 20: Respondents' self-assessment on interest in healthy living and natural nutrition

- Shopping Habits and Preferences: In terms of frequency of vegetable shopping, 52.5% of the participants stated that they shop once a week, while 28.8% stated that they shop two or three times a week. Chain markets (76.3%) and neighborhood markets (42.4%) stand out among the preferred shopping channels. The most important determining factors in purchasing decisions are freshness (88.1%), organicity (47.5%) and price (44.1%).

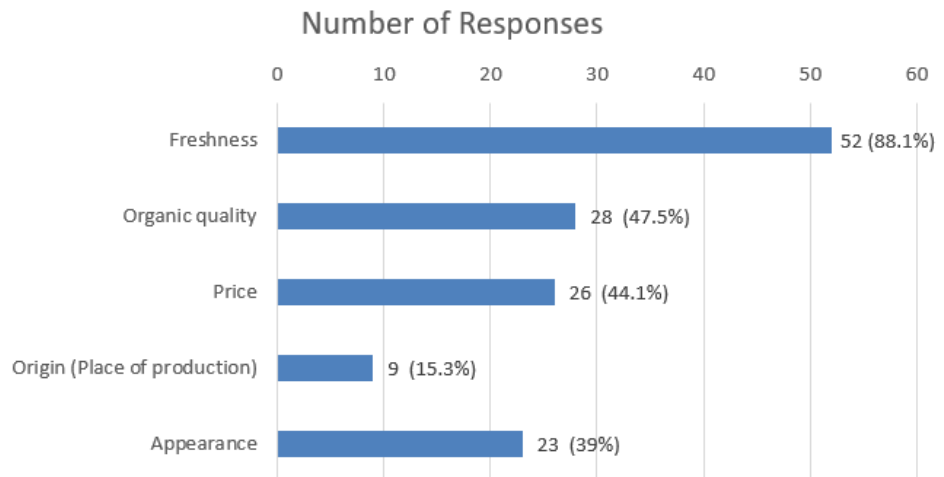


Figure 21: Key criteria considered by respondents when purchasing vegetables

- Attitudes Towards Organic Products and Willingness to Pay: The level of trust in whether organic products are truly organic was found to be relatively low; only 6.8% of the participants stated that they had full confidence, while 57.6% stated that they did not. On the other hand, the idea of products obtained directly from the producer through a transparent production process was largely approached positively (78.0%). While 93.2% of the participants stated that they would support a project for organic production, 49.1% said they were open to online procurement of organic products, and 42.4% said they were ready to pay 40–80% more than the market price for organic products whose

reliability they were sure of. In addition, the rate of those who stated that social responsibility values were effective in their consumption decisions was 62.7%.

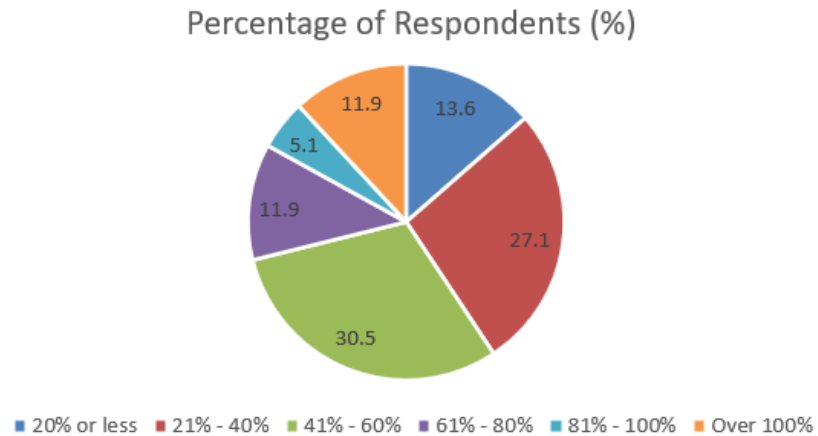


Figure 22: *Respondents' willingness to pay above the market price for reliable organic products*

- Consumer Motivations: The main motivations for organic product consumption include “protecting one’s own health” (72.9%), “protecting family health” (66.1%) and “supporting local producers” (59.3%). The main factors that make organic products attractive are product reliability (76.3%), transparency of the production process (62.7%) and easy access (52.5%).

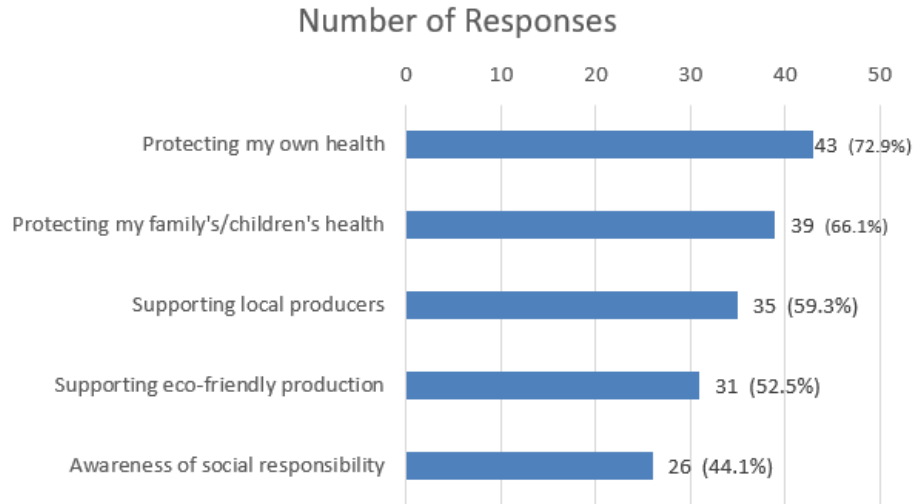


Figure 23: *Key motivations for organic product consumption among respondents*

6.4 Statistical Data Analysis and Interpretation of Findings

6.4.1 Normality Tests and Distribution Analysis

In order to examine the distribution properties of the data, visual and statistical normality tests were first performed. Histogram and QQ-plot graphics were used in visual analyses, and the Shapiro-Wilk test was preferred in statistical analyses. All analyses were performed using R Studio 2024.03.0 Build 386 version. The R codes used are given in Appendix 12.4.

In the QQ-plot graphics, it was observed that the data systematically deviated from the theoretical normal distribution line. Significant deviations were observed especially in the extreme values, indicating that the distribution was significantly different from normal.

In the Shapiro-Wilk test applied to support these visual analyses, the fact that the p-values for all variables were below 0.05 revealed that the data did not show a statistically normal distribution. Table 10 shows the Shapiro Wilk Normality Test results.

Variable	W Value	p-value	Distribution Interpretation
Interest in healthy lifestyle	0.889	6.29E-05	Not normally distributed
Preference for organic products	0.876	2.25E-05	Not normally distributed
Trust in organic products	0.878	2.79E-05	Not normally distributed
Interest in transparent producers	0.798	1.47E-07	Not normally distributed
Influence of social responsibility	0.823	6.52E-07	Not normally distributed
Willingness to purchase online	0.887	5.29E-05	Not normally distributed
Importance of production transparency	0.799	1.53E-07	Not normally distributed
Willingness to support organic projects	0.66	2.03E-10	Not normally distributed

Table 10: *Shapiro-Wilk Normality Test results*

As a result, it was determined that the data did not meet the normal distribution assumption as a result of both visual and statistical analyses. Therefore, non-parametric test methods were preferred in the continuation of the analyses. Mann-Whitney U and Kruskal-Wallis tests were used in difference analyses; Spearman Correlation Coefficient was used in relationship analyses.

6.4.2 Descriptive Statistics

Since the assumption of normality was not met, the median, interquartile range (IQR), minimum and maximum values were used as descriptive statistics. These statistical summaries are of critical importance in understanding the central tendencies and

distributions of the data. Table 11 provides descriptive statistics of the basic variables included in the survey.

Variable	Minimum	Median	Maximum	IQR
Interest in healthy lifestyle	1	4	5	1
Preference for organic products	1	3	5	2
Trust in organic products	1	3	5	2
Interest in transparent producers	1	4	5	2
Influence of social responsibility	1	4	5	1
Willingness to purchase online	1	3	5	2
Importance of production transparency	2	5	5	1
Willingness to support organic projects	2	5	5	1

Table 11: Descriptive statistics of key survey variables

According to the findings, the majority of the participants show high interest in healthy living (Median = 4), find the transparency of the production process important (Median = 5) and are inclined to support organic production projects (Median = 5). On the other hand, a relative dispersion was observed in the levels of trust and preference for organic products.

6.4.3 Difference Tests and Hypothesis Tests

Since the data did not show a normal distribution, nonparametric methods were used to test whether there were significant differences between demographic groups

(gender, age, income and education level) and participant attitudes. All analyses were performed in the R Studio environment and the codes used are given in Appendix 12.4.

- Differences by Gender: A significant difference was found between female and male participants only in terms of the variable “trust that organic products are truly organic” ($p = 0.0016$). No significant difference was observed in other variables ($p > 0.05$). This finding shows that although women and men generally approach organic products in a similar way, there is a significant difference in terms of their trust levels. Female participants declared higher trust in this regard.
- Differences by Age Range: No significant difference was observed between different age groups in terms of any variable. All p -values were greater than 0.05. This finding shows that although some trend changes are observed as age progresses, these differences are not statistically significant.
- Differences According to Income Level: No significant difference was found in any variable according to the monthly income levels of the participants. This shows that different income levels do not have a significant effect on organic product preference, trust or support. According to this result, attitudes do not

change as income levels change; organic product interest seems to be a trend independent of income.

- Differences According to Education Level: Statistically significant differences were found in three different variables according to education level.

Variable	p-value
Interest in healthy living and natural nutrition	0.008
Tendency to prefer organic products	0.01
Willingness to purchase organic products from transparent producers	0.006

Table 12: Significant differences according to educational attainment

The findings in Table 12 show that as the level of education increases, the participants' interest in healthy living, their preference for organic products and their desire for transparent producers increase significantly. It is observed that individuals with a master's degree or higher give higher scores on these issues.

A set of hypotheses was developed to test whether demographic characteristics have a statistically significant effect on the participants' attitudes towards organic products and healthy living. Table 13 summarizes all hypotheses tested in this study, including the formulated null and alternative hypotheses, the applied nonparametric tests, the obtained

p-values, and the final conclusions. In all cases, statistically significant differences were found, leading to the rejection of the null hypotheses.

Hypothesis No	Null Hypothesis (H ₀)	Alternative Hypothesis (H ₁)	Test Type	p-value	Conclusion
H1	There is no significant difference in trust in organic products between males and females.	There is a significant difference in trust in organic products between males and females.	Mann-Whitney U	0.0016	H ₀ rejected
H2	There is no significant difference in interest in healthy living across different education levels.	There is a significant difference in interest in healthy living across different education levels.	Kruskal-Wallis	0.008	H ₀ rejected
H3	There is no significant difference in preference for organic products across education levels.	There is a significant difference in preference for organic products across education levels.	Kruskal-Wallis	0.01	H ₀ rejected
H4	There is no significant difference in willingness to buy from transparent producers across education levels.	There is a significant difference in willingness to buy from transparent producers across education levels.	Kruskal-Wallis	0.006	H ₀ rejected

Table 13: Hypotheses tested, related assumptions, and results

6.4.4 Correlation Analysis and Interpretation of Findings

Since the data set did not provide normal distribution conditions, the relationship analysis between variables was performed with the nonparametric Spearman rank difference correlation coefficient (ρ). This analysis aims to determine the direction and strength of the relationships between the participants' interest in healthy living, organic product preference, social responsibility sensitivity and attitudes towards transparent production processes. All analyses were performed in the R Studio environment, and the heat map visualizing the correlation coefficients is presented in Figure 24. The codes are given in Appendix 12.4.

According to the results obtained, a moderately positive relationship ($\rho = 0.38$) was determined between the interest in healthy living and organic product preference. Weakly positive relationships were observed between the same variable and organic product trust ($\rho = 0.29$), transparent producer interest ($\rho = 0.31$) and social responsibility impact ($\rho = 0.27$). Very weak positive correlations were found between variables such as positive approach to online sales ($\rho = 0.14$), importance given to transparency of

production process ($\rho = 0.12$) and support for organic production ($\rho = 0.23$) and interest in healthy living. These findings show that interest in healthy living has a positive but limited effect on the general tendencies of individuals.

A moderate correlation was found between organic product preference and transparent producer interest ($\rho = 0.48$) and social responsibility effect ($\rho = 0.40$); and a weak-moderate positive correlation was found with support for organic production ($\rho = 0.37$). These results indicate that individuals who prefer organic products are also more sensitive to transparent production processes and social responsibility elements.

A high level of positive relationships was found between transparent producer interest and social responsibility effect ($\rho = 0.59$), and a moderate level of positive relationships were found between positive approach to online sales ($\rho = 0.43$), transparency of the production process ($\rho = 0.38$) and support for organic production ($\rho = 0.42$). In particular, the high correlation observed between transparent producer interest and social responsibility sensitivity is an important finding that strengthens the social impact potential of the project.

A moderate level of significant relationship was also found between social responsibility effect and support for organic production ($\rho = 0.42$). This result shows that social values such as women's employment and environmentally friendly production affect the participants' motivation to support organic production.

As a result of all these analyses, it was observed that the highest correlation was between transparent producer interest and social responsibility effect. In addition, it was determined that individuals who care about healthy living generally tend to prefer organic products and show sensitivity to transparency, but the trust in whether organic products are really reliable is generally low and its relationship with other variables remains weak.

This highlights the need for marketing strategies to emphasize not only product quality but also the integrity of production processes and alignment with social values.

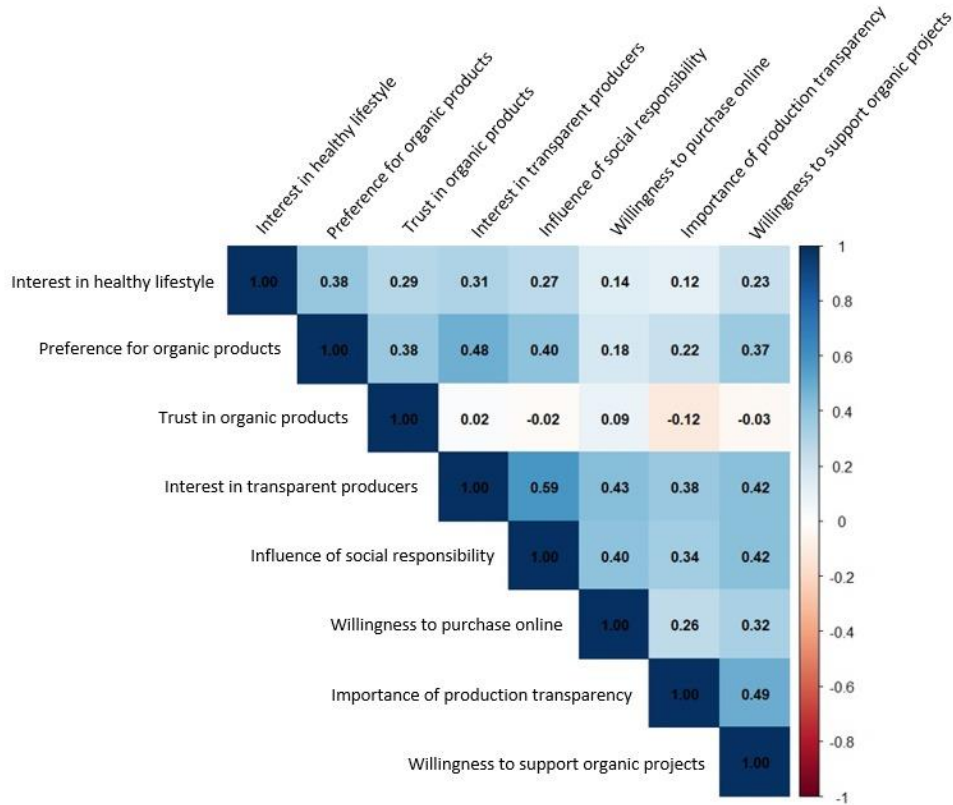


Figure 24: *Spearman Correlation Coefficient heatmap*

6.5 General Conclusion and Implications

This study aimed to assess the consumer potential and behavioral orientation toward the “Senin Bahçen” project, which promotes organic agriculture through transparent, socially responsible, and participatory production models. Based on the survey conducted with 59 participants, comprehensive statistical analyses were performed

using R Studio, including normality testing, difference testing, hypothesis formulation, and Spearman correlation analysis.

The results indicate that the target audience for the project largely consists of individuals aged between 25–44, residing in urban centers, and belonging to the upper-middle-income group. There is a significant interest in healthy living, accompanied by a moderate tendency to prefer organic products. However, trust in the authenticity of organic labels is generally low, with only 6.8% of respondents expressing full confidence. Despite this, a strong majority showed a favorable attitude toward transparency in production and the idea of purchasing directly from producers.

The difference tests revealed that education level significantly influences health awareness, preference for organic products, and demand for transparent sourcing. Additionally, gender was found to significantly affect the level of trust in organic product authenticity. Correlation analysis further supported these findings, highlighting strong associations between transparency-oriented attitudes and values related to social responsibility.

From a strategic standpoint, the findings suggest that while demand for organic products exists, it is closely tied to consumers' perception of trust and social alignment. Thus, marketing communications should emphasize not only product quality but also transparency of production processes and ethical values such as local empowerment, women's participation, and ecological responsibility. The willingness to pay a premium for trusted organic products (up to 40–80% higher than market price, as stated by 42.4% of participants) also indicates the feasibility of a value-based pricing strategy—provided that the trust barrier is addressed effectively.

Furthermore, the project's potential social impact is supported by the significant correlation between the desire to support organic production and participants' sensitivity

to social causes. This implies that the “Senin Bahçen” model is well-positioned not only to satisfy consumer needs but also to deliver meaningful societal contributions.

6.6 Insight from In-Depth Interviews

In addition to the quantitative data obtained through the survey, five preliminary interviews were conducted within the scope of the “Senin Bahçen” project in order to understand consumer perceptions on themes such as organic agriculture, transparency in the production process and reliability in more depth. These interviews were conducted with individuals between the ages of 28-58, who are urban individuals with high sensitivity to healthy living and sustainable consumption and who are suitable for the target customer profile.

A total of fourteen questions were asked in the interviews to understand the value judgments, motivation sources, concerns and behavioral tendencies of the participants. However, due to the limited number of participants (n=4), a systematic content analysis or scoring method was not applied; the interview data was only evaluated from a qualitative perspective. In this context, the findings obtained do not claim to be generalizable, but they were evaluated as supporting the survey results and contributing to the conceptual development process.

The prominent themes in the interviews include sensitivity to food safety and health issues, the importance given to transparency in the production process and the relationship of trust. In addition, emotional factors such as the desire to protect family health, longing for traditional agriculture, and awareness of social responsibility (e.g. supporting women producers, cooperatives, etc.) were also frequently mentioned. Participants also drew attention to the difficulty of accessing natural and clean food in urban life and positively evaluated models that enable direct contact with producers.

“We support small producers instead of the rich. The problems you mentioned are actually Türkiye’s problems.” (Interviewer 1, male, age 58)

“It is not easy to find natural products in a place like Istanbul. If I am happy with it, of course I will do it.” (Interviewer 2, male, age 45)

“Even if I cannot consume it all myself, I will send it to my family and friends.” (Interviewer 3, male, age 28)

“Access to non-toxic food is very important today.” (Interviewer 4, female, age 47)

“In order to support regional producers, I make it a point to buy the products of peasant women who are already producing.” (Interviewer 5, female, age 54)

These quotes highlight a deep trust gap in the organic market, as well as a strong emotional attachment to clean and traceable food sources. Although the number of interviews was limited, the fact that these views coincide with the survey findings indicates a significant consistency in user expectations.

6.6.1 Interview Coding Scheme

An interview coding and theme analysis was carried out to systematic evaluate the qualitative information obtained from in-depth interviews. This analysis process made it possible to identify important motivators, issues, and criteria for making decisions that support the Senin Bahçen project's main goals, which include advancing organic production based on trust, assisting women in the workforce, guaranteeing production transparency, and providing social and environmental benefits. The emerging themes, their frequency of recurrence, and their perceived significance in relation to consumer expectations are compiled in Table 14 and Table 15.

	Sentence	Label 1	Label 2
Interview 1	I make sure that it is fresh and natural. Apart from that, price is also a factor.	Healthy Nutrition	Price Sensitivity
	I normally buy organic products from friends... But I do not buy the ones in markets.	Need for Trust	Price Sensitivity
	Of course these are good things, we support small producers instead of rich people.	Small Producer Support	Social Benefit
	If the price is not too excessive, I would like to have organic boxes sent to my home.	Price Sensitivity	Interest in Subscription Model
	I'd pay 50% or even 100% more for vegetables I know were produced in my own field.	Willingness to Pay High Prices	Customized Production
	I want to make sure that I am not being scammed... transparency... monitoring.	Need for Trust	Transparency
Interview 2	If I buy it and am satisfied... I would recommend it... supports women's labor...	Social Benefit	Tendency to Recommend
	I prefer products that look natural and are not too cheap.	Healthy Nutrition	
	I don't trust organic-labeled products at all... news of pesticides...	Need for Trust	Price Sensitivity
	If I knew a natural producer, I'd buy everything from them.	Need for Trust	Direct Producer
Interview 3	Transparency is very important... I need to follow the process.	Transparency	Need for Trust
	I usually prefer the ones that are average priced and look natural.	Healthy Nutrition	Price Sensitivity
	They are supposedly all natural, all chemical-free – I don't trust them at all.	Need for Trust	
	I volunteered for a project on women's employment.	Social Benefit	
	I would definitely look favorably if I can trust that it is organic.	Need for Trust	
	Even if I couldn't consume it all myself, I would send it to family and friends.	Social Benefit	Tendency to Recommend
Interview 4	I don't care about price. If it's healthy and beneficial, I'll pay.	Willingness to Pay High Prices	Healthy Nutrition
	I want to trust organic certificates... even in Europe there are residues.	Need for Trust	Certification
	I shop from cooperatives and neighborhood-based bulk buying groups.	Social Benefit	
	Product diversity and price are important.	Price Sensitivity	Product Diversity
	Access to non-toxic food is very important today.	Healthy Nutrition	
Interview 5	I especially try to find producers who do not use chemicals in production.	Need for Trust	Healthy Nutrition
	Factors such as the proximity of the regions where so-called organic products are grown to the industry shook my confidence.	Need for Trust	Certification
	I make it a point to buy what the village women produce.	Social Benefit	
	Cargo penetration and carbon footprint are the issues I am most sensitive to.	Environmental Concern	Certification
	Transparency is important, whatever is being done should be shared with the reasons.	Transparency	Need for Trust

Table 14: Interview Coding Scheme

Theme	Frequency	Perceived Importance
Need For Trust	11	High
Price Sensitivity	6	High
Healthy Nutrition	6	High
Social Benefit	6	High
Transparency	3	Medium
Certification	3	Medium
Tendency to Recommend	2	Medium
Willingness to Pay High Prices	2	Medium
Small Producer Support	1	Low
Interest in Subscription Model	1	Low
Customized Production	1	Low
Direct Producer	1	Low
Product Diversity	1	Low
Environmental Concern	1	Low

Table 15: Thematic Frequency Table

7. BUSINESS MODEL AND OPERATIONAL STRUCTURE

7.1 Business Model Canvas

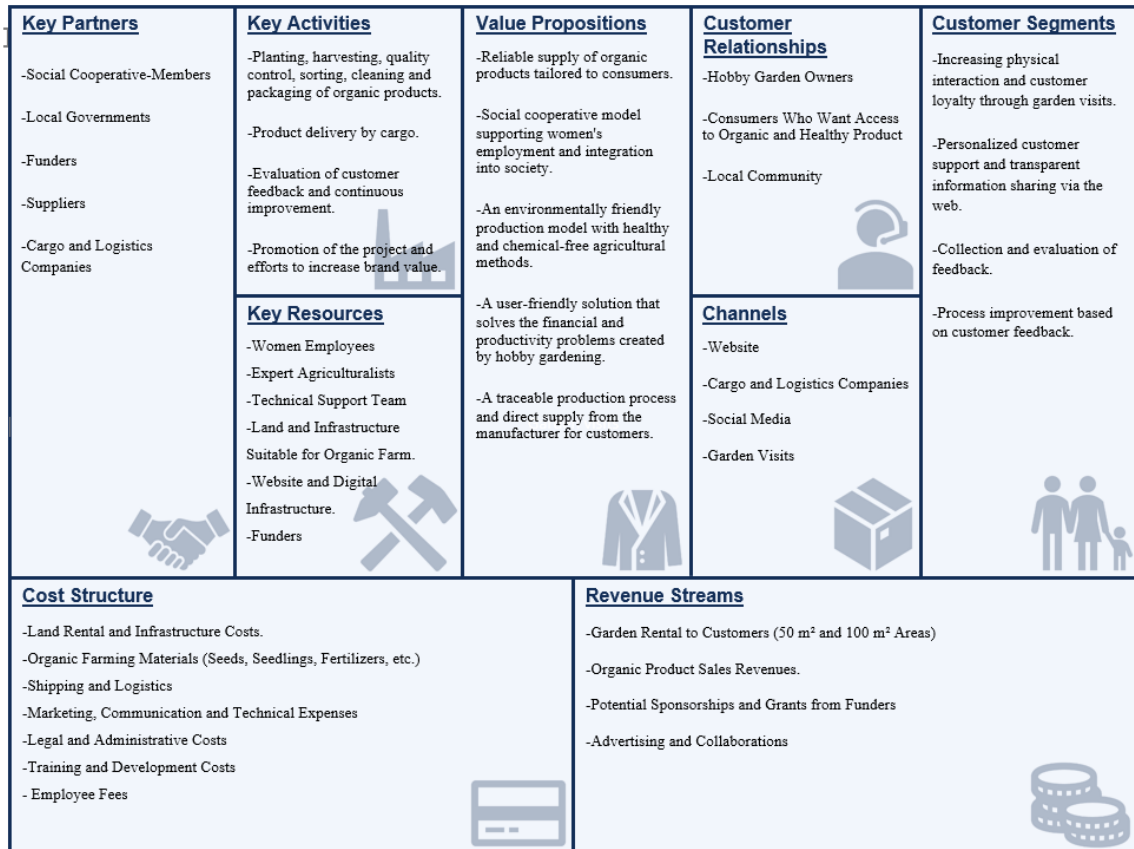


Figure 25: Business Model Canvas

7.2 Social Business Model Canvas

Key Resources	Key Activities	Type of Intervention	Segments	Value Proposition
- Woman Employees - Technical Support Team - Land and Infrastructure - Website & Digital Infrastructure - Funders and Sponsors	- Planting, harvesting, sorting, and packaging organic products. - Delivering products via logistics partners. - Managing garden rentals for hobby gardeners. - Collecting & acting on customer and community feedback. - Promoting the social cooperative model.	- Product: Organic Vegetables - Service: Rental of personal garden plots (50 m², 100 m²) - Experience: Hobby gardening experience for urban citizens. - Program: Employment and empowerment of rural woman through agriculture.	- Rural woman gaining employment and economic empowerment. - Urban individuals connecting with nature and sustainable farming.	- Empowering woman through employment and co-ownership. - Promoting environmentally sustainable, chemical-free agriculture. - Strengthening local food systems and rural economies.
Partners + Key Stakeholders	- Organizing training sessions for woman employees.	Channels	<i>Beneficiary</i> - Health-conscious consumers seeking organic food. - Families and individuals interested in hobby gardening.	<i>Social Value Proposition</i> - Number of woman employed. - Volume of organic produce delivered - Garden usage rates and customer satisfaction. - Environmental indicators (chemical use avoided, composting used)
- Local Governments. - Social Cooperative Members. - Funders & Grant Organizations. - Logistics & Supply Companies. - NGOs Supporting Woman Employment		- Website and Digital Platforms - Social Media Campaigns - Garden Visits and Guided Tours. - Logistics Partners for Delivery.	<i>Customer</i>	<i>Impact Measures</i> - Access to fresh, organic, trustworthy food. - Transparent production process & farm-to-table experience. - Enjoyable & therapeutic gardening activity.
Cost Structure		Surplus	Revenue	
- Land Rental & Irrigation Infrastructure. - Organic Seeds, Fertilizers and Compost. - Logistics & Cargo Operations. - Marketing, Communication & Branding. - Salaries for Woman Workers. - Training, Education & Workshop Materials.		- Reinvestment in woman's training. - Development of Infrastructure - Organizing educational gardening programs for children & families. - Funding additional garden areas & cooperative expansions.	- Garden Rentals. (50 m² or 100 m²) - Sponsorships & Grants - Advertising & Brand Collaborations	

Figure 26: Social Business Model Canvas

The Business Model Canvas prepared at the beginning of the project expressed the basic components of the business model. However, since the project does not only generate economic benefits but also creates social value, the Social Business Model Canvas approach was adopted in the later stages of the project. Thus, by adding additional titles such as “Social Value Proposition”, “Beneficiary”, “Impact Measures” and “Surplus”, social impacts such as increasing women's employment, promoting environmentally friendly production methods and contributing to the local economy have become more visible. A more inclusive segmentation has been made, in which individuals and groups that directly benefit are also identified. With this shift, it is emphasized that the project is not only a product or service offering, but also a structure that produces social benefits. As a result, the Social Business Model Canvas approach reflects the purpose of the project and the value it creates in a more holistic way and demonstrates a strong alignment with the social enterprise perspective.

7.3 Limitations

There are some operational, climatic and structural limitations due to the nature of the agricultural activities carried out under the project. These limitations are given below under headings:

- **Product Variety Limitation:** Due to the use of ancestor seeds and the fact that production is only seasonal, some types of vegetables cannot be produced. It may also not be possible to meet the demand for different product groups (e.g. fruit, legumes, etc.).
- **Climatic Limitation:** The production process is carried out depending on the climatic conditions of Aydın region. However, product yield cannot be guaranteed in unexpected and extreme weather conditions.
- **Delivery Time Limitation:** Product deliveries are made only after the start of the harvest period and on a weekly basis. Interim deliveries are not made during non-harvest periods.
- **Production Area Capacity Limitation:** Production areas available to customers are limited to 50 to 100 square meters. This directly affects the amount of product that can be produced per customer.
- **Operational Limitation:** Since the planned and healthy growth of the cooperative structure is prioritized, there may be delays in responding to this demand if demand increases faster than expected. This is a limitation in terms of production quality and sustaining the understanding of conscious organic agriculture.
- **Logistics Limitation:** Logistical delays that may occur in post-harvest cargo deliveries are among the limitations that may occur beyond the control of the project.

- **Yield and Quality Limitation:** The yield obtained due to organic production methods may be lower compared to conventional agriculture. In addition, periodic variations in product quality can be observed.

These limitations have been taken into account in the planning and operation processes of the project and necessary risk measures have been developed.

7.4 Capacity Development Plan

A long-term capacity building plan has been developed for the sustainability and scalability of the project. This plan covers technical and operational improvements that can be implemented in the short, medium and long term. The plan serves the objectives of efficient use of resources, increasing customer satisfaction and improving production efficiency.

- **High Priority (0–2 Year):** It is planned to gradually increase the arable land to 6, 8, 10 and 12 acres. In the same period, soil moisture sensors and an automatic irrigation system will be integrated to minimize water consumption and make irrigation processes more efficient. In addition, four cameras to be placed at different points of the land will enable live broadcasting, enabling customers to follow their own gardens instantly on the website.
- **Medium Priority (2–5 Year):** Precautions will be taken against frost risk and overheating with early intervention systems equipped with soil temperature sensors. Thanks to weekly drone shots, each customer's garden will be visualized from above and customers will be provided with the opportunity to follow up visually. In addition, it is aimed to extend the production season with different vegetable/fruit plans covering not only the summer season but also the spring and fall seasons.

- Low Priority (5+ Year): It is planned to integrate sensors that monitor soil pH levels and intervene with natural regulators (organic fertilizers, lime, etc.) when necessary. In addition, it is aimed to increase the level of knowledge of stakeholders through online or physical training workshops on topics such as the use of ancestor seeds and organic farming techniques.

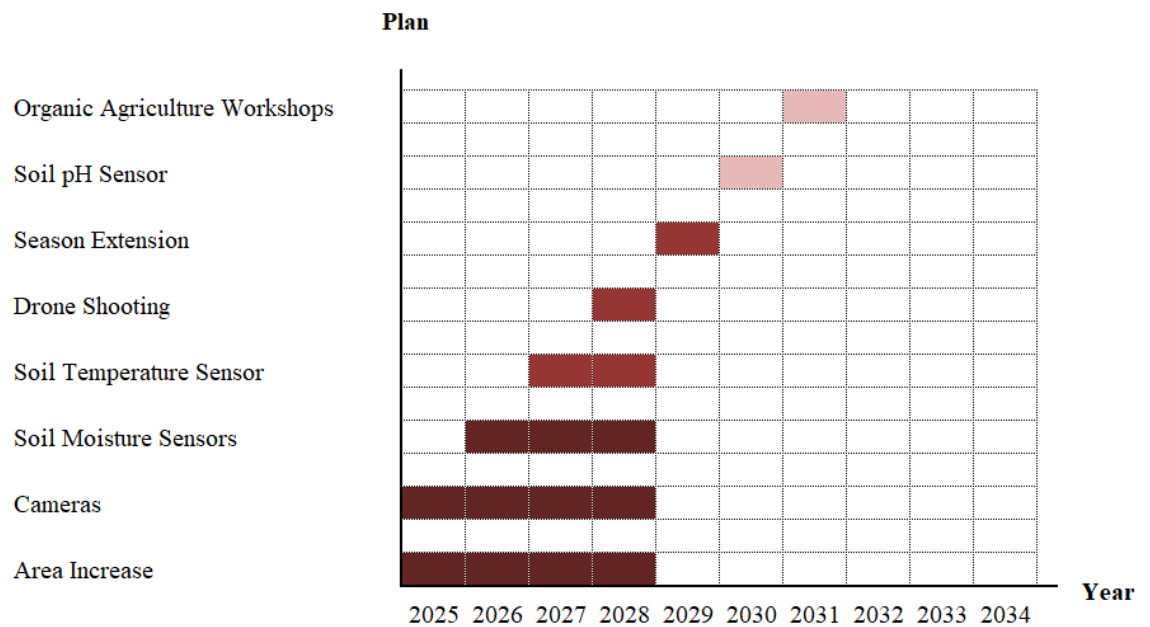


Figure 27: Capacity Development Plan

In the Figure 27, the capacity building plan is presented. This capacity development plan aims not only to strengthen the technical infrastructure, but also to increase the quality of service to the client and the social impact of the project.

7.5 Technical Specifications and Agricultural Production Plan

Due to the limited availability of publicly documented and scientifically validated agronomic data specific to ancestor (Ata) seeds—particularly in terms of planting spacing and yield per seedling—this study initially referred to data sets and guidelines developed

for hybrid seed varieties. These sources provide standardized recommendations widely accepted in conventional agricultural practices.

Subsequently, estimations for ancestor varieties were derived through expert consultation and adjusted in accordance with the unique characteristics of such seeds, including their typically lower yield potential and greater variability. Thus, the figures presented in this study represent an informed approximation based on hybrid seed benchmarks, refined through agronomic expertise and contextual judgment.

The planning of agricultural production and resource utilization within the scope of the project was based on technical data. In this section, parameters such as irrigation requirement, area per seedling, crop yield and total production capacity are presented in detail.

7.5.1 Seedling Area and Crop Yield

The following data are based on figures provided by *Haiifa Group* (n.d.). The vegetables to be produced in the project were distributed in a total area of 4,000 square meters according to different demand rates. For each vegetable type, the area per seedling was calculated according to the recommended planting spacing, and the total number of seedlings and expected yield were determined accordingly. For example, for tomato, an area of 0.51 m² was foreseen and a total of 2,353 seedlings were targeted to yield approximately 6,588 kg of produce. Similarly, for cucumber, the yield per seedling is 4.00 kg, and when the waste rate is deducted, a net yield of 3.20 kg is calculated and a total yield of 3,200 kg is targeted.

When calculating the yield per seedling, a 20% waste rate was taken into account and the efficiency on a m² basis was analyzed based on “net harvest” values (e.g. 13.33 kg/m² for beans and 5.49 kg/m² for tomatoes). A total of 21,705 kg of product is planned to be obtained. Technical details are given in the Table 16.

	Expected Demand (m2)	Planting Spacing (cm)	Area per Seedling (m2)	Harvest per Seedling (kg)	Wastage Rate	Harvesting After Wastage (kg)	Efficiency (kg/m2)	Cultivation Area (%)	Cultivation Area (m2)	Total Seedlings	Estimated Harvest (kg)
Tomato	30%	60 x 85	0,51	3,50	0,20	2,80	5,49	30%	1.200	2.353	6.588
Cucumber	15%	50 x 120	0,60	4,00	0,20	3,20	5,33	15%	600	1.000	3.200
Pepper	15%	45 x 65	0,29	1,50	0,20	1,20	4,10	15%	600	2.051	2.462
Eggplant	10%	65 x 90	0,59	4,00	0,20	3,20	5,47	10%	400	684	2.188
Bean	10%	20 x 45	0,09	1,50	0,20	1,20	13,33	10%	400	4.444	5.333
Zucchini	5%	110 x 110	1,21	3,00	0,20	2,40	1,98	5%	200	165	397
Stuffed Pepper	5%	55 x 75	0,41	2,00	0,20	1,60	3,88	5%	200	485	776
Gherkin	5%	100 x 135	1,35	3,00	0,20	2,40	1,78	5%	200	148	356
Okra	5%	45 x 70	0,32	0,80	0,20	0,64	2,03	5%	200	635	406
TOTAL	100%							100%	4.000		21.705

Table 16: Technical production details

7.5.2 Irrigation Planning

In the production process, a total agricultural activity period of 4 months (May-August) is foreseen. In this period, there are 123 days in total, and it is estimated that 77 days of active irrigation will be carried out depending on the frequency of irrigation determined according to climatic conditions and plant needs. According to the General Directorate of Agricultural Research and Policies (TAGEM, 2017, p. 535), the average monthly water consumption in this period is calculated as 300 tons in May, 400 tons in June, and 420 tons in July and August. Total water consumption is 1,540 tons in 4 months. This weather-dependent consumption is supported by the water efficiency measures planned in the project.

7.5.3 Organic Production Certificate

Producers must apply to a control and certification agency approved by the Ministry of Agriculture and Forestry and receive an Organic Agriculture Certificate in order to legally authorize organic agriculture production activities. Application, field inspection, certification, and annual control phases are all part of the certificate procedure, which needs to be updated annually (Ministry of Agriculture and Forestry, 2021). As of 2025, small-scale crop producing businesses in Türkiye must pay an average fee of 6,000

to 8,000 TL to many certification agencies. These fees cover annual control and reporting expenses as well as application and registration fees. For small businesses that produce on fewer than ten decares, the total cost of the organic agriculture certification services provided by CERES Control and Certification Services is around 6,000 TL (CERES, 2025). Within 30 to 60 days of the application, a certificate is issued, and yearly assessments ensure that the certificate remains valid (CTR Organik, 2025).

7.5.4 Agricultural Insurance Premium

The State-Supported Agricultural Insurance System (TARSİM), which was put into place to safeguard Türkiye's agricultural output from unanticipated dangers including pests, natural disasters, and climate risks, gives farmers the chance to insure their crops. In this regard, growers can cover their crops against a number of dangers, including drought, frost, hail, storms, fire, and extreme rainfall, thanks to crop insurance. The state pays 50% of the insurance premium under Law No. 5363 on Agricultural Insurance; for fruit crop frost coverage, this percentage can reach 67% (TARSİM, 2025a). This support system promotes agricultural sustainability and makes risk management easier, particularly for small and medium-sized producers.

The type of crop, the size of the production area, the location, and the type of coverage chosen all affect crop insurance premiums as of 2025. For instance, the average insurance premium to be paid after state assistance is roughly TL 1,200 for a tomato grower in a 10-decare region (TARSİM, 2025b). However, contracted farmers, crops that are subject to production planning, and areas with water constraints are eligible for additional discounts that range from 10% to 15%. To avoid placing a burden on farmers' finances, insurance premiums are paid in Turkish Lira and a range of installment plans are available. TARSİM is a crucial financial tool that promotes agricultural production sustainability in the face of climate change because of its structure.

7.5.5 Irrigation Union

Since Aydın province is where the Senin Bahçen project is intended to be executed, it is critical to make efficient use of the area's agricultural infrastructure. In this regard, making use of the Aydın Ovası Sağ Sahil Sulama Birliği's irrigation infrastructure will be a wise move in terms of cost planning and sustainable water use. Irrigation unions supply agricultural production areas with controlled and compensated water under the authority of the Irrigation Unions Law No. 6172 (Ministry of Agriculture and Forestry, 2023).

On the basis of similar unions' nationwide norms, a rough financial foundation for the project can be created. In general, irrigation unions charge producers 150 TL to 450 TL per acre for irrigation services as of 2023, according to the Ministry of Agriculture and Forestry's 2023 Minimum Tariff for Water Use Service Fees (Ministry of Agriculture and Forestry, 2023). According to Seyhan Sağ Sahil Sulama Birliği (2025), for instance, the tariff for corn irrigation is 240 TL and for cotton irrigation it is 408 TL per acre. Given that our project will start with an expected 4 acres of production space, it is reasonable to assume that the Aydın Sağ Sahil Sulama Birliği Subscription will cost about 750 TL.

7.5.6 Cooperative Establishment

The Cooperatives Law No. 1163 governs the process of forming a social cooperative in Türkiye and defines a number of financial and legal requirements that must be met. The Articles of Association Draft, which lays up the cooperative's guiding ideas, is among the most significant of these duties. After being approved by the Ministry of Trade, this document, which contains fundamental legal components including the cooperative's name, mission, area of operation, capital structure, shareholding terms, management, and oversight organizations, comes into effect (Ministry of Trade, 2023). At

least seven founding members must sign the agreement, which must be registered before a notary public upon the cooperative's formation (Cooperatives Law, 1163/2).

Notary fees are among the costs incurred during the incorporation procedure (Notary Fees). According to TÜRMOB (2024), notary fees for establishing a cooperative range from 2,000 to 4,000 TL on average, depending on operations such as authorization paperwork, signature certifications, and articles of organization approval. Additionally, the cooperative's legal books must be approved through the Book Certifications process. All commercial books, including general ledgers, journals, and decision books, must be notarized. The corporate identity process must also be finished, and branded papers, like letterheads and stamps, must be ready to be utilized in the cooperative's official documents. Both accounting and external correspondence can use these legally recognized documents (TESK, 2022).

Cooperative training, which is given to management personnel and founding members, is another factor that influences the formation of social cooperatives. Basic knowledge about the cooperative's vision, partnership law, financial administration, and partnership relations is covered in this program, which is organized by the Turkish Cooperative Society and a few universities (Anadolu University, 2023). In terms of institutional capacity, documentation of this training is advantageous, particularly when submitting grant and funding applications. Lastly, in order to register the cooperative with the Trade Registry, a minimum initial capital commitment (Initial Capital Contribution) is needed. This capital, which is often computed as a minimum of 100–500 TL per member, depends on the cooperative's field of activity and membership size (Law on Cooperatives, 1163/19). To finalize the establishment, the capital must be deposited into the cooperative's bank account and the receipt must be turned in to the register.

7.5.7 Licenses and Production Permits

Producers must register with the District Directorates of Agriculture and Forestry in order to legally begin agricultural producing activities in Türkiye. Data such as production area, product kind, producer information, and enterprise type are all recorded into a central database as part of this registration, which is done under the authority of the "Çiftçi Kayıt Sistemi (ÇKS)." In order to prove producer identity and get public support, ÇKS registration is required (Ministry of Agriculture and Forestry, 2023). This registration should also be combined with the "Organik Tarım Bilgi Sistemi (OTBİS)" because the Senin Bahçen project is based on organic agriculture. Additionally, a pump permit or water usage license needs to be acquired if production will use groundwater. Technical information including pump flow rate, depth, and position are included in this permit, which is issued by the General Directorate of State Hydraulic Works (DSİ). According to Law No. 167 on Groundwater, unlicensed water use is punishable by administrative sanctions (DSİ, 2024).

A commercial license, which must be acquired from the appropriate municipality, is necessary for the commercialization of the production activity through sales. Municipal Law No. 5393 of 2005 regulates municipal administrative obligations, which serve as the foundation for this license. To get the license, requirements such the business's physical conditions, cleanliness, fire safety actions, and plan approvals are examined (Ministry of Interior, 2022). A Food Production and Sales Permit is also necessary because the crops that will be produced are intended to be processed directly and sold. The Ministry of Agriculture and Forestry's provincial/district organizations grant this authorization in accordance with Law No. 5996 on Veterinary Services, Plant Health, Food, and Feed. According to the Ministry of Agriculture and Forestry (2024), the food business needs to be registered and its production must follow hygienic guidelines like HACCP. For the

project to function sustainably and auditably within the bounds of the law, certain permissions are essential.

7.5.8 Cooperative Training

Cooperatives have to educate their members and staff through education in addition to completing the legal establishment procedure in order to be managed sustainably. In order to give women cooperative members an elementary knowledge of cooperative structure, legal responsibilities, and democratic governance principles, Anadolu University will offer an online "Cooperative Education Program." The course includes a wide range of subjects in 12 hours, including the concept, background, governance structure, legal framework, financial obligations, and social impact of cooperatives (Anadolu University, 2024). Particularly for rural women who are new to the workforce, the program offers an accessible, easy, and broad learning experience by combining academic knowledge with real-world situations.

In addition to providing information, this training serves to help female members having more active positions within the cooperative and taking part in decision-making processes. According to the literature, women's training in cooperatives improves their status in society and productivity, and by enhancing their economic independence, it directly supports regional development (Spear, 2000). Furthermore, in future applications for funding and public support, the e-certificate that is produced at the conclusion of the training acts as a significant indicator of institutional capability and boosts the institutional representation of individuals. The systematic continuation of such certified trainings is adopted as a strategic objective in order to increase the social benefit of the Senin Bahçen model.

7.5.9 Product Shipping

The logistical structure of the cooperative product delivery process is based on timely, dependable, and economical delivery. Following discussions with Yurtiçi Kargo's Nazilli Branch, it was agreed that clients would receive 30-kilogram boxes containing 21,705 kilograms of products, which is our normal yearly production quantity. The company will provide the cooperative with the packages to be used in the logistics process at no cost to the cooperative under the terms of the established partnership, thereby avoiding extra expenses related to packaging. Each package costs 549.60 TL to transport, and the estimated yearly logistics cost for shipping 724 boxes is 397,635.60 TL (Yurtiçi Kargo Nazilli Branch Official, personal communication, May 2, 2025). This pricing removes seasonal cost changes by using a box-based contract and a fixed fee policy.

The legal guarantee of potential risks during the transportation process is just as important to the success of this distribution model as pricing and planning. The cooperative's preparation of the goods to be shipped in compliance with Yurtiçi Kargo's packaging and packaging standards sets out exactly who is responsible for what in the event of any damage during transportation. The duty to pay for potential damages during transportation may arise if the products are packaged securely before transportation and the invoice accurately states the price. The customer must notify the damage to the cargo officer in writing after noticing it during delivery. Completing these steps and proving that the carrier was in fault for the harm are requirements for the legality of compensation claims (Yurtiçi Kargo, 2025; Turkish Commercial Code, 2011).

In addition to physically delivering crops to customers, this integrated logistics solution gives an advantage in terms of customer happiness, product safety, and cooperative image. Quick and undamaged delivery of crops boosts the cooperative model's scalability to urban areas and strengthens consumer confidence in the cooperative.

This promotes a sustainable cycle of production and consumption as well as rural development.

7.6 Conceptual Alternatives Analysis

To assess the design and strategic positioning of the "Senin Bahçen" project, three different production concepts were developed and comparatively analyzed. These alternatives reflect various compromises between organicity, environmental sustainability, production efficiency and costs. The reference concept represents the basic model of the project, while the other scenarios provide alternative paths based on different input configurations.

7.6.1 Comparative Concept Evaluation

The comparative assessment was conducted on six basic criteria: fertilization, irrigation, seed type, weeding (mulching), logistic suitability and cost. In the scoring system, "0" reflects no decisive difference compared to the reference model, "+" reflects a relative advantage and "-" reflects a relative disadvantage. The total net score was calculated according to the relative position of each alternative compared to the reference model. The detailed comparative assessment is presented in Table 17.

Selection Criteria	Senin Bahçen (Reference)	Concept 2	Concept 3	Concept 4
Fertilization	0	0	-	-
Irrigation	0	-	0	-
Seed Type	0	-	0	-
Weeding (Mulching)	0	0	-	-
Logistical Suitability	0	+	0	+
Cost	0	0	+	+
Net Score	0	-1	-1	-2
Rank	1	2	3	4
Organic Production	Organic	Organic	Non-organic	Non-organic

Table 17: Conceptual alternatives comparison for "Senin Bahçen"

7.6.2 Scenario Descriptions

- Reference Concept; Senin Bahçen (Organic, Balanced, Community-Based Model): The reference model represents the design of the “Senin Bahçen” project, which combines fully organic production standards with ancestor seeds, drip irrigation, natural fertilizers, and mulching techniques. The project adheres to organic certification standards while balancing environmental sustainability, community involvement, and economic viability.
- Concept 2; Organic Production Alternative with Lower Input Quality: Concept 2 offers an organic production model that includes natural fertilizer and mulching methods without using chemical pesticides for fertilization and weed control. However, in this concept, local hybrid seeds are used instead of ancestor seeds and wild irrigation is preferred. These preferences both lead to excessive water use and have negative consequences in terms of sustainability. Despite preserving the organic quality, these two factors reduce the environmental suitability of the model. On the other hand, the production of more durable vegetables due to the use of hybrid seeds provides a logistical advantage. However, the production cost is high due to irrigation inefficiency and the expensiveness of organic production. This concept exemplifies scenarios that can maintain organic status despite some entrepreneurial compromises.
- Concept 3; Cost-Oriented, Technically Strong, but Inorganic Model: Concept 3 seems to be a model that emphasizes environmental sustainability with the use of ancestor seeds and drip irrigation system. However, the chemical fertilizer and pesticide applications in the model, which are incompatible with organic production principles, constitute a significant weakness in terms of environmental sustainability, contrary to what is apparent. In addition, from a logistical perspective, it is disadvantageous because the products obtained

from ancestor seeds are more sensitive than those obtained from hybrid seeds. This concept deviates from organic production targets, is a technically balanced but controversial scenario in terms of environmental adequacy, and draws attention with aspects such as the use of local seeds and water efficiency.

- Concept 4; Conventional, High Efficiency and Low Cost Scenario: Concept 4 is a completely conventional production model that includes imported seeds, chemical fertilizers and pesticides, and wild irrigation practices. This significantly reduces production costs, and provides logistic advantages with product varieties that are resistant to transportation. However, the model does not meet the environmental and ethical standards of sustainable agriculture. This concept exemplifies yield and cost-oriented strategies that are closer to the logic of industrial agriculture, and stands in sharp contrast to organic and community-based approaches. In this respect, it reveals how economic priorities can conflict with social and environmental values.

7.6.3 Strategic Evaluation of Conceptual Alternatives

The comparative analysis of alternative concepts clearly shows that only the reference model is fully compatible with the multidimensional sustainability objectives of the project. All scenarios present technically feasible paths, but the reference model uniquely meets environmental sustainability through organic production without any chemical inputs and preservation of biodiversity. In contrast, Concept 2 maintains organic status but reduces ecological efficiency; Concept 3 achieves technical efficiency but violates organic principles through chemical input dependency; Concept 4 maximizes cost efficiency at the expense of environmental responsibility and social mission integrity. Therefore, the comprehensive assessment highlights the superiority of the reference model as the most consistent and sustainable approach for the intended mission and long-term viability of the "Senin Bahçen" project.

7.7 Quality Function Deployment

7.7.1 Customer Requirements Identification

The process of preparing a Quality Function Deployment (QFD) started with the identification of customers' basic needs and expectations. Through extensive surveys and in-depth interviews with target consumers, a customer segment was identified and 13 different customer requirements were identified. These requirements reflect various dimensions such as trust, price sensitivity, dietary preferences, social impact, transparency, certification and technological expectations. Each requirement was assigned an importance level based on customer feedback to prioritize their importance in the design process. Table 18 shows the Requirement List.

Requirement No	Customer Requirement	Definition / Explanation	Source	Importance Level	Weighted Importance
1	Need for Trust	Customer needs to be assured about product authenticity.	Survey + Interview	5	9,80%
2	Price Sensitivity	Customers are concerned about the high cost of organic products.	Survey + Interview	3	5,88%
3	Healthy Nutrition	Desire for access to safe, chemical-free and nutritious food.	Survey + Interview	4	7,84%
4	Social Benefit	Support for models that empower women and contribute to social good.	Survey + Interview	5	9,80%
5	Transparency	Demand to observe and verify how products are grown and handled.	Survey + Interview	5	9,80%
6	Certification	Preference for products with formal organic or ecological certification.	Interview	4	7,84%
7	Small Producer Support	Preference for buying from local or small-scale producers.	Interview	4	7,84%
8	Customized Production	Interest in selecting specific crops or personalizing produce.	Interview	2	3,92%
9	Direct Producer	Demand for buying directly from the farmer or cooperative.	Survey + Interview	5	9,80%
10	Product Diversity	Preference for a variety of fruits and vegetables.	Survey	3	5,88%
11	Sustainability	Sensitivity to ecological and environmentally friendly practices.	Survey + Interview	4	7,84%
12	Online Accessibility	Desire to access services via a website or app easily.	Survey	3	5,88%
13	Digital Process Tracking	Interest in monitoring production via digital tools.	Survey + Interview	4	7,84%

Table 18: Requirement list

7.7.2 Linking Customer Needs to Design Metrics

Having identified customer requirements, the next step is to translate these requirements into measurable design metrics. Each metric corresponds to one or more customer requirements and establishes a clear link between what the customer values and the technical aspects that can be controlled during product and service development. This mapping provided a systematic approach to ensure that design efforts directly address customer expectations. Table 19 shows the Need-Metric List.

Metric No	Metric	Related Needs
1	Organic Production Certificate	1,3,6
2	Price Affordability	2
3	Cooperative Structure	4
4	On-Site Experience	1,5
5	Employment of Local Women	4,7
6	Customized Production	8
7	Direct Interaction with Farmers	9
8	Seed Diversity	10
9	Environmentally Friendly Production Methods	11
10	Digital Platform Availability	12,13
11	Process Monitoring	13
12	Photographic Update	5
13	Certificate of Analysis	1,3,6
14	Customizable Production Area Option	8
15	Drip Irrigation	11
16	Customer Support	12
17	Ancestor Seed Usage	11

Table 19: Need-Metric list

7.7.3 Relationship Matrix Development

In this stage, the strength of the relationships between customer requirements and the relevant design metrics was assessed. Using weighted scores (9 for strong relationships, 3 for moderate relationships, 1 for weak relationships), the matrix provides a quantitative assessment of how strongly each metric influences the fulfillment of a specific customer requirement. This step guides prioritization by highlighting which design features have the most significant impact on customer satisfaction. Table 20 shows the Need-Metric Matrix.

		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
		Organic Production Certificate	Price Affordability	Cooperative Structure	On-Site Experience	Employment of Local Women	Customized Production	Direct Interaction with Farmers	Seed Diversity	Environmentally Friendly Production Methods	Digital Platform Availability	Process Monitoring	Photographic Update	Certificate of Analysis	Customizable Production Area Option	Drip Irrigation	Customer Support	Ancestor Seed Usage
1	Need for Trust	9			9			1						9			1	
2	Price Sensitivity		9	1														
3	Healthy Nutrition	9								3				9				3
4	Social Benefit			9		9												
5	Transparency	1			9			1				3	9	1				
6	Certification	9												9				
7	Small Producer Support			3		9												
8	Customized Production						9		1						9			
9	Direct Producer				1			9										
10	Product Diversity						1		9									
11	Sustainability	3								9						9		9
12	Online Accessibility										9	3	3				9	
13	Digital Process Tracking										9	9	3					
	Total Priority	134	27	60	95	81	21	55	29	48	63	60	66	122	18	36	32	48

Table 20: Need-Metric matrix

7.7.4 House of Quality

The final stage synthesizes all previous analyses within the framework of the House of Quality. This comprehensive matrix not only reflects the weighted relationships, but also examines the interdependencies between the design metrics themselves (positive, moderate or negative correlations). The House of Quality allows for a holistic assessment, aligning customer needs, technical capabilities and strategic priorities, while also identifying synergies between design features. Table 21 shows the House of Quality.

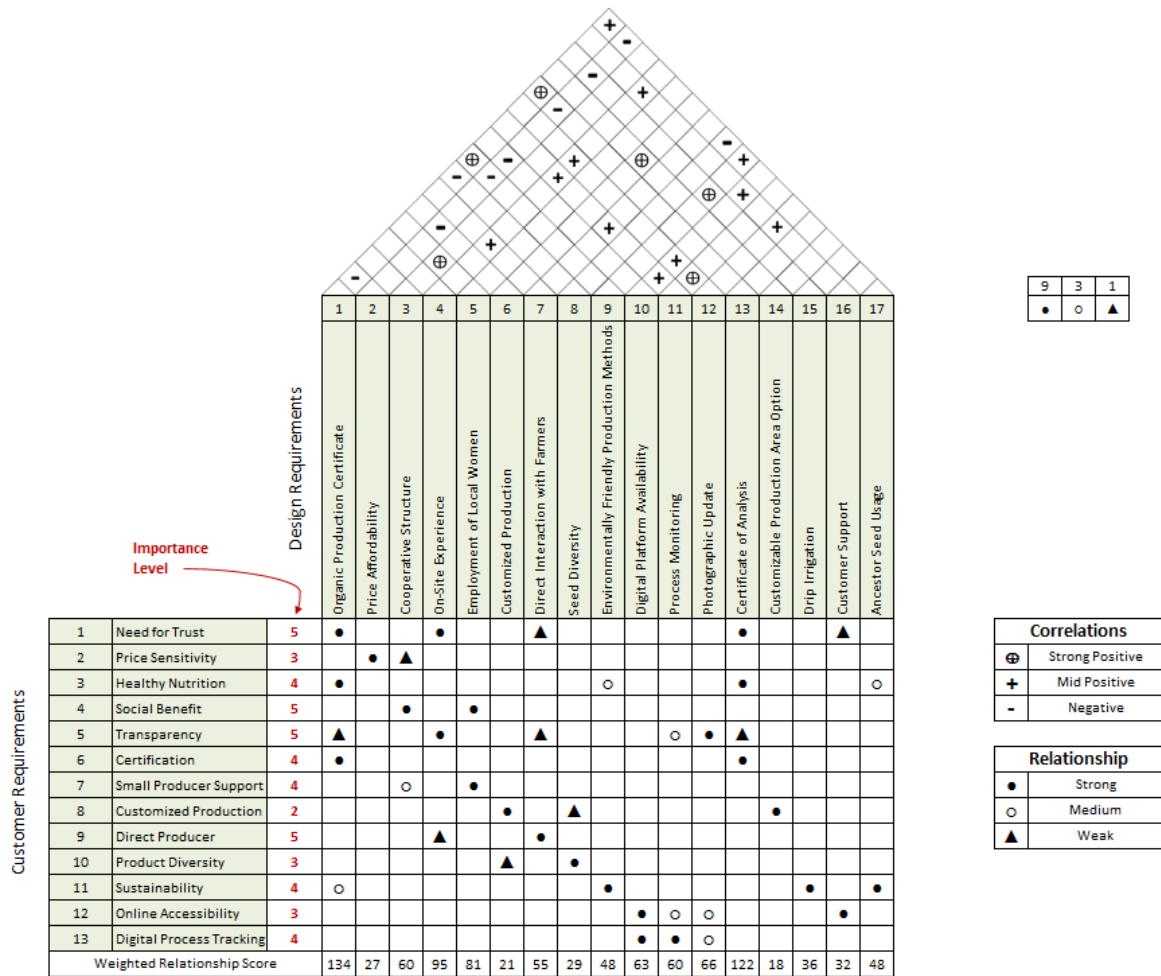


Table 21: House of Quality

7.7.5 Conflict and Challenge Analysis

The House of Quality analysis identified a large number of negative correlations. These indicate potential interactions whereby if one design metric is improved, the other metric may be negatively affected. These conflicts should be carefully managed to ensure an optimal balance between customer satisfaction and operational feasibility.

- Organic Production Certificate vs. Price Affordability: Obtaining official organic certification requires additional inspection, documentation and

compliance costs. These processes increase production costs and create challenges to maintain competitive pricing.

- Organic Production Certificate vs. Seed Diversity: Organic certification requires the use of specific seed varieties approved by certification bodies, thus limiting the flexibility to use some seeds.
- Price Affordability vs. Customized Production: Customized production means that products vary according to individual customer preferences. This increases the complexity and cost of production and conflicts with the provision of affordable prices.
- Price Affordability vs. Environmentally Friendly Production Methods: Environmentally friendly production techniques such as the reduction of chemical inputs or the use of renewable energy require higher initial investment, leading to higher product pricing.
- Price Affordability vs. Digital Platform Availability: The development of advanced digital platforms for customer access, tracking and transparency requires additional technological investments that will entail higher product costing.
- Price Affordability vs. Certificate of Analysis: Laboratory analyses to provide certificates confirming product quality and safety incur extra costs, which conflicts with the availability of affordable prices.
- Price Affordability vs. Drip Irrigation: The implementation of drip irrigation systems improves water efficiency, but requires additional capital investment. This increases production costs and makes an appropriate price strategy difficult.
- Price Affordability vs. Ancestor Seed Usage: The use of ancestor seed varieties limits crop yields and requires more specialized farming practices. This increases production costs and makes an appropriate price strategy difficult.

- Seed Diversity vs. Ancestor Seed Usage: The use of ancestor seeds conflicts with efforts to offer a wider variety of seeds due to their specific adaptation requirements.

7.8 Cooperative and Community Engagement

7.8.1 Cooperative Governance and Institutional Structure

The “Senin Bahçen” project is a social cooperative based on social benefit. The project is not only focused on production, but also on fair income distribution and the empowerment of women's labor. Women producers in the project will also be members of the cooperative and will have an active voice in the management processes. This will support the empowerment of our women producers by ensuring that they are not only workers, but also economic and managerial partners of the enterprise.

In line with the basic principles of the cooperative, the production process will not only be for economic gain, but also for social solidarity and local development. Throughout the production and packaging process, women's labor will be essential and women's participation in social and economic life will be directly supported. Thanks to this structure, rural women will be employed and gain visibility in social life.

Within cooperative structure, five women farmers engaged in agricultural production are direct members of the cooperative and share in the profits generated by its activities. Managerial responsibilities are collectively undertaken by all seven cooperative members. This group of seven equal partners possesses decision-making authority and all cooperative activities are carried out based on the principle of collective governance. This structure promotes both participatory engagement and institutionalized economic solidarity.

The cooperative's financial management has likewise been designed in accordance with principles of sustainability and risk management. Each year, the net profit—after deducting relevant expenses, reserve funds, and other liabilities—is distributed among the partners. In addition, taking into account the risks of natural disasters and climatic uncertainties, 20% of the annual income is allocated to a contingency reserve fund. This fund is intended to ensure full reimbursement to customers in the event that adverse weather conditions, such as frost, damage the harvest. As such, the reserve functions as a vital mechanism for enhancing the cooperative's financial resilience.

Furthermore, 10% of the annual profit is earmarked for long-term investments as part of the cooperative's capacity development plan. These funds support technology-driven and innovative initiatives such as the gradual expansion of the current 4-decare cultivation area, the installation of temperature and humidity sensors in the fields, the implementation of surveillance systems, and the development of AI-powered agricultural decision-support systems. This approach contributes both to increased productivity and to strengthening the cooperative's competitiveness through digital transformation.

The institutional structure of the cooperative consists of three core governing bodies: the General Assembly, the Board of Directors, and the Audit Committee. The General Assembly, comprising all seven members, functions as the highest decision-making authority. Each member holds one vote, and decisions are made in accordance with democratic participation principles. Strategic decisions, such as the approval of annual reports, profit allocation, and board elections, are finalized within this body.

The Board of Directors is composed of three members, including Naci Hakan İlğün, Nihat Devrim Şen, and one woman farmer, thereby ensuring the representation of both producer-members and those with executive responsibilities. The Board is responsible for the cooperative's daily operations, including production processes and

financial and administrative management. It convenes regularly to plan operational activities and makes decisions based on a majority vote.

The Audit Committee consists of a single auditor responsible for independently monitoring the cooperative's financial transactions and the activities of the Board of Directors. The auditor conducts at least one comprehensive audit per year and reports directly to the General Assembly. This institutional framework guarantees that the cooperative operates in a transparent, participatory, and sustainable manner.

7.8.2 Interaction with Society

The project is a multi-stakeholder model that involves not only rural women with the potential to become producers, but also urban consumers and local authorities. Consumers will be able to digitally monitor the production process, visit the farming fields and, if they wish, pick the produce themselves. This physical and digital transparency aims to build trust between the cooperative and its customers.

Cooperation protocols with local municipalities and civil society organizations are also planned. Collaborations with these institutions will enable the allocation of production space and public awareness activities to be carried out effectively. Thus, the cooperative structure will function not only as an economic production area, but also as a social platform that increases local participation and belonging.

7.8.3 Participant Feedback

The “Senin Bahçen” model adopted a user-oriented approach from the very beginning of the design process and analyzed the needs of the target audience through both quantitative (survey) and qualitative (in-depth interviews) methods. In this way, not only market analysis but also analyses to understand consumer motivations and value judgments were carried out and important outputs were obtained. For example, themes

such as transparency, support for women's labor and eco-friendly production emerged strongly from the surveys and interviews.

The project will experience a continuous feedback loop. Consumers will be able to follow the processes and directly submit their suggestions, complaints and evaluations through digital channels. Thus, the model will be continuously improved and sustainability will be reinforced by keeping the interaction with the community dynamic.

7.8.4 Social Capital and Creating Belonging

Due to the nature of the cooperative structure and participatory approach, strong social capital is expected to be built among the individuals involved in the project. Both the bonds that producers will form through their partnership in the cooperative and the experience of consumers being directly involved in the production process will ensure that they embrace this model not only economically but also emotionally.

Through this project, which prioritizes social benefit, the consumer will become not only the buyer of the product, but also a part of the transformation. Factors such as the products coming from our customer's own garden, supporting women's labor, and being produced with environmentally friendly methods create an emotional bond with the consumer and create a social belonging beyond brand loyalty. Thus, both customer loyalty and social solidarity will be increased in the long run.

7.9 Marketing Plan

The “Senin Bahçen” model created in this study is designed as a holistic structure that brings together social benefit elements such as organic farming principles, women's labor, local development and production transparency. It is aimed to evaluate this model not only as an agricultural product supply system but also as a production and distribution network with high social impact. In this framework, unlike traditional approaches to

planning marketing strategies, a multifaceted methodology that focuses on both emotional and functional value propositions has been adopted.

7.9.1 Target Audience Identification and Segmentation

As a result of market research, surveys and qualitative interviews, it has been determined that the target audience of the project consists of individuals living in big cities, who care about healthy nutrition and natural life, aged 25-55, with a high level of education and belonging to the middle-high income group. The main motivation factors of this consumer group were determined to be reliable food supply, transparency at the production stage and the potential to provide social benefits. Especially in the preference of organic products, the opportunity to communicate directly with producers and the traceability of the production process is one of the main factors affecting shopping decisions.

However, it is understood that the group identified as the target audience not only has access to organic products, but also has a close connection with social responsibility values. Social impact factors such as female labor force, environmentally friendly production and cooperatives are among the factors that directly affect the purchasing habits of this group. At this point, it was thought that the value proposition should be developed with an understanding that not only provides organic food but also aims to create social impact. A strategic choice was made to shape the brand positioning around the themes of “direct access to the producer” and “cooperative agricultural model supporting women's labor”.

7.9.2 Value Proposition and Communication Strategy

The products and services offered will not only provide physical benefits, but also make a meaningful contribution to society. Three main axes will be emphasized in marketing communication:

- Transparency and Trust: Product safety will be emphasized through live broadcasting of the production process, packaging systems that can be monitored with QR codes and field visits.
- Social Benefit: Themes related to women's employment, cooperatives and environmentally friendly production approaches will be promoted through story-driven content, aiming to create an emotional connection with consumers.
- Personalized Experience: Each consumer will be able to produce in a garden allocated with their own name, transforming the user experience into a sense of ownership. This feature is planned to be supported with creative content through digital platforms.

7.9.3 Distribution and Channel Selection

Sales and operation processes are expected to be carried out entirely through a digital platform. It is planned to carry out garden rental, production tracking and product delivery processes through the website named “Senin Bahçen”. Promotional activities will be carried out through social media platforms (Instagram, X, LinkedIn) to increase the visibility of the project. Within the framework of this social enterprise project, it was deemed appropriate to adopt a “storytelling” centered communication language with the target audience.

7.9.4 Pricing Strategy

A value-oriented pricing approach was adopted and it was decided that the price level set would be above the market averages. According to the results of the survey, 42.4% of the respondents indicated that they were willing to pay between 40% and 80% more than the market price for organic products that they believed to be safe. In this context, the price differential was aimed to be supported by factors such as production transparency, social responsibility and quality for consumers.

7.9.5 Promotion and Launch Activities

In the promotion phase, collaborations with social media influencers who produce content on healthy living and sustainability are planned. In addition, it is expected to increase interaction with reference-oriented campaigns that will share the experiences of pilot users. Joint promotional events will be organized with universities and women's cooperatives to ensure that the social benefits of the project reach a wider audience.

7.10 Social Feasibility Analysis

This section evaluates the social feasibility of the “Senin Bahçen” project, which aims to promote sustainable organic agriculture, empower rural women economically, and foster community development through a cooperative structure. The analysis considers target group alignment, stakeholder expectations, social acceptability, and long-term social sustainability.

7.10.1 Social Objectives and Target Group Compatibility

The primary social objective of the project is to contribute to the economic empowerment of women in rural areas through their active participation in organic agricultural production within a cooperative governance framework. This model directly addresses systemic challenges such as low female labor force participation, income inequality, and limited employment opportunities for women in rural Türkiye.

Survey data indicate that the target customer group demonstrates significant interest not only in organic food consumption but also in contributing to projects that generate social value, particularly women’s employment and local development. Furthermore, in-depth interviews confirm that prospective customers are motivated by both product quality and the project’s social mission, indicating a strong alignment between project design and societal expectations.

7.10.2 Social Acceptance and Community Participation

The cooperative structure adopted by the project promotes social solidarity, inclusion, and democratic participation. By actively involving local women not only as employees but also as cooperative members with voting rights and management responsibilities, the project creates pathways for both economic and social empowerment. In addition, opportunities for customer interaction such as farm visits, transparent digital monitoring of production processes, and personalized garden ownership serve to foster trust and build an emotional connection between producers and consumers. This dual engagement strategy strengthens social acceptance both within the local community and among the urban customer base.

The collaborative engagement of public institutions, local governments, civil society organizations, and academic institutions further reinforces the project's social embeddedness and public legitimacy.

7.10.3 Socioeconomic Development and Gender Equality Contribution

The project contributes to several dimensions of regional socioeconomic development.

- **Income Generation:** Employment of rural women enables additional household income, reducing poverty and promoting financial independence.
- **Gender Equality:** The project provides women with employment opportunities in agricultural production, logistics, marketing, and cooperative governance, directly contributing to narrowing gender-based labor market disparities.
- **Capacity Building:** Educational and training programs offered within the cooperative model enhance women's technical skills, leadership competencies, and social entrepreneurship capacities.

- Social Inclusion: The cooperative framework supports the integration of women into the formal economy, strengthens their social networks, and increases their visibility in local decision-making processes.

These social outcomes directly align with Türkiye’s national gender equality objectives and the United Nations Sustainable Development Goals, particularly SDG 1 (No Poverty), SDG 5 (Gender Equality), SDG 8 (Decent Work and Economic Growth), and SDG 10 (Reduced Inequalities).

7.10.4 Potential Social Risks and Mitigation Strategies

Potential social risks identified for the project and corresponding mitigation measures include.

- Social Prejudices: Traditional gender roles and cultural barriers in rural areas may initially hinder women's participation. Proactive community outreach, awareness programs, and collaboration with local stakeholders will be implemented to promote acceptance.
- Skill Gaps: Limited prior experience in organic farming and cooperative management may challenge the successful operation of the cooperative. Comprehensive training programs and continuous technical assistance will address skill development needs.
- Social Equity: Ensuring equitable distribution of cooperative revenues and governance participation rights will be strictly governed by transparent, democratic decision-making mechanisms embedded in the cooperative charter.

The “Senin Bahçen” project demonstrates strong social feasibility by successfully integrating disadvantaged women into the organic agricultural value chain while simultaneously meeting increasing consumer demand for transparent and socially responsible food production. The project structure fosters social cohesion, empowers

women economically and socially, and contributes to regional development in alignment with both national policy priorities and global sustainable development goals.

7.11 Legal Feasibility Analysis

This section explains the legal feasibility of the “Senin Bahçen” project, which aims to empower local women in Aydın through organic agriculture within a social cooperative structure. The project is designed in full compliance with Turkish cooperative legislation, agricultural regulations, food safety standards, certification requirements, and taxation obligations.

7.11.1 Legal Status and Cooperative Structure

The project will be established as a “Women's Initiative Production and Business Cooperative” in accordance with the Cooperatives Law No. 1163 (Grand National Assembly of Türkiye [GNAT], 1969). According to Article 1 of the law, cooperatives are legal entities established by natural or legal persons to meet their economic and social needs. The cooperative will be founded with at least seven founding members. All founding members must be citizens of the Republic of Türkiye and legally capable of exercising civil rights (GNAT, 1969, Article 2). The articles of association will be prepared in accordance with the model text published by the Ministry of Trade (2023) and registered with the Trade Registry Office.

7.11.2 Certification and Agricultural Compliance

Organic production activities will be conducted in accordance with the Organic Agriculture Law No. 5262 (GNAT, 2004) and related secondary legislation. In this context, a contract will be signed with a certification body authorized by the Ministry of Agriculture and Forestry (2024a); production sites will be inspected, and organic production certificates will be obtained. In addition, for the sale of retail products, a Business Registration Certificate will be obtained in compliance with the Food Hygiene

Regulation (Official Gazette, 2011). All organic products will be labeled and packaged in accordance with the Turkish organic agriculture logo.

Additionally, the cooperative's production areas will be leased from private landowners. Lease agreements will be executed through notary-approved contracts and registered at the Land Registry Office. In case of utilizing public lands, necessary permits and allocation procedures will be carried out before the relevant public authorities (municipalities, treasury, etc.).

7.11.3 Tax Liabilities and Financial Obligations

The cooperative will be structured to fulfill the following financial and tax obligations.

- In accordance with the Corporate Tax Law No. 5520 (GNAT, 2006) and the specific provisions on cooperative tax exemptions, corporate tax declarations will be submitted. The ultimate corporate tax liability will depend on the scope of the cooperative's activities.
- Value-Added Tax (VAT): Since the cooperative primarily sells fresh vegetables, these products will be subject to a 1% VAT rate as classified under basic food items according to the Presidential Decree No. 2007/13033 and its subsequent amendments. Should the product range include further processed items, a 10% VAT rate will apply in line with applicable regulations.
- Income tax withholding (via Withholding and Premium Service Returns) and social security contributions for employees will be reported and paid regularly.
- Pursuant to the Cooperatives Law No. 1163, annual activity reports and financial statements will be submitted to the Provincial Directorate of Trade.

The cooperative will operate as a non-profit social enterprise model. Revenues will be distributed among the cooperative's women members after covering operational expenses and retaining financial reserves for sustainability. Since corporate tax obligations may arise depending on activity structure, professional tax consultancy services will be utilized to assess potential exemptions and liabilities.

7.11.4 Public Support and Government Incentives

The cooperative will benefit from various public support and incentive programs as listed below.

- KOSGEB (2024): Grant applications will be submitted under the Women Entrepreneurship Support Program.
- IPARD Program (2024) and Ministry of Agriculture and Forestry rural development grants will be pursued.
- İŞKUR (2024): Employment and training support programs will be utilized for women workers.
- Ministry of Agriculture and Forestry (2024b): In-kind and cash support mechanisms for cooperatives will be utilized.

7.11.5 Legal Risks and Preventive Strategies

Potential legal risks and preventive measures during project implementation are outlined below.

- Since the legal definition of “social cooperative” is not yet regulated under Turkish law, operations will be carried out under the legally recognized “Women’s Initiative Production and Business Cooperative” status.
- Land lease transactions will be executed with notary-certified contracts and registered with the land registry.

- Organic production, food safety, and certification processes will be regularly audited by authorized certification bodies.
- All financial and legal transactions will be carried out under the supervision of professional financial and legal advisors.

The “Senin Bahçen” project will be implemented in full compliance with Turkish legislation through the legal framework of a Women’s Initiative Production and Business Cooperative under Law No. 1163. By fulfilling all legal obligations, securing the necessary permits, and utilizing public incentive mechanisms, the project will operate as a legally sustainable and scalable business model.

7.12 Alternative Scenario and Flexibility Policy

During the project design process, alternative models that provide flexibility for various scenarios were also considered, not only depending on ideal conditions. In particular, we focused on feasible structures for situations where official certificates for organic agriculture cannot be obtained or sustainability is in danger. Environmental or governance issues that may arise during the organic production certification process - for example, environmental factors such as the construction of an airport in the region - may harm the organic nature of production. In such cases, the aim is to transition to a model that does not have organic certification but is environmentally sensitive and complies with the principles of sustainable production.

In addition, alternative methods have been developed to make the project sustainable in case financing is not available. In this context, services will be provided in the same social cooperative structure as production activities without organic certification but without pesticides, hormones and chemical fertilizers. This will provide access to a wider group of investors and allow the project to adapt to various budgetary and regulatory requirements.

In terms of seed supply, multiple supplier models were used and back-up suppliers were identified to ensure continuous security and production continuity. In case of delays or access problems that may be encountered in the supply of ancestor seed, alternative sources that will meet quality requirements have been included in the planning. In this way, it is aimed to ensure the continuity of production flow and reduce risks.

8. FINANCIAL ANALYSIS

8.1 Cost Analysis

In order to assess the financial structure and feasibility of the project, all cost items were classified and analyzed as fixed and variable costs. Below, all cost items from production infrastructure to marketing, from training expenses to website setup are detailed. Current purchasing sources for all cost items listed in this table are included in the References section, except for the land and warehouse rental fees, which were obtained via direct communication with a local real estate agent due to the lack of publicly available listings.

- Fixed Costs: The total fixed cost was calculated as ₦86,191.88. These items include agricultural land rental (5 decares), a 500 m² storage area, drip irrigation infrastructure (main line pipe, irrigation pipes, valves, pumps, filter systems), production certification costs, legal and operational costs for the establishment of the cooperative, and technical equipment. In addition, technological and operational infrastructure investments such as computers, camera systems, packaging tables, border fences, website setup and SEO optimization are also included. The annual depreciation expenses of fixed cost items were also calculated according to their specific life cycles, and thus the total annual fixed cost was determined as ₦86,191.88.
- Variable Costs: Total variable cost is calculated as ₦432,285.60. These costs are directly related to production and logistics operations throughout the season. Among the prominent items are the irrigation fee to be paid to the irrigation union (77 days x ₦450) and the transportation cost of the products (₦18.32/kg x 21,705 kg).

- Total Cost: The total annual cost of the project is calculated as ₺518,477.48, which is the sum of fixed and variable costs. Details of the cost items are in the Table 22 and 22.1.

ID	Fixed Costs	Unit Cost	Amount	Total Cost	Duration of Use (Year)	Annual Fixed Cost	References
1	Agricultural Land (1 Decare)	₺ 3.500,00	5	₺ 17.500,00	1	₺ 17.500,00	https://www.sahibinden.com
2	Storage Facility (500 m²)	₺ 20.000,00	1	₺ 20.000,00	1	₺ 20.000,00	https://www.sahibinden.com
3	Permit / Licensing Fees						
3.1	Organic Production Certificate	₺ 6.000,00	1	₺ 6.000,00	1	₺ 6.000,00	https://www.nissert.com.tr/2025/organik
4	Agricultural Insurance Premium	₺ 1.200,00	5	₺ 6.000,00	1	₺ 6.000,00	https://www.tarsim.gov.tr
5	Irrigation Union Subscription Fee (Aydın Ovası Sağ Sahil Sulama Birliği)	₺ 750,00	1	₺ 750,00	1	₺ 750,00	https://www.ayto.org.tr/tr/ilan/119
6	Drip Irrigation System						
6.1	Main Line Pipe (32 mm x 100 m)	₺ 3.012,75	4	₺ 12.051,00	5	₺ 2.410,20	https://iconbanyo.com/urun/eye-yildiz-3
6.2	Drip Tubing (17 mm x 4000 m)	₺ 4.250,00	2	₺ 8.500,00	5	₺ 1.700,00	https://www.intfarming.com/tarim-aletle
6.3	Drip Valves / Fittings	₺ 7,07	100	₺ 707,00	5	₺ 141,40	https://www.trendyol.com/emek/vana-50
6.4	Water Tank (20 Tons)	₺ 45.990,90	1	₺ 45.990,90	10	₺ 4.599,09	https://www.intfarming.com/tarim-aletle
6.5	Water Pump	₺ 3.649,00	1	₺ 3.649,00	7	₺ 521,29	https://www.trendyol.com/mur-cell/2-hp
6.6	Pressure Regulator	₺ 1.061,00	1	₺ 1.061,00	7	₺ 151,57	https://www.hepsiburada.com/seray-kali
6.7	Filter	₺ 3.400,00	1	₺ 3.400,00	7	₺ 485,71	https://www.trendyol.com/firtina-damla
7	Mulch Film	₺ 48.960,00	1	₺ 48.960,00	3	₺ 16.320,00	https://www.intfarming.com/tarim-aletle
8	Packing Tables	₺ 2.169,00	6	₺ 13.014,00	10	₺ 1.301,40	https://www.trendyol.com/sandalie/ratt
9	Computer	₺ 23.999,00	1	₺ 23.999,00	7	₺ 3.428,43	https://www.mediamarkt.com.tr/tr/prod
10	Boundary Fencing Wire	₺ 1.799,00	15	₺ 26.985,00	15	₺ 1.799,00	https://www.trendyol.com/moonshop/pa
11	Boundary Fence Poles	₺ 97,50	70	₺ 6.825,00	5	₺ 1.365,00	https://www.n11.com/urun/10-adet-5-7-c
12	Panel Fence Garden Gate	₺ 3.550,00	1	₺ 3.550,00	15	₺ 236,67	https://www.hepsiburada.com/panel-cit
13	Backpack Sprayer (for Organic Spraying)	₺ 1.093,40	5	₺ 5.467,00	5	₺ 1.093,40	https://www.kocbas.com.tr/veta-16t-kolli
14	Cooperative Establishment Costs						
14.1	Articles of Association Draft	₺ 5.000,00	1	₺ 5.000,00	-	₺ -	https://www.samsunto.org.tr/tescil/koo
14.2	Notary Fees	₺ 4.000,00	1	₺ 4.000,00	-	₺ -	https://www.samsunto.org.tr/tescil/koo
14.3	Book Certifications	₺ 1.000,00	1	₺ 1.000,00	-	₺ -	https://www.samsunto.org.tr/tescil/koo
14.4	Stamp and Branded Documents	₺ 1.000,00	1	₺ 1.000,00	-	₺ -	https://www.samsunto.org.tr/tescil/koo
14.5	Cooperative Training	₺ 2.000,00	1	₺ 2.000,00	-	₺ -	https://www.samsunto.org.tr/tescil/koo
14.6	Initial Capital Contribution	₺ 5.000,00	1	₺ 5.000,00	-	₺ -	https://www.samsunto.org.tr/tescil/koo
15	Licenses & Production Permits						
15.1	District Agriculture Office Production Permit	₺ 300,00	1	₺ 300,00	-	₺ -	https://www.samsunto.org.tr/tescil/koo
15.2	Well Permit / Water Use License	₺ 3.000,00	1	₺ 3.000,00	-	₺ -	https://www.samsunto.org.tr/tescil/koo
15.3	Commercial License	₺ 5.000,00	1	₺ 5.000,00	-	₺ -	https://www.samsunto.org.tr/tescil/koo
15.4	Food Production / Sales Permit	₺ 2.000,00	1	₺ 2.000,00	-	₺ -	https://www.samsunto.org.tr/tescil/koo
16	Website						
16.1	Setup / Development	₺ 15.000,00	1	₺ 15.000,00	-	₺ -	https://armut.com/?gad_source=1&gad_
16.2	Domain Name	₺ 450,00	1	₺ 450,00	1	₺ 450,00	https://armut.com/?gad_source=1&gad_
16.3	Hosting	₺ 1.000,00	1	₺ 1.000,00	1	₺ 1.000,00	https://armut.com/?gad_source=1&gad_
16.4	SEO Optimization	₺ 3.000,00	1	₺ 3.000,00	-	₺ -	https://armut.com/?gad_source=1&gad_

Table 22: Detailed cost items

17	Branding & Marketing Costs						
17.1	Logo Design	₺	2,000,00	1	₺	2,000,00	-
17.2	Organic Agriculture Fairs	₺	10,000,00	1	₺	10,000,00	1
17	Influencer Marketing (Micro)	₺	3,000,00	10	₺	30,000,00	1
17.4	Banner Advertising (Cost Per Click)	₺	0,70	2000	₺	1,400,00	1
17.5	Targeted Instagram Ads (CPC)	₺	0,80	2500	₺	2,000,00	1
17.6	Brochure Printing	₺	2,40	500	₺	1,200,00	1
18	Basic Cooperative Training	₺	1,000,00	7	₺	7,000,00	1
19	Basic Farming Hand Tools						
19.1	Pickaxe	₺	485,10	3	₺	1,455,30	8
19.2	Hoe	₺	459,00	3	₺	1,377,00	8
19.3	Shovel	₺	334,90	3	₺	1,004,70	8
19.4	Rake	₺	898,99	3	₺	2,696,97	6
19.5	Pruning Saw	₺	356,99	3	₺	1,070,97	3
20	Harvesting, Packaging & Transport Equipment						
20.1	Harvest Scissors	₺	419,00	5	₺	2,095,00	5
20.2	Plastic Crates	₺	299,00	50	₺	14,950,00	7
20.3	Electronic Scale (300 kg Capacity)	₺	2,400,00	2	₺	4,800,00	7
20.4	Load Transport Vehicle	₺	1,970,85	2	₺	3,941,70	15
21	Support Materials						
21.1	Gloves	₺	149,95	6	₺	899,70	0,5
22	Seeds						
22.1	Tomato (2 Varieties)	₺	0,10	2000	₺	199,78	-
22.2	Pepper (5 Varieties)	₺	0,10	5000	₺	499,45	-
22.3	Okra (2 Varieties)	₺	0,10	2000	₺	199,78	-
22.4	Eggplant (2 Varieties)	₺	0,10	2000	₺	199,78	-
22.5	Bean (2 Varieties)	₺	0,80	800	₺	638,40	-
22.6	Zucchini (2 Varieties)	₺	2,00	300	₺	598,50	-
22.7	Cucumber (1 Variety)	₺	0,10	1000	₺	99,89	-
22.8	Gherkin (1 Variety)	₺	0,10	1000	₺	98,90	-
22.9	Stuffed Pepper (2 Varieties)	₺	0,10	2000	₺	199,78	-
Variable Costs		Amount	Unit	Expectation	Total Variable Costs		
Irrigation Union Subscription Fee (Aydin Ovasi Sag Sahil Sulama Birliigi)		₺	450,00	/ Hour	92 Days	₺	34.650,00
Product Shipping Cost		₺	18,32	/ KG	21705	₺	397.635,60
TOPLAM						₺	432.285,60
Initial Investment		₺					342.534,50
Annual Fixed Cost		₺					86.191,88
Annual Variable Cost		₺					432.285,60
Annual Total Cost		₺					518.477,48

Table 22.1: Detailed cost items (cont'd)

8.2 Revenue Analysis

The income model created within the scope of the project is based on the rental of 50/100 square meter garden areas. This model not only creates an experience space for customers to grow their own produce, but also provides a supply of organic products.

In line with the customer demand analysis, it is predicted that approximately 80% of the participants will prefer 50 m² and 20% will prefer 100 m² gardens. According to these rates, a total of 67 customers are targeted. 50 m² gardens will be offered to 54 customers for ₺17,500+VAT (%1) per year, while 100 m² gardens will be rented to 13 customers for ₺30,000+VAT (%1) per year.

Of the total land to be rented, 80% of the customer base has been allocated an area of 50 square meters and 20% has been allocated an area of 100 square meters. According

to this structure, the total annual income to be obtained from 50 m² gardens is ₦945,000, while the income to be obtained from 100 m² gardens is ₦390,000, and the total annual income is calculated as ₦1,335,000. This income model not only provides a return on investment, but also allows for the financial sustainability of the structure, which targets social benefit and environmental sustainability together. The expected annual revenues are shown in Table 23.

	Estimated Demand	Estimated Number of Customers	Annual Rental Price	Total Annual Revenue
50 m ²	80%	54	₦ 17.500	₦ 945.000
100 m ²	20%	13	₦ 30.000	₦ 390.000
TOTAL	100%	67		₦ 1.335.000

Table 23: Income projection

8.3 Financial Tables

8.3.1 Cash Flow Scenarios

The following data are based on figures provided by *Investing.com* (n.d.) In order to assess the financial feasibility of the project, a 10-year cash flow projection was first created. This projection was modeled over three different capacity utilization scenarios (Good - 100% occupancy, Medium - 83% occupancy, Poor - 66% occupancy). Similarly, three different discount rate scenarios (Good - 39.50%, Medium - 44.08%, Poor - 49.00%) were defined to reflect project risk and market conditions. The combination of these two different triple scenarios resulted in a total of nine different cash flow scenarios. The detailed results are presented in Table 24.

Year	Cash Flow (Best) (100%)		Cash Flow (Base) (83%)		Cash Flow (Worst) (66%)		Discount Rate Factor (Best) (i = 39,50)	Discount Rate Factor (Base) (i = 44,08)	Discount Rate Factor (Worst) (i = 49,00)
0	₺	342.534,50	₺	342.534,50	₺	342.534,50	1,0000	1,0000	1,0000
1	₺	816.522,52	₺	677.713,69	₺	538.904,86	0,7168	0,6941	0,6711
2	₺	1.175.792,43	₺	975.907,72	₺	776.023,00	0,5139	0,4817	0,4504
3	₺	1.693.141,10	₺	1.405.307,11	₺	1.117.473,12	0,3684	0,3343	0,3023
4	₺	2.438.123,18	₺	2.023.642,24	₺	1.609.161,30	0,2641	0,2321	0,2029
5	₺	3.510.897,38	₺	2.914.044,83	₺	2.317.192,27	0,1893	0,1611	0,1362
6	₺	5.055.692,23	₺	4.196.224,55	₺	3.336.756,87	0,1357	0,1118	0,0914
7	₺	7.280.196,81	₺	6.042.563,35	₺	4.804.929,89	0,0973	0,0776	0,0613
8	₺	10.483.483,40	₺	8.701.291,22	₺	6.919.099,04	0,0697	0,0538	0,0412
9	₺	15.096.216,10	₺	12.529.859,36	₺	9.963.502,62	0,0500	0,0374	0,0276
10	₺	21.738.551,18	₺	18.042.997,48	₺	14.347.443,78	0,0358	0,0259	0,0185

Table 24: 3 scenario cash flow

The discounted cash flows corresponding to these scenarios were calculated separately for each year. In these calculations, present values were determined for each scenario using the current discount factor. The average of these nine scenarios calculated for each year was taken to obtain the annual “Expected cashflow”. Details of the Cash Flows are in the Table 25.

Year	Discounted Cashflows										Expected Cashflow
0	₺ 342.534,5000	₺ 342.534,5000	₺ 342.534,5000	₺ 342.534,5000	₺ 342.534,5000	₺ 342.534,5000	₺ 342.534,5000	₺ 342.534,5000	₺ 342.534,5000	₺ 342.534,5000	₺ 342.534,5000
1	₺ 585.320,8029	₺ 566.714,6863	₺ 548.001,6913	₺ 485.816,2664	₺ 470.373,1896	₺ 454.841,4038	₺ 386.311,7299	₺ 374.031,6929	₺ 361.681,1162	₺ 515.350,6611	
2	₺ 604.202,1191	₺ 566.400,0196	₺ 529.612,3728	₺ 501.487,7588	₺ 470.112,0163	₺ 439.578,2694	₺ 398.773,7986	₺ 373.824,1029	₺ 349.544,1660	₺ 516.909,0225	
3	₺ 623.692,5100	₺ 566.085,5276	₺ 511.840,1455	₺ 517.664,7833	₺ 469.850,9879	₺ 424.827,3208	₺ 411.637,0566	₺ 373.616,4482	₺ 337.814,4960	₺ 518.746,4271	
4	₺ 643.811,6233	₺ 565.771,2103	₺ 494.664,3017	₺ 534.363,6473	₺ 469.590,1046	₺ 410.571,3704	₺ 424.915,6713	₺ 373.408,9988	₺ 326.478,4391	₺ 520.864,2971	
5	₺ 664.579,7401	₺ 565.457,0675	₺ 478.064,8285	₺ 551.601,1843	₺ 469.329,3660	₺ 396.793,8076	₺ 438.622,8825	₺ 373.201,6645	₺ 315.522,7868	₺ 523.264,4886	
6	₺ 686.017,7963	₺ 565.143,0991	₺ 462.022,3846	₺ 569.394,7709	₺ 469.068,7723	₺ 383.478,5792	₺ 452.771,7455	₺ 372.994,4454	₺ 304.934,7738	₺ 525.948,8836	
7	₺ 708.147,4026	₺ 564.829,3050	₺ 446.518,2777	₺ 587.762,3442	₺ 468.808,3232	₺ 370.610,1705	₺ 467.377,2857	₺ 372.787,3413	₺ 294.702,0633	₺ 528.820,0697	
8	₺ 730.990,8672	₺ 564.515,6852	₺ 431.534,4429	₺ 606.722,4198	₺ 468.548,0187	₺ 358.173,5876	₺ 482.453,9274	₺ 372.580,3522	₺ 284.812,7632	₺ 532.180,6640	
9	₺ 754.571,2178	₺ 564.202,2395	₺ 417.053,4213	₺ 626.294,1107	₺ 468.287,8588	₺ 346.154,3397	₺ 498.017,0037	₺ 372.373,4781	₺ 275.255,2581	₺ 535.733,6601	
10	₺ 778.912,2248	₺ 563.888,9679	₺ 403.058,3401	₺ 646.497,1466	₺ 468.027,8433	₺ 334.538,4222	₺ 514.082,0684	₺ 372.166,7188	₺ 266.018,5044	₺ 539.582,3606	

Table 25: 9 scenario discounted cash flow

8.3.2 Discount Rate

In determining the 3-scenario discount rate figures, we proceeded by decomposing them into components in line with the approach frequently used in the finance literature. Türkiye's 10-year government bond yield was taken as 28.69% and the risk-free rate was determined. With the country's CDS premium of 3.39% and the project-specific risk premium of 12%, the total discount rate was determined as 44.08%. While this rate is

considered as the medium scenario, rates of 39.50% and 49.00% were used in the optimistic and pessimistic scenarios, respectively. The following table is based on figures provided by *Investing.com* (n.d.) Details of the Discount Rate items are in the Table 26.

Component	Current Rate	Explanation	Best Scenario	Base Scenario	Worst Scenario
Risk-free Rate	28,69%	Turkey 10-Year Government Bond Rate (28.69%)	25,00%	28,69%	32,00%
Country Risk Premium	3,39%	Turkey's CDS Premium (3.39%)	2,50%	3,39%	5,00%
Project Spesific Risk Premium	12,00%	Project Specific Risk Premium (12%)	12,00%	12,00%	12,00%
Discount Rate	44,08%	Total Discount Rate (44.08)	39,50%	44,08%	49,00%

Table 26: Discount rate calculation method

8.3.3 NPV, ROI, IRR and Payback Period

Running a Monte Carlo Simulation with 10,000 iterations, the probabilities of these nine cash flow scenarios were estimated. The expected values were calculated by multiplying the probability of each scenario by the NPV, IRR, ROI and Payback Period of the scenario. Based on these analyses, the expected Net Present Value (NPV) of the project is TL 5,290,031, the expected Internal Rate of Return (IRR) is 1.62%, the expected Return on Investment (ROI) is 154%, and the expected Payback Period is 0.44 years. Details are in the Table 27.

Cashflow Scenario	Best	Best	Best	Base	Base	Base	Worst	Worst	Worst
Discount Rate Scenario	Best	Base	Worst	Best	Base	Worst	Best	Base	Worst
Probability (Monte Carlo)	0,1813	0,2972	0,1221	0,0732	0,1245	0,0517	0,0429	0,0766	0,0305
NPV	₺ 6.437.711,80	₺ 5.310.473,31	₺ 4.379.835,71	₺ 5.285.069,93	₺ 4.349.461,98	₺ 3.577.032,77	₺ 4.132.428,06	₺ 3.388.450,65	₺ 2.774.229,84
IRR	1,74	1,65	1,57	1,45	1,37	1,29	1,16	1,09	1,02
ROI	1879%	1550%	1279%	1543%	1270%	1044%	1206%	989%	810%
Payback Period	0,42	0,42	0,42	0,51	0,51	0,51	0,64	0,64	0,64

Table 27: Scenarios and financial analysis of the project

8.3.4 Index Scoring Model

3 scenario capacity utilization rates were determined using the Index Scoring Model, a model based not only on technical assumptions but also on market realities. In this model, a weighted scoring is made with scores and coefficients based on a total of 10 criteria such as rental tendency, competitive situation, economic environment, price suitability, product attractiveness and promotional power. As a result of these calculations, 83% capacity utilization rate is determined as the medium scenario value. Index Scoring Model is in the Table 28.

Index Scoring Model	ID	Factor	Coefficient	Score	Weighted Score
	1	Willingness to Rent	0,30	0,92	0,27
	2	Competition	0,20	1,00	0,20
	3	Economy	0,10	0,70	0,07
	5	Affordability	0,10	0,60	0,06
	6	Attractiveness	0,10	0,90	0,09
	4	Marketing	0,05	0,60	0,03
	7	Accessibility	0,05	0,70	0,04
	9	Alternative Use	0,05	0,70	0,04
	10	Environmental Sensitivity	0,05	0,80	0,04
Base Scenario					83%

Table 28: Index Scoring Model

8.3.5 Seasonal Vegetable Price Forecasts

A basket analysis was also conducted to analyze product pricing and the value offered to customers. Based on data from the Turkish Statistical Institute, the average consumption rates of 9 main vegetable products were determined and used as a demand coefficient.

Since the harvest and delivery period of the project is predominantly scheduled for July, the annual food inflation rate in Türkiye for July 2024, recorded at 58.91%, was taken into consideration to forecast 2025 prices (Trading Economics, 2024). In addition, wholesale vegetable prices published by the Istanbul Metropolitan Municipality for the

same month were used as a baseline for price projections (Istanbul Metropolitan Municipality, 2024). By applying the inflation rate to these baseline prices, the estimated weighted average basket price for the 2025 summer season was calculated as 78.04 TL/kg. In contrast, the weighted average price of the vegetable basket provided to customers under the project remains at 64.57 TL/kg. Accordingly, the market value of an average 271 kg produce package from a 50 m² garden amounts to approximately 21,173.90 TL, whereas the same package is offered to customers within the project at a competitive price of ₺17,500+VAT (%1). Details are in the Table 29.

Seasonal Vegetable Price Forecasts					
Product	Demand Coefficient	2024 Wholesale Prices	Food Inflation Expectations	Expected 2025 Prices	Weighted Basket Share
Tomato	30%	₺ 20,00	58,91%	₺ 31,78	₺ 9,53
Cucumber	15%	₺ 20,00	58,91%	₺ 31,78	₺ 4,77
Pepper	15%	₺ 55,00	58,91%	₺ 87,40	₺ 13,11
Eggplant	10%	₺ 40,00	58,91%	₺ 63,56	₺ 6,36
Bean	10%	₺ 100,00	58,91%	₺ 158,91	₺ 15,89
Zucchini	5%	₺ 30,00	58,91%	₺ 47,67	₺ 2,38
Stuffed Pepper	5%	₺ 60,00	58,91%	₺ 95,35	₺ 4,77
Gherkin	5%	₺ 7,00	58,91%	₺ 11,12	₺ 0,56
Okra	5%	₺ 110,00	58,91%	₺ 174,80	₺ 8,74
BASKET	100%	₺ 41,60	70%	₺ 78,04	₺ 66,11
The Total Price the Consumer would Normally Pay for Products (TOTAL 270 KG):					₺ 21.173,90
Our Price					₺ 17.500,00

Table 29: Seasonal vegetable price forecasts

8.3.6 Assurance Fund and Customer Risk Compensation Policy

In order to ensure the sustainability and trust of this business model, an assurance system was designed taking into account the risks that may be encountered due to the nature of agriculture. A “reserve fund” has been established to protect customers against product losses due to adverse weather conditions such as frost, excessive rainfall or drought. This fund will be set aside and kept at a rate of 25% of the company's net profit

each year (approximately 204,130 TL). Thus, the financial cost that may arise in extraordinary situations where production cannot be realized or is severely disrupted will be covered.

In these potential disaster situations, customers will be offered two different solution options. As the first option, customers can receive the same service with a 25% discount the following year. The second option is a full refund of the customer's deposit. This practice reflects the importance the company attaches to transparency, ethical responsibility and customer satisfaction; it also aims to create a business model based on trust in the sector.

9. ALIGNMENT OF OUR PROJECT WITH SUSTAINABLE DEVELOPMENT GOALS OF UNITED NATIONS

As stated by Gözüml (2024) “Women's employment is recognized worldwide as an important way to support economic growth and social development. Reducing gender inequality and increasing women's participation in the labor force are considered as part of the sustainable development goals” (p. 114).

- No Poverty (No.1) : Women's economic independence plays an important role in reducing poverty globally.
- Gender Equality (No.5) : Supporting women's participation in the labor force and ensuring gender equality in employment conditions are directly related to the goal of achieving gender equality.
- Decent Work and Economic Growth (No.8) : Steps taken to provide employment for women on equal terms with men lead to economic growth and decent work conditions.
- Reduced Inequalities (No.10) : When we ensure the presence of women in socio-economic life, social and economic inequalities will also decrease.
- Responsible Consumption and Production (No.12) : The production and consumption of organic products promotes sustainable agricultural practices and minimizes damage to the environment.
- Life on Land (No.15) : With organic farming, the use of chemical fertilizers and pesticides is reduced. This prevents soil contamination, increases biodiversity, and prevents the soil from being exposed to harmful substances that reduce yields in the long term.



Figure 28: Sustainable Development Goals related to our business model

10. IMPLEMENTATION ROADMAP

In order to successfully implement and sustainably scale up the project, a comprehensive Implementation Roadmap has been developed. The Implementation Roadmap outlines the sequential phases of project implementation, covering both the initial set-up and capacity building phases. The Implementation Roadmap is structured to ensure a systematic and controlled rollout of the project components by clarifying the preparation, regulatory compliance, technology integration and market entry activities. Figure 29 shows the Implementation Roadmap of the Senin Bahçen Project.

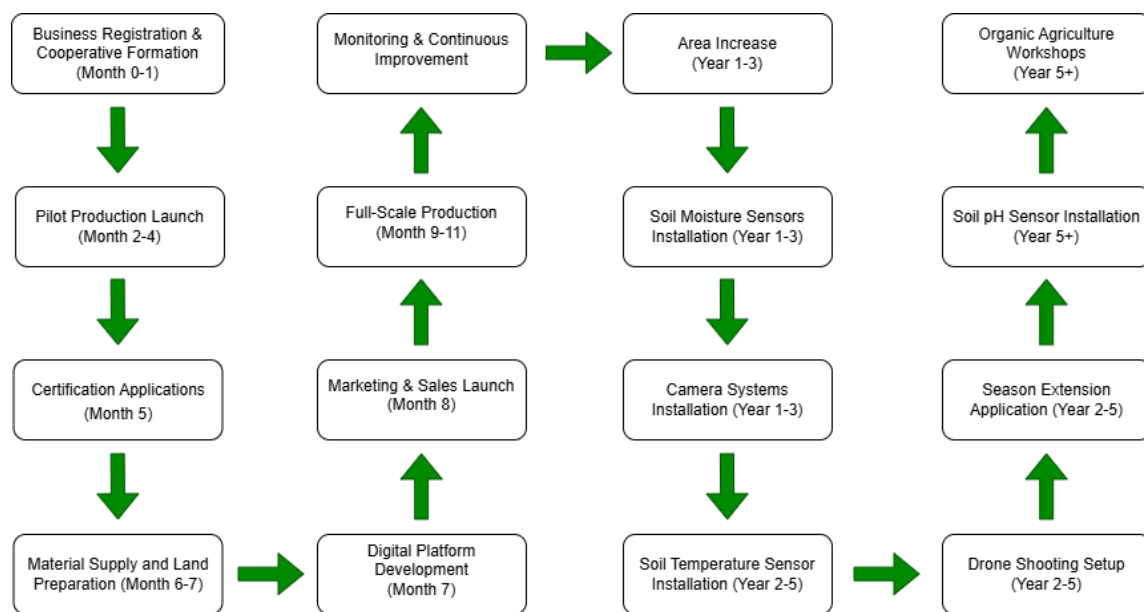


Figure 29: Implementation Roadmap

11. FINAL EVALUATION

This study has developed and thoroughly examined the "Senin Bahçen" project, a sustainable and inclusive agricultural business model that integrates organic farming, social cooperative structures, and women's empowerment in rural Türkiye. Throughout the research process, both extensive quantitative and qualitative methodologies were employed to evaluate the feasibility, social impact, and financial sustainability of the proposed model.

The project has successfully addressed several critical challenges identified in the literature, including the limitations of conventional agriculture, the underrepresentation of women in rural employment, and the increasing consumer demand for transparent and reliable organic food sources. By adopting a cooperative structure that actively includes women as both producers and decision-makers, the model offers a viable pathway to enhance women's economic independence while simultaneously contributing to regional development and social cohesion.

From a market perspective, survey findings demonstrated significant consumer interest in transparency, social responsibility, and trust in organic production. Although trust in existing organic certification remains relatively low, the prospect of direct producer-consumer relationships combined with transparent monitoring systems significantly enhances consumer confidence and willingness to pay premium prices for reliable organic products. In-depth interviews further revealed strong emotional and ethical motivations among potential consumers, highlighting the project's alignment with emerging consumption values that go beyond price sensitivity.

Operationally, the model integrates advanced production planning based on ancestor seeds, environmentally friendly irrigation systems, and digital transparency tools, which together differentiate "Senin Bahçen" from existing alternatives in the Turkish organic agriculture sector. Comparative benchmarking against established platforms such as ÇiftçidenEve, Ovacık Doğal, and Ekozel Organik demonstrated the uniqueness of the

proposed demand-specific production and rental-based garden model, which offers customers a personalized agricultural experience while supporting women's labor and social solidarity.

The financial analyses conducted under multiple capacity utilization and risk scenarios have also confirmed the economic viability of the model. The high Net Present Value (NPV), together with strong ROI, IRR, and short payback period projections, supports the project's long-term financial sustainability even under conservative assumptions. The inclusion of a customer assurance fund and capacity development plan further enhances the project's resilience against agricultural risks and climate uncertainties.

Moreover, the alignment of the project with the United Nations Sustainable Development Goals underscores its broader social significance, particularly in advancing gender equality, poverty reduction, decent work opportunities, and sustainable consumption patterns.

In conclusion, the "Senin Bahçen" model demonstrates a holistic approach that integrates environmental sustainability, economic feasibility, and social empowerment into a coherent and scalable business model. Its cooperative governance structure, customer-centric transparency mechanisms, and emphasis on women's employment position the project as a pioneering initiative capable of contributing meaningfully to both local development and global sustainability objectives. The findings of this study may serve as a valuable reference for future social entrepreneurship initiatives, policymakers, and stakeholders seeking innovative solutions for inclusive and sustainable rural development.

12. APPENDIX

12.1 Taxonomy Table

Research Topic	Research	Author(s)	Date	Country	Method	Conclusion
Social Cooperative	Strong Cooperatives for a Social Solidarity Economy	Act Human	2021	Türkiye	Literature Review Case Study Analysis Policy Recommendations	Emphasizes the importance of cooperatives to strengthen the social solidarity economy.
	World Standards of Social Cooperatives	CICOPA	2018	Global (International Analysis)	Comparative Analysis Criteria Development Data Collection	Describes the international standards of social cooperatives and compare the practices of cooperatives combining social and economic objectives in different countries.
	Social Economy and Social Cooperatives in Turkey: SWOT Analysis of the Current Situation	Demircan Yıldırım, P	2023	Türkiye	SWOT Analysis Statistical Data Analysis Literature Review	Analyze the strengths, weaknesses, opportunities and threats of social cooperative practices in Turkey.
Woman Employment	2015: Advancing Gender Equality: The Co-operative Way	International Labour Organization (ILO)	2015	Global (International Analysis)	Literature Review Field Studies Policy Analysis	Examine the economic and social benefits of social cooperative models that aim to increase women's participation in the workforce.
	2017: Cooperatives and the World of Work		2017			
	2022: Empowering Women in Turkey through Cooperatives: Legal and Structural Strategies		2022	Türkiye		
	Social Norms as a Barrier to Women's Employment in Developing Countries	Jayachandran S.	2021	India & South Asia	Literature Review Econometric Analysis Case Analysis	Analyzes the economic and cultural effects of social norms that prevent women from participating in the workforce in developing countries.
	Women Employment in Türkiye: Challenges and Opportunities	Karabulut, M. & Şahin, Z.	2024	Türkiye	Literature Review Statistical Analysis Policy Recommendations	Analyzes in detail the current situation of women's employment in Turkey, the problems and opportunities faced.

Table 30: Taxonomy table

Organic Agriculture	Organic Agriculture Statistics	TurkStat	2023	Türkiye	Statistical Data Analysis Official Data Sources	Provides up-to-date and reliable data on organic agriculture production in Turkey, the amount of land used and the market share of organic products.
	The World of Organic Agriculture 2023	FIBL	2023	Global (International Analysis)	Global Data Analytics Multi-Country Comparisons Annual Reporting	Analyzes in detail the development, market size and trends of organic agriculture at the global level.
Social Impact & Local Development	Social Cooperatives: European Union Examples in terms of Legislation and Developments in Türkiye	TÜSEV	2018	Türkiye & European Union	Comparative Analysis Literature Review Policy Recommendations	Analyzes legal regulations and structural differences by comparing social cooperative models in the European Union and Turkey.
	Genç İşi Kooperatifi: Who Are We?	Genç İşi Kooperatif	2015	Türkiye	Case Analysis Practical Experience Sharing Local Sampling	Explains the establishment process, objectives and contributions to society of Genç İşi Kooperatifi, a youth-oriented social cooperative in Turkey.

Table 30.1: Taxonomy table (cont'd)

12.2 Original Survey Form (in Turkish)

This survey was originally conducted in Turkish. The full questionnaire is presented below in its original format.

Senin Bahçen Projesi – Pazar Araştırması

Merhaba, biz İstanbul Teknik Üniversitesi, İşletme Mühendisliği son sınıf öğrencileri Devrim ve Hakan. Bitirme projemiz için organik tarım üzerine bir sosyal girişim projesi yürütüyoruz. Kentte yaşayan bireylerin ve ailelerin sağlıklı, güvenilir ve tamamen organik tarım ürünlerine ulaşmalarını kolaylaştırmayı hedefliyoruz ve bu hedefimizi kadın istihdamı ve bölgesel kalkınmaya destek sağlayan, doğaya verilecek her türlü zararı ortadan kaldırmayı hedefleyen bir üretim modeli içerisinde gerçekleştirmeyi amaçlıyoruz. Bunun için aşağıda yer alan soruları objektif ve samimi bir şekilde cevaplamanızı rica ediyoruz. Görüşmemiz anonim kalacak ve sadece proje araştırmasında kullanılacaktır. Yanıtlarınız için şimdiden teşekkür ederiz. Hazırsanız başlayalım.

1. Email *

2. Cinsiyetiniz nedir? *

- ☐ Kadın
- ☐ Erkek
- ☐ Diğer
- ☐ Belirtmek İstemiyorum

3. Yaş aralığınız nedir? *

- ☐ 18-24
- ☐ 25-34
- ☐ 35-44
- ☐ 45-55
- ☐ 55 ve üzeri

4. Eğitim durumunuz nedir? *

(Örnek: Lisans öğrencisi iseniz, Lisans seçeneğini seçiniz.)

- ☐ Ortaöğretim
- ☐ Lise
- ☐ Ön Lisans
- ☐ Lisans
- ☐ Yüksek Lisans ve üstü

5. Çalışma durumunuz nedir? *

- ☐ Öğrenciyim.
- ☐ Yarı zamanlı çalışıyorum.
- ☐ Tam zamanlı çalışıyorum.
- ☐ Çalışmıyorum.
- ☐ Emekliyim.

6. Aylık geliriniz yaklaşık olarak ne kadardır? *

- ☐ 20.000 TL ve altı
- ☐ 20.001 - 40.000 TL
- ☐ 40.001 - 60.000 TL
- ☐ 60.001 - 80.000 TL
- ☐ 80.001 - 100.000 TL
- ☐ 100.000 TL ve üzeri

7. Yaşadığınız bölgeyi tanımlar mısınız? *

- ☐ Şehir Merkezi
- ☐ Şehir Çevresi (Banliyö vs.)
- ☐ Kırsal Alan

8. Aşağıdaki platformlardan hangilerini aktif olarak kullanıyorsunuz? *

(Birden fazla seçebilirsiniz.)

- ☐ Instagram
- ☐ Facebook
- ☐ LinkedIn
- ☐ X / Twitter
- ☐ Forumlar / Bloglar
- ☐ Hiçbiri

9. Sağlıklı yaşam ve doğal beslenme konularında kendinizi nasıl değerlendirirsiniz?

* (1-Kesinlikle ilgili değilim, 5- Kesinlikle ilgiliyim.)

1	2	3	4	5
☐	☐	☐	☐	☐

10. Sağlıklı yaşam ile ilgili aşağıdaki aktivitelerden hangilerini düzenli olarak yapıyorsunuz? * (Birden fazla seçebilirsiniz.)

- ☐ Düzenli spor yapıyorum.
- ☐ Sağlıklı beslenmeye özen gösteriyorum.
- ☐ Organik/doğal ürünler tercih ediyorum.
- ☐ Bitkisel/vejetaryen ürünler tercih ediyorum.
- ☐ Meditasyon/yoga gibi aktiviteler
- ☐ yapıyorum. Hiçbiri

11. Gıda alışverişinizi ağırlıklı olarak nereden yapıyorsunuz? *

(Birden fazla seçebilirsiniz.)

- ☐ Zincir Marketler
- ☐ Semt Pazarları
- ☐ Organik Ürün Marketleri
- ☐ Online Platformlar
- ☐ Doğrudan Üreticiden

12. Sebze alışverişinizi ortalama hangi sıklıkla yapıyorsunuz? *

- ☐ Günlük
- ☐ Haftada 2-3 kez
- ☐ Haftada 1 kez
- ☐ 2 haftada 1 kez
- ☐ Ayda 1 kez
- ☐ Nadiren

13. Sebze alırken sizin için en önemli kriterler nelerdir? *

(Birden fazla seçebilirsiniz.)

- ☐ Tazelik
- ☐ Organiklik
- ☐ Fiyat
- ☐ Menşei (Üretim Yeri)
- ☐ Görünüş

14. Sebze alımlarınızda organik ürünleri tercih ediyor musunuz? *

(1- Kesinlikle tercih etmiyorum, 5- Kesinlikle tercih ediyorum.)

1	2	3	4	5
☐	☐	☐	☐	☐

15. Organik diye pazarlanan ürünlerin gerçekten organik olduğuna ne kadar güveniyorsunuz? *

(1- Kesinlikle güvenmiyorum, 5- Kesinlikle güveniyorum.)

1	2	3	4	5
☐	☐	☐	☐	☐

16. Bir üreticiden doğrudan ve şeffaf bir üretim süreci ile organik ürün alabilme imkanınız olsa ilgilenir misiniz? *

(1- Kesinlikle ilgilenmem, 5- Kesinlikle ilgilenirim.)

1	2	3	4	5
☐	☐	☐	☐	☐

17. Sosyal sorumluluk değerleri (kadın istihdamı, doğa dostu üretim, kooperatifçilik, yerel kalkınmaya destek vb.) tüketim kararlarınızda ne kadar etkili olur? * (1- Kesinlikle etkili olmaz, 5- Kesinlikle etkili olur.)

1	2	3	4	5
☐	☐	☐	☐	☐

18. Güvenilirliğinden emin olduğunuz organik ürünler için, güncel piyasa fiyatının % kaç üstünü ödemeye razı olursunuz? *

- ☐ %20 ve altı
- ☐ %20 - 40
- ☐ %40 - 60
- ☐ %60 - 80
- ☐ %80 - 100
- ☐ %100 ve üzeri

19. Organik ürünlerin online bir şekilde teminine sıcak bakıyor musunuz? *
(1- Kesinlikle sıcak bakmıyorum, 5- Kesinlikle sıcak bakıyorum.)

1	2	3	4	5
☐	☐	☐	☐	☐

20. Organik ürünlerin üretim sürecinin (toprak işleme, ekim, hasat) şeffaflığını takip edebilmek sizce önemli midir? *
(1- Kesinlikle önemsizdir, 5- Kesinlikle önemlidir.)

1	2	3	4	5
☐	☐	☐	☐	☐

21. Organik üretime yönelik bir projeyi destekler misiniz? *

(1- Kesinlikle desteklemem, 5- Kesinlikle desteklerim.)

1	2	3	4	5
☐	☐	☐	☐	☐

22. Organik ürün tüketiminizde aşağıdaki motivasyonlardan hangisi daha ön plandadır? * (Birden fazla seçebilirsiniz.)

- ☐ Kendi sağlığını korumak
- ☐ Ailemin/çocuklarımın sağlığını korumak
- ☐ Yerel üreticiyi desteklemek
- ☐ Çevre dostu üretimi desteklemek
- ☐ Sosyal sorumluluk bilinci

23. Size göre organik ürün alımını cazip kılacak en önemli faktörler hangisidir? * (Birden fazla seçebilirsiniz.)

- ☐ Ürünün Güvenilirliği
- ☐ Sosyal Etki (Kadın İstihdamı, Doğa Dostu Üretim vs.)
- ☐ Kolay Erişim (Online Sipariş vb.)
- ☐ Üretim Sürecinin Şeffaflığı

Google Forms

12.3 Survey Result Summaries and Response Table

This appendix includes the complete survey response data in tabular format. The file contains raw response distributions exported from Google Forms and processed in Microsoft Excel. This data has served as the basis for the statistical analyses presented in Section 5.4.



Senin Bahçen Projesi
- Pazar Araştırması Ya

12.4 R Studio Code Archive

This appendix includes the R scripts used throughout the analysis process of the survey data. All tests—such as Shapiro-Wilk normality test, Mann-Whitney U, Kruskal-Wallis difference tests, and Spearman correlation—were performed using R Studio. The following code block presents the full script in chronological order.



RStudioScript.txt

12.5 Full Transcripts of In-Depth Interviews

- Participant 1 Interview (Male, 58 years old)

Hello, we are Devrim and Hakan, final year Management Engineering students at Istanbul Technical University. For our graduation project, we are conducting a social enterprise project on organic agriculture. We aim to make it easier for individuals and families living in the city to access healthy, reliable and completely organic agricultural products and we aim to realize this goal within a production model that supports women's employment and regional development and aims to eliminate any harm to nature. For this

purpose, we kindly ask you to answer the following questions in an objective and sincere manner. Our interview will remain anonymous and will only be used for project research. Thank you in advance for your answers. Let's start if you are ready.

Question 1: What is your name and surname?

Answer 1: Anonymized.

Question 2: How old are you?

Answer 2: 58

Question 3: What is your educational background?

Answer 3: I graduated from Bilkent Business Administration.

Question 4: What is your profession?

Answer 4: I have my own company where I import and export food raw materials, food preservatives and food improvers.

Question 5: Where and how often do you shop for vegetables? Please explain.

Answer 5: We usually buy from local markets or grocery stores once a week.

Question 6: What are your most important criteria when shopping for vegetables? Please explain.

Answer 6: I make sure that it is fresh and natural. Apart from that, price is also a factor.

Question 7: Do you regularly buy products labeled as natural or organic? If yes, what is your level of confidence in the naturalness and organicity of these products? Please explain.

Answer 7: I normally buy organic products from friends and relatives in the village or from my friend in Çatalca. But I do not buy the products that are sold at high prices as organic in the shelves of the markets. They market products with pesticides and chemicals as organic.

Question 8: Would it be appealing to you to buy directly from a producer you trust, whose organic nature you can be sure of? Please explain.

Answer 8: If it is as you say, it would be attractive, why not.

Question 9: How important is it to you that the producer you will purchase from provides social benefits such as providing women's employment within a social cooperative structure, not harming the environment, and supporting regional producers? Please explain.

Answer 9: Of course, these are good things, we support small producers instead of rich people. The problems you mentioned are actually Türkiye's problems. The issue is not just organic food.

Question 10: Would you look favorably upon a model that offers all the social benefits mentioned in the previous question, where vegetables are produced in season, with ancestor seeds, and without using any chemical inputs, completely organically, in line with your demands (vegetable variety, type, quantity, etc.)? Please explain.

Answer 10: Will the products I want be produced completely organically, in the quantity I want, and sent to me? (Yes) Of course that would be nice.

Question 11: For the model we mentioned in the previous question, would you participate in a subscription model where you can follow the production process on the website or on-site, and even be directly involved in this process, where you rent a 50 or 100 m² garden and the products produced in your garden are shipped to you weekly after the harvest process begins? Please explain.

Answer 11: I do not have time to participate in the processes, as you said, I cannot deal with it for the purpose of a hobby garden. However, if the price is not too excessive, I would like to have organic boxes of the vegetables I want sent to my home with such a subscription.

Question 12: How much more than the current market price would you be willing to pay for organic products produced within the model mentioned? Please explain. (Example: I would be willing to pay approximately 50% more than the market price of products.)

Answer 12: When you say you're going to buy organic produce today, you're already paying twice the market price. On top of that, I'd pay 50% or even 100% more for a package of vegetables that I'm sure are organic and that I know were produced in my own field.

Question 13: In addition to those mentioned, what would you expect from such an enterprise model? (process monitoring, transparency, communication style, etc.) Share your suggestions.

Answer 13: As you said, if I am going to pay a large amount of money from the beginning, I want to make sure that I am not being scammed. You need to provide this to the customers. Apart from this, you say that there will be transparency and process monitoring. I guess I would be convinced under these conditions. Where will this field be? *(It will be in Kuyucak, Aydın.)* We pass by there when we go to Muğla in the summers,

I know it. Since I rented a place there, I would like to stop by and see it while passing by, it would be nice.

Question 14: Would you recommend such a project to your friends or social circle? Please explain.

Answer 14: If I buy it myself and am satisfied with the process, if the products reach me without any problems as you said, and if I am satisfied with the products, of course I would recommend it. If it is going to be a project that supports women's labor and has the social impacts you mentioned, you win. Of course, I would recommend it to my circle.

- Participant 2 Interview (Male, 45 years old)

Hello, we are Devrim and Hakan, final year Management Engineering students at Istanbul Technical University. For our graduation project, we are conducting a social enterprise project on organic agriculture. We aim to make it easier for individuals and families living in the city to access healthy, reliable and completely organic agricultural products and we aim to realize this goal within a production model that supports women's employment and regional development and aims to eliminate any harm to nature. For this purpose, we kindly ask you to answer the following questions in an objective and sincere manner. Our interview will remain anonymous and will only be used for project research. Thank you in advance for your answers. Let's start if you are ready.

Question 1: What is your name and surname?

Answer 1: Anonymized.

Question 2: How old are you?

Answer 2: I am 45 years old.

Question 3: What is your educational background?

Answer 3: I have a master's degree and I am currently doing my doctorate.

Question 4: What is your profession?

Answer 4: I am an industrial engineer and a pilot.

Question 5: Where and how often do you shop for vegetables? Please explain.

Answer 5: Due to my busy work life, I usually don't have the chance to go to the bazaar, if my wife can go, we buy from the bazaar. If she can't go, we usually have to buy from the markets. If we can go to the market, we buy weekly, but if we can't, we don't buy that much, we buy as needed, 2-3 times a week.

Question 6: What are your most important criteria when shopping for vegetables? Please explain.

Answer 6: I generally try not to buy products that look too artificial, I prefer to buy products that look more natural. If it is too cheap compared to the market, there may be a prejudice against the product, so I prefer products that look the most natural to my eye and have an average price.

Question 7: Do you regularly buy products labeled as natural or organic? If yes, what is your level of confidence in the naturalness and organicity of these products? Please explain.

Answer 7: I mean, it says natural on almost all products. Some also say organic, but their prices are higher than others. Frankly, it wouldn't be wrong to say that I don't trust them

at all, both because of this and because of the news we see periodically, such as “products with pesticides returned from the border or detected to be sold”.

Question 8: Would it be appealing to you to buy directly from a producer you trust, whose organic nature you can be sure of? Please explain.

Answer 8: Of course, it would be. As I mentioned in the previous question, I almost never trust the products on the market. If there was a marketer I knew and was sure was doing natural production, I would probably do all my shopping there. Even when I couldn't go to the market, I would have my wife or friends buy it for me, but I still wouldn't buy it from the market.

Question 9: How important is it to you that the producer you will purchase from provides social benefits such as providing women's employment within a social cooperative structure, not harming the environment, and supporting regional producers? Please explain.

Answer 9: I am someone who camps often and I can say that I am an environmentalist and nature lover. In addition, it would definitely make me feel better to know that the products I buy also provide such social support.

Question 10: Would you look favorably upon a model that offers all the social benefits mentioned in the previous question, where vegetables are produced in season, with ancestor seeds, and without using any chemical inputs, completely organically, in line with your demands (vegetable variety, type, quantity, etc.)? Please explain.

Answer 10: It is important to me that it is made with ancestor seeds. I think these seeds should be protected. (Will the products I want be produced?) I found a special greengrocer for myself, why wouldn't I be keen on it.

Question 11: For the model we mentioned in the previous question, would you participate in a subscription model where you can follow the production process on the website or on-site, and even be directly involved in this process, where you rent a 50 or 100 m² garden and the products produced in your garden are shipped to you weekly after the harvest process begins? Please explain.

Answer 11: We already get 2-3 deliveries a day, it would be nice if someone would bring natural vegetables.

Question 12: How much more than the current market price would you be willing to pay for organic products produced within the model mentioned? Please explain. (Example: I would be willing to pay approximately 50% more than the market price of products.)

Answer 12: Products sold as organic are usually above the market price anyway. As you mentioned, if it is completely organic and comes to me, I can give more than 50%.

Question 13: In addition to those mentioned, what would you expect from such an enterprise model? (process monitoring, transparency, communication style, etc.) Share your suggestions.

Answer 13: Transparency is very important, of course, what you have said is very nice, but I also need to be sure that it is true. In order to be sure, I need to be aware of the process and be able to follow the process.

Question 14: Would you recommend such a project to your friends or social circle? Please explain.

Answer 14: The fact that the food products sold are unhealthy and contain chemicals is already a topic that my friends and I talk about. It is not easy to find natural products in a

place like Istanbul. For that reason, if I am happy with it, of course I would do it, why wouldn't I?

- Participant 3 Interview (Male, 28 years old)

Hello, we are Devrim and Hakan, final year Management Engineering students at Istanbul Technical University. For our graduation project, we are conducting a social enterprise project on organic agriculture. We aim to make it easier for individuals and families living in the city to access healthy, reliable and completely organic agricultural products and we aim to realize this goal within a production model that supports women's employment and regional development and aims to eliminate any harm to nature. For this purpose, we kindly ask you to answer the following questions in an objective and sincere manner. Our interview will remain anonymous and will only be used for project research. Thank you in advance for your answers. Let's start if you are ready.

Question 1: What is your name and surname?

Answer 1: Anonymized.

Question 2: How old are you?

Answer 2: I am 28 years old.

Question 3: What is your educational background?

Answer 3: I graduated from Istanbul Technical University, Department of Electronics and Communication Engineering.

Question 4: What is your profession?

Answer 4: I am an electronics engineer.

Question 5: Where and how often do you shop for vegetables? Please explain.

Answer 5: I usually buy it from the grocery store near my house about once every 10 days. I buy it from the market during the hours or days when the grocery store is closed.

Question 6: What are your most important criteria when shopping for vegetables? Please explain.

Answer 6: I usually try to buy the ones that look nice and natural, I guess that's my first criterion. My second criterion is usually the price. Some of them are very cheap and this cheapness makes me suspicious, to be honest I don't prefer them. I usually prefer the ones that are average priced and look natural.

Question 7: Do you regularly buy products labeled as natural or organic? If yes, what is your level of confidence in the naturalness and organicity of these products?

Answer 7: Many of the products I buy say they are natural or organic, so I can say that I buy them regularly. But it would be right to say that I don't trust them at all. As I said, they are supposedly all natural, all chemical-free.

Question 8: Would it be appealing to you to buy directly from a producer you trust, whose organic nature you can be sure of? Please explain.

Answer 8: If I could be really sure that it was organic, it would come. In fact, there is a bit of a situation like that when you buy it from the greengrocer I mentioned. Frankly, I prefer it there because they say that they don't buy it from the market, but that the distributor brings it from the village.

Question 9: How important is it to you that the producer you will purchase from provides social benefits such as providing women's employment within a social cooperative structure, not harming the environment, and supporting regional producers? Please explain.

Answer 9: I am someone who cares about these issues in my own life. I have previously volunteered for a project on women's employment. Also, it is equally important for nature to be protected in these times when we are disregarding nature and preparing our own end. These issues are very effective in my decision.

Question 10: Would you look favorably upon a model that offers all the social benefits mentioned in the previous question, where vegetables are produced in season, with ancestor seeds, and without using any chemical inputs, completely organically, in line with your demands (vegetable variety, type, quantity, etc.)? Please explain.

Answer 10: As I mentioned in the previous question, social contributions are important to me. On top of that, organic production is done and if I can trust that it is organic, I would definitely look favorably on it. I am not much into eating that kind of food, but I could make a small gesture by planting vegetables that my mother likes and sending them to her.

Question 11: For the model we mentioned in the previous question, would you participate in a subscription model where you can follow the production process on the website or on-site, and even be directly involved in this process, where you rent a 50 or 100 m² garden and the products produced in your garden are shipped to you weekly after the harvest process begins? Please explain.

Answer 11: Since I work very hard, I usually can't make much effort to reach the best and natural food. But you say that the products you choose will be produced in your garden

and delivered to your door, and this business will have social benefits. As I said, even if I couldn't consume it all myself, I would send it to my family and friends.

Question 12: How much more than the current market price would you be willing to pay for organic products produced within the model mentioned? Please explain. (Example: I would be willing to pay approximately 50% more than the market price of products.)

Answer 12: I guess I wouldn't pay much attention to the price. We haven't been able to trust what we eat lately, so I think I'll pay whatever it costs as long as it's healthy. Also, if I knew that the money I paid would have other positive outcomes, I could pay more (50%).

Question 13: In addition to those mentioned, what would you expect from such an enterprise model? (process monitoring, transparency, communication style, etc.) Share your suggestions.

Answer 13: I have to be able to trust the producer in this regard. I have to make sure that the products are organic and that the mentioned social benefits are provided. Of course, I would also like to be able to follow the process and be informed.

Question 14: Would you recommend such a project to your friends or social circle? Please explain.

Answer 14: After I took part in the project and was pleased with it, I would of course recommend it. In fact, I am quick to embrace such things, and if I liked it, I would probably talk about it everywhere.

- Participant 4 Interview (Female, 47 years old)

Hello, we are Devrim and Hakan, final year Management Engineering students at Istanbul Technical University. For our graduation project, we are conducting a social enterprise project on organic agriculture. We aim to make it easier for individuals and families living in the city to access healthy, reliable and completely organic agricultural products and we aim to realize this goal within a production model that supports women's employment and regional development and aims to eliminate any harm to nature. For this purpose, we kindly ask you to answer the following questions in an objective and sincere manner. Our interview will remain anonymous and will only be used for project research. Thank you in advance for your answers. Let's start if you are ready.

Question 1: What is your name and surname?

Answer 1: Anonymized.

Question 2: How old are you?

Answer 2: 47

Question 3: What is your educational background?

Answer 3: University graduate.

Question 4: What is your profession?

Answer 4: Textile production, marketing.

Question 5: Where and how often do you shop for vegetables? Please explain.

Answer 5: I buy organic fruits and vegetables from the Tazedirekt online shopping platform once or twice a week.

Question 6: What are your most important criteria when shopping for vegetables? Please explain.

Answer 6: Organic, from a reliable producer.

Question 7: Do you regularly buy products labeled as natural or organic? If yes, what is your level of confidence in the naturalness and organicity of these products?

Answer 7: Frankly, I want to trust organic certificates. At least, I think they will be less toxic than the products of uncontrolled producers sold in chain markets or bazaars. Even in Europe, organic products can contain pesticide residues. I try to choose the safest one I can.

Question 8: Would it be appealing to you to buy directly from a producer you trust, whose organic nature you can be sure of? Please explain.

Answer 8: Of course, it would. There are cooperatives that I shop directly from, or product-based producers (cheese, flour, noodles, butter, etc.). Here, product diversity and logistics criteria come into play. For example, if I were to shop weekly, my goal would be to buy my fruit and vegetables from as few producers as possible in order to reduce shipping costs. Or, purchasing in bulk with neighborhood-based organizations such as food cooperatives is more profitable and sustainable for both producers and consumers.

Question 9: How important is it to you that the producer you will purchase from provides social benefits such as providing women's employment within a social cooperative structure, not harming the environment, and supporting regional producers? Please explain.

Answer 9: Of course, all social benefits are very important for our country, they are not the only criteria, but these features determine my selection priority.

Question 10: Would you look favorably upon a model that offers all the social benefits mentioned in the previous question, where vegetables are produced in season, with ancestor seeds, and without using any chemical inputs, completely organically, in line with your demands (vegetable variety, type, quantity, etc.)? Please explain.

Answer 10: Yes, I would. As I said, pricing and product variety are important factors here. For example, Şaban Burhan Çiftliği sells directly and also through Tazedirekt. When I compared the prices, I didn't see much difference. When I shop from Tazedirekt, it reaches me much faster and I can add different products from other producers in the same basket. That's why I prefer to buy from Tazedirekt most of the time. But when I shop directly from the producer of Degirmeci Dayı, organic flour, butter, noodles, it is much more profitable, these products are very expensive at Tazedirekt. Since they are non-perishable products, when I buy in bulk from the producer every few months, I can access a more economical and safe product without paying shipping fees.

Question 11: For the model we mentioned in the previous question, would you participate in a subscription model where you can follow the production process on the website or on-site, and even be directly involved in this process, where you rent a 50 or 100 m² garden and the products produced in your garden are shipped to you weekly after the harvest process begins? Please explain.

Answer 11: I would love to do this, but I don't think I can directly participate in the conditions of Istanbul and our fast-paced life, to be honest. Instead of a model where a 50 m² garden aims to feed a single family, it seems to me that a 1 acre cultivated area aims to feed 20 families (I am writing this as an estimated figure). (Food community model)

How many types of products will grow in a 50 m2 garden, will there be parsley every week, for example, etc. reasons.

Question 12: How much more than the current market price would you be willing to pay for organic products produced within the model mentioned? Please explain. (Example: I would be willing to pay approximately 50% more than the market price of products.)

Answer 12: A price between 30% and 50% higher than chain markets may be acceptable.

Question 13: In addition to those mentioned, what would you expect from such an enterprise model? (process monitoring, transparency, communication style, etc.) Share your suggestions.

Answer 13: I want to be confident in their competence. That's why I prefer both entrepreneurs and the people they employ to be people who have received training in organic farming and who improve themselves by following innovations.

Question 14: Would you recommend such a project to your friends or social circle? Please explain.

Answer 14: Yes, I do. Access to non-toxic food is very important today.

- Participant 5 Interview (Female, 54 years old)

Hello, we are Devrim and Hakan, final year Management Engineering students at Istanbul Technical University. For our graduation project, we are conducting a social enterprise project on organic agriculture. We aim to make it easier for individuals and families living in the city to access healthy, reliable and completely organic agricultural products and we aim to realize this goal within a production model that supports women's

employment and regional development and aims to eliminate any harm to nature. For this purpose, we kindly ask you to answer the following questions in an objective and sincere manner. Our interview will remain anonymous and will only be used for project research. Thank you in advance for your answers. Let's start if you are ready.

Question 1: What is your name and surname?

Answer 1: Anonymized.

Question 2: How old are you?

Answer 2: 54

Question 3: What is your educational background?

Answer 3: I graduated from ITU Management Engineering and Auzef Istanbul University Child Development.

Question 4: What is your profession?

Answer 4: In the first half of my life, I worked in finance companies as an Management Engineer. Now I am a child development specialist and a forest kindergarten guide from alternative education models.

Question 5: Where and how often do you shop for vegetables? Please explain.

Answer 5: I only shop at the market; I meet all my vegetable and fruit needs from Seferihisar's local producer market and village markets.

Question 6: What are your most important criteria when shopping for vegetables? Please explain.

Answer 6: I make sure to buy from local producers and consume vegetables and fruits on time. I have not consumed greenhouse products for about 5 years. For example, I never buy and eat tomatoes and cucumbers in winter, and I consume whatever is available in each season. For the past 5 years, I have been living close to the countryside, so I buy from the producers around me, and I try to find producers who do not use chemicals in the production process. I try to grow my own tomatoes, peppers and similar products by producing seedlings from ancestor seeds.

Question 7: Do you regularly buy products labeled as natural or organic? If yes, what is your level of confidence in the naturalness and organicity of these products?

Answer 7: I used to care more about and believe in organicism, but the process I went through shook my trust in organic products and organic certificates. Factors such as the fact that the regions where so-called organic products are grown are close to the industry are the reason for this.

Question 8: Would it be appealing to you to buy directly from a producer you trust, whose organic nature you can be sure of? Please explain.

Answer 8: For me, it is important that the criteria for organic product certificates are transparent, that the soil planted, the seed used, the pesticides used, the region mapped and explained in order to avoid suspicion. As someone who is engaged in agriculture and who has discussed the issue in detail with the farmers I have met and tried to learn about it, I have observed that the yield is very low when agriculture is practiced with zero pesticides.

Question 9: How important is it to you that the producer you will purchase from provides social benefits such as providing women's employment within a social cooperative

structure, not harming the environment, and supporting regional producers? Please explain.

Answer 9: Of course, it is important. In order to support regional producers, I make sure to buy the products of peasant women who are already producing.

Question 10: Would you look favorably upon a model that offers all the social benefits mentioned in the previous question, where vegetables are produced in season, with ancestor seeds, and without using any chemical inputs, completely organically, in line with your demands (vegetable variety, type, quantity, etc.)? Please explain.

Answer 10: I have also experimented with ancestor seeds, agriculture is a very demanding business. With zero chemicals, the amount of production is very limited, you don't spray, but when the neighboring fields do, your product is damaged. Especially until it reaches big cities, the prices go up considerably. I think regional production and placing it on the market in the same region should be a priority.

Question 11: For the model we mentioned in the previous question, would you participate in a subscription model where you can follow the production process on the website or on-site, and even be directly involved in this process, where you rent a 50 or 100 m² garden and the products produced in your garden are shipped to you weekly after the harvest process begins? Please explain.

Answer 11: My situation is a bit special, I'm happy to say that I'm past the stage of buying fruits and vegetables by cargo. Cargo prevalence and carbon footprint are the issues I am most sensitive to. That's why I moved away from the city as an Istanbul person, but of course it is a good model for those who live in the city. I have heard of similar examples before, it would be a great experience for children, you can emphasize this in your model,

and if there is a problem in the product, learning the reasons from the producer will support the ecological literacy of all of us.

Question 12: How much more than the current market price would you be willing to pay for organic products produced within the model mentioned? Please explain. (Example: I would be willing to pay approximately 50% more than the market price of products.)

Answer 12: Maybe not 50% but 30-35% would be acceptable for me. This is how the producers or village markets I shop at now price their products.

Question 13: In addition to those mentioned, what would you expect from such an enterprise model? (process monitoring, transparency, communication style, etc.) Share your suggestions.

Answer 13: Transparency is important, whatever is being done should be shared with reasons. The crop may be unexpectedly unproductive that year, the reasons for this should also be shared. After all, we are dependent on nature. If there is a frost or an unexpected insect outbreak, etc., it should be explained. There are dozens of topics like this that we as city dwellers do not know. But information will be very valuable in order to learn, to be open to learning and not to lose touch with nature.

Question 14: Would you recommend such a project to your friends or social circle? Please explain.

Answer 14: I would definitely recommend it. I am happy to announce and be an ambassador for projects that I trust. When it comes to healthy eating, you hit the nail on the head! Good luck!

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